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17 July 1978
SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

MANUALS, TECHNICAL: EQUIPMENTS AND SYSTEMS

CONTENT REQUIREMENTS FOR

This specification is approved for use by all interested Commands of the Departments of the Navy and the Marine Corps and is available for use by all other Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth Naval Sea Systems Command (NAVSEA) and Naval Electronic Systems Command (NAVELEX) content requirements for manuals necessary for installation, operation, organizational-, intermediate-, and depot-level maintenance, repair, and parts support (without services of manufacturer's representatives) of equipment and systems.

1.1.1 Scope of coverage. Unless otherwise specified, type I and type II manuals shall include all information necessary to perform all (organizational, intermediate, and depot-level) maintenance. No manual shall omit maintenance or logistics information for any equipment or component thereof designated repairable, regardless of the location of the designated repair point (ship, tender, or shore facility) unless the contract requires that the information be included in a separate document to be provided to the Government and reference is made thereto.

1.2 Classification. Manuals shall be of the following types, as specified (see 6.2 and 6.6):

Type I - Hull, mechanical and electrical equipment manuals.
Type II - Electronic and interior communications equipment manuals.
Type IIS - Service test electronic and interior communications equipment manuals.
Type IIX - Experimental electronic and interior communications equipment manuals.
Type III - Systems manuals.

1.3 Data items. The following technical manual data items are detailed by this specification (see 6.2):

(a) Book plan.
(b) Quality program.
(c) Manuscript (review) (see 3.11.3 and 3.13.4).
(d) Preliminary manuals (see 6.5.10).
(e) Reproducible copy.
    (1) Preliminary manual.
    (2) Final manual (basic).
    (3) Change (permanent and interim).
    (4) Revision.
(f) Printed manuals.
(g) Status reports.
(h) Photolithographic negatives.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-M-8910 - Manual, Technical, Illustrated Parts Breakdown, Preparation of.
MIL-M-9868 - Microfilming of Engineering Documents, 35mm, Requirements for.
MIL-M-9868/1 - Microfilming of Engineering Documents (35mm), for Naval Ship System.
MIL-D-23140 - Drawings, Installation Control, for Shipboard Electronic Equipment.
MIL-M-24100 - Manuals, Technical: Functionally Oriented Maintenance Manuals (FOMM) for Equipment and Systems.
MIL-M-24365 - Maintenance Engineering Analysis: Establishment of, and Procedures and Formats for Associated Documentation; General Specification For.
MIL-P-24534 - Planned Maintenance Subsystem, Development of Maintenance Requirements Cards, Maintenance Index Pages, and Associated Documentations.
MIL-M-38748 - Microfiche for Engineering/Technical Data, Reports, Studies and Related Data, Requirements For.
MIL-M-38784 - Manuals, Technical: General Style and Format Requirements for.
MIL-P-38790 - Printing Production of Technical Manuals; General Requirements for.
MIL-M-38807 - Manual, Technical, Illustrated Parts Breakdown, Preparation of.

STANDARD

MILITARY

MIL-STD-12 - Abbreviations For Use on Drawings, Specifications, Standards and in Technical Documents.
MIL-STD-1364 - Standard General Purpose Electronic Test Equipment.
MIL-STD-17-1 - Mechanical Symbols (Other than Aeronautical, Aerospacecraft and Spacecraft Use) (Part I).
MIL-STD-17-2 - Mechanical Symbols for Aeronautical, Aerospace and Spacecraft Use (Part II).

PUBLICATIONS

MILITARY

DOD 5220.22M - Industrial Security Manual for Safeguarding Classified Information.
NAVSEA 0967-LP-000-0120 - Handbook of Electronic Circuits.
NAVSEA 0969-LP-019-7000 - Application Guide for Electronic Test Equipment.
NAVSEA S9086-00-STM-000/NSTM - Naval Ships Technical Manual.
NAVSEA S9086-CZ-STM-000/CH90 - Inspections, Tests, Record, and Reports.
NAVEXOS P-35 - Publications and Printing Regulations.

HANDBOOKS

MILITARY

H4-1 - Cataloging Handbook, Federal Supply Code for Manufacturers, United States and Canada, Name to Code.
H4-2 - Cataloging Handbook, Federal Supply Code for Manufacturers, United States and Canada, Code to Name.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings Rules and Regulations

(Application for copies should be addressed to the ATA Tariff Section, 1616 P Street, N.W. Washington, DC 20036).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
X3.5-70 - Flowchart Symbols and Their Usage in Information Processing.
Y14.15-66 - Electrical and Electronics Diagrams.
Y32.2-70 - Graphic Symbols for Electrical and Electronics Diagrams, (Including Reference Designation Class Designation).
Y32.10-67 - Graphic Symbols for Fluid Power Diagrams, Including Supplements.
Y32.14-73 - Logic Diagrams; (Two-State Devices) Graphic Symbols for.
Y32.16-75 - Reference Designations for Electrical and Electronics Parts and Equipments.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Precedence. In the preparation of technical manuals, when conflict exists between the requirements of this specification and its referenced documents, this specification shall take precedence. When conflict exists between the requirements of the contract and this specification or its referenced documents, the contract shall take precedence.

3.2 Security classification. The security classification of manuals shall be assigned by the command or agency concerned. Marking, handling, and production of all classified material shall be in accordance with the DOD 5220.22M Industrial Security Manual for Safeguarding Classified Information. DD Form 254 (Contract Security Classification Specification), which constitutes a part of the contract for all classified material is designed to indicate, by a combination of a check list and narrative comment, the classified areas of information involved in the contract or subcontract, and particularly to identify the specific items of information which require security classification protection.

3.3 Level of writing. The level of writing and development of text for types I, II, IIS, and III manuals shall be in accordance with MIL-M-38784 and the following:

(a) As a general guide, the level of writing should be for a high school graduate having specialized training as a technician in Navy training courses.
(b) Summary portions in chapter 1 shall be written to the level of command and supervisory personnel.
(c) Operating instructions shall be written to the level of an operator having previous experience in the operation of similar or related equipment.
(d) The level of writing for other portions of the manual shall be to that of the least experienced or technically skilled Navy technician who is intended to utilize the manual for training on, or operation or maintenance of the system/subsystem/equipment. The grade level (i.e., E-4, E-5, E-6, etc.) varies with the nature and complexity of the hardware; the level ultimately chosen will be derived from the Ship Manning Document or similar planning information. The Government will furnish the contractor with more detailed guidance in the course of outline/book plan reviews (where required) or other early in-process review stages.
(e) This technical manual will be used by personnel with various educational backgrounds. While they are able to learn similar material when presented in a classroom environment, it should be assumed that individual reading skills are no higher than 9th grade. Unless a comprehension standard is cited in the contract or order which sets a different criteria for readability (which is a factor of comprehension) reading level (GL) shall be established by use of the recalculated Flesh Reading Ease Formula where GL=0.39 (words/sentences) + 11.8 (syllables/word)-15.59.

3.3.1 Type IIX. Except when the manual will be used by the Navy for operational evaluation, IIX manuals shall be written to the level of an engineer familiar with the type of equipment involved. The level of writing for IIX manuals to be used during operational evaluation shall be the same as that of IIS manuals.

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3.3.2 Accessibility and referencing of data. Cross referencing techniques in text and illustrations shall provide rapid access to all maintenance data (equipment and equipment circuits) in such a manner that the technician may proceed through the maintenance action without interruption or dead-end. Referencing shall be as specified in 3.7.4.

3.3.3 Warnings, cautions, and notes.

3.3.3.1 Warnings and cautions. WARNING AND CAUTION statements shall be included for emphasis in all applicable installation, operating, maintenance, and repair procedures and data in accordance with MIL-M-38784, and as follows:

(a) A WARNING statement shall be used to call particular attention to a step of a procedure which, if not strictly followed, could result in serious injury or death of personnel.
(b) A CAUTION statement shall be used to call particular attention to a step of a procedure which if not strictly followed, could result in damage, to or destruction of equipment.
(c) WARNING and CAUTION statements shall immediately precede the text of those procedural steps where failure to strictly follow the particular step of a procedure would be likely to result in personal injury or death or damage to or destruction of equipment. The WARNING or CAUTION shall state the consequences of failure to strictly follow the procedural step to which it applies.
(d) The use of WARNING or CAUTION statements shall be limited to significant cases so that too extensive use of these statements will not make technicians and operators apathetic of them and thus reduce their value.
(e) WARNING or CAUTION statements shall not contain procedural steps or alternate procedural steps, nor shall they be numbered.
(f) When a WARNING or CAUTION consists of two or more paragraphs, the heading "WARNING" or "CAUTION" shall not be repeated above each paragraph. The format of either text (e.g. indentation of italics) should suffice to identify the following paragraphs.
(g) If it is necessary to precede a paragraph by both a WARNING and CAUTION, the WARNING shall precede the CAUTION.

3.3.3.2 Notes.

(a) Notes shall not be used with or in lieu of procedural steps.
(b) Notes, shall be confined solely for use in narrative or illustrative non-procedural data. Notes shall never be used in lieu of WARNINGS or CAUTIONS, nor shall they be used to modify an action.
(c) Notes in narrative or illustrative non-procedural data shall be in accordance with MIL-M-38784.

3.4 Issues and arrangement.

3.4.1 Issues. The issues of a technical manual shall be as specified in the contract or order and as specified in 3.4.1.1 through 3.4.1.5.2.

3.4.1.1 Basic manual. The basic issue of a technical manual shall be the original edition prepared in full conformance to this specification and in accordance with the requirements of the applicable contract.

3.4.1.2 Preliminary manual. A preliminary manual shall be prepared for type I, type II, or type III manuals, as specified (see 6.5.10.1 and 6.5.10.2).

3.4.1.3 Commercial manual. A commercial manual shall be a manufacturer's existing manual containing technical information covering assembly, installation, operation, servicing, overhaul, and parts identification, normally furnished by a manufacturer to purchasers of his equipment. As required, this manual shall be supplemented (see 3.4.3.3). (The commercial manual is sometimes referred to as an "off-the-shelf" manual.) When no commercial manual exists, or it requires extensive change, a basic manual conforming to all requirements of 3.4.1.1 should be supplied.

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3.4.1.4 Supplement manual. A supplement manual shall be a document that complements the data in a basic manual. Supplements shall only be used to provide information and data concerning the addition of a unit or assembly to an equipment, such that all equipments will not be modified, and where its use with the equipment technical manual is effective. It shall not be used in lieu of a change or revision to the manual when such information should be integrated into the manual.

3.4.1.5 Revisions. Revisions shall be of two types, as specified in the contract or order, and shall be prepared in accordance with MIL-M-38784 and this specification (see 3.10.4 and 6.5.13).

3.4.1.5.1 Updated revision. An updated revision shall incorporate configuration modifications and all previous data issued as changes to the existing manual, and shall be prepared in accordance with the content and format arrangement of the basic manual.

3.4.1.5.2 Complete revision. A complete revision shall be a completely rewritten manual and shall comply with all of the content and format requirements of this specification.

3.4.2 Arrangement. Manuals shall contain the following data, as applicable, to provide instructions for installation, operation, and maintenance and shall be arranged as follows:

Front matter
Technical content
Appendixes
Index
User activity comment sheets

3.4.2.1 Front matter. The front matter shall consist of the following, as applicable:

Cover
Title page
List of effective pages
Change record

Types I and III
Approval and procurement record page

Content assurance pages
Table of contents
List of illustrations
List of tables

3.4.2.1.1 Cover and title page. The cover and title page shall be prepared in accordance with MIL-M-38784 and shall include, if applicable, a supersedure notice (see 3.10.2). Manuals procured by the NAVSEA or the NAVELEX or any of its subordinate or field activities, shall show the respective systems command in the authority notice but shall include the seal of the systems command below the subtitle. Title pages for permanent changes shall be in accordance with 3.10.3. The cover and title page shall include the manufacturer's name. Where the manual is applicable to more than one manufacturer, the prime manufacturer shall be identified. For multi-volume manuals the volume number shall be indicated on the top-right hand corner; all volumes shall carry the same nomenclature (prime title). Chapter numbers and titles of chapters contained in each volume shall be indicated under the manual prime title (e.g., chapters 1, 3, 4, 6; general information, functional description, etc., respectively).

3.4.2.1.2 List of effective pages. The list of effective pages shall be prepared in accordance with MIL-M-38784 (see 3.10.2 and 3.10.3).

3.4.2.1.3 Change record. A permanent change record prepared in accordance with MIL-M-38784 shall follow the list of effective pages.

3.4.2.1.4 Approval and procurement record page (types I and III only). As required, an approval and procurement record (APR) page shall be included and conform in content to figure 1.

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3.4.2.1.5 Content assurance pages. Content assurance pages shall follow the title page or APR page and shall conform to figure 2. In multi-volume manuals, these pages shall be included in each volume covering the content of that volume.

3.4.2.1.6 Table of contents, list of illustrations, and list of tables. These items shall be prepared in accordance with MIL-M-38784. In multi-volume manuals, volume I shall contain a complete table of contents and list of illustrations covering all volumes. Other volumes shall contain their own listings.

3.4.3 Technical content. The technical content shall be as specified in 3.4.3.1 through 3.4.3.4.

3.4.3.1 Basic manual. The technical content for the basic manual shall include as applicable the following:

Type I manual - (see 3.5 through 3.5.10).
Type II, IIS, IIX - (see 3.5 through 3.5.11.2).
(Note: For paragraphs not applicable to type IIS and type IIX manuals (see 3.5.11.1 and 3.5.11.2).
Type III manual - (see 3.6 through 3.6.10.11.2).

3.4.3.2 Preliminary manual. A preliminary manual will be prepared for type I, type II, and type III manuals (see 6.5.10.1 and 6.5.10.2) as specified in 3.11.2 or the contract or order.

3.4.3.3 Commercial manual. A commercial manual shall include the basic data as specified in 3.5.2 through 3.5.10 and may be arranged in any logical format or layout. A supplement shall be provided with each existing printed commercial manual to provide any required data not included in the commercial manual (see 3.4.3.2).

3.4.3.4 Supplement manual. A supplement manual shall be prepared in accordance with MIL-M-38784 as applicable, and shall contain all data specified in 3.5.2 through 3.5.10 that is not included in the commercial or technical manual it supplements. Special attention shall be given to interface information, by detailed reference to the basic technical manual or by repetition of the data. Format requirements shall parallel and augment the basic manual. Production requirements shall conform to the applicable requirements of 3.8, providing that the basic manual with the supplement is effective for use.

3.4.4 Appendixes. Appendixes shall conform to MIL-M-38784 and the following (see 6.2):

(a) For classified equipment, a copy of DD Form 254 applicable to the equipment shall be included as an appendix to each volume of the manual.
(b) Manual changes shall not be issued in the form of an appendix.

3.4.5 Index. An alphabetical index, prepared in accordance with MIL-M-38784, shall be prepared for all manuals containing over one hundred pages.

3.4.6 User activity comment sheets. The manual shall include three Technical Manual Deficiency Evaluation Reports (TMDR) or user activity comment sheets (see figures 3 and 4), with header identification filled in as applicable, in each separately bound volume. These sheets shall be located immediately following the last page in the manual.

3.5 Technical contents for equipment manuals (types I, II, IIS, and IIX). (See also the equipment military specification invoked in the contract or order for any special technical content requirements for a specific equipment.)

3.5.1 Applicability. All paragraphs apply to type I, and type II manuals except those designated as applying only to type I or type II manuals. Certain paragraphs between 3.5.2 through 3.5.10 are applicable to types I and II manuals and are set forth in single column (across the page) format. Requirements unique to type I manuals are identified as type I and are set in the left hand column only, with a heading indicating the type of manual to which it applies. Requirements for type II manuals are identified similarly and are set in the right hand column only.

3.5.1.1 Type I (single component), type IIS, and type IIX manuals. All requirements for type I manuals apply for type I (single component) manuals, and type II requirements apply to types IIS and IIX manuals, except those requirements designated below which need not be included, providing that the purpose of the technical manual for which procured, is not adversely affected.

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Type of manual	Requirements which may be excepted	
I Single component manuals (See 6.5.1)	3.5.3.3 3.5.4.2 3.5.4.3 and 3.5.4.3.1 3.5.5.1 (b) through (e) 3.5.5.2.2 through 3.5.5.2.5 3.5.7.1(c) 3.5.7.2.2 3.5.7.2.3.2 through 3.5.7.2.3.4	3.5.7.2.4 through 3.5.7.2.12 3.5.9.1 3.5.9.2 3.5.9.5 3.5.10.1 through 3.5.10.3
IIS Service test equipment manuals	3.5.5.2(c) 3.5.6.2 3.5.6.4(h) 3.5.7.2.1.1 3.5.7.2.1.2	3.5.7.2.3.3 3.5.7.2.12 3.5.9.3.2.1(c) 3.5.9.5 3.5.10.9.1
IIX Experimental equipment manuals	3.5.5.2 3.5.6.2 3.5.6.4(h) 3.5.7.2.1.1 3.5.7.2.1.2 3.5.7.2.1.3 3.5.7.2.1.4 3.5.7.2.2	3.5.7.2.3.3 3.5.7.2.3.4 3.5.7.2.10 3.5.7.2.12 3.5.8.3 3.5.9.3.2 3.5.9.5 3.5.10.9.1

Note: Unless otherwise noted, subparagraphs of excepted paragraphs are also excepted.

3.5.2 Chapters. The contents shall be arranged in chapters in accordance with the following:

Chapter 1 - General Information and Safety Precautions.
Chapter 2 - Operation.
Chapter 3 - Functional Description.
Chapter 4 - Scheduled Maintenance.
Chapter 5 - Troubleshooting.
Chapter 6 - Corrective Maintenance.
Chapter 7 - Parts List.
Chapter 8 - Installation.

3.5.3 Chapter 1, General information and safety precautions. All safety precautions necessary for the protection of personnel and the ship shall be included or cross-referenced as the initial paragraph(s) ahead of the introduction. These precautions shall be prepared in accordance with 3.3.3.1. Otherwise, the content of this chapter shall be such that command level, supervisory personnel, and other users having a general interest in the equipment can easily and rapidly determine the purpose, physical and functional characteristics, and the operational capabilities of the equipment.

3.5.3.1 Introduction. The introduction shall provide an explanation of the purpose, scope, supersedure data, and applicability of the technical manual, including the models, serial numbers, and configurations covered. The interface relationship of the technical manual to other referenced publications and the relationship of the equipment to referenced systems or other equipment shall also be included. Warranty/guarantee information shall be included, as applicable.

3.5.3.2 Equipment description. The equipment description shall describe the intended use (why, where, when, how, and with what), capabilities, and limitations of the equipment. Text covering physical description or structural arrangement shall be brief, with special attention given to avoiding unnecessary or repetitious details that are easily illustrated. The equipment, or all units of the equipment shall be clearly illustrated and identified. If the technical manual covers more than one equipment configuration, a table defining the differences shall be included. Assembly and detail drawings as appropriate shall be included. Title blocks and notes may be included on drawings.

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3.5.3.3 Equipment illustration. For installations consisting of more than one unit/assembly, a pictorial illustration representing the equipment, or all units comprising the equipment shall be included and shall be designated figure 1-1 (see figure 5). The illustration shall show the major units of the equipment, relative size of each unit, basic interconnections between units, and their relationship with other equipment. The illustration shall be a left-hand full page or foldout (never backed up) and shall be assigned the folio (blank/1-0).

3.5.3.4 Reference data. Reference data, equivalent to the following, shall be included in tabular form:

(a) Descriptive (identification plate data) which identifies manufacturer, type, model, component identification number (CID), and Repairable Identification Code (RIC), if available, as applicable.
(b) Functional characteristics, such as: power requirements, horsepower, pressure, capacity, modes of operation, power output, frequency, pulse characteristics, sensitivity, selectivity, including tolerances, where applicable.
(c) Capabilities and limitations, such as: pounds of thrust, knots, turning radius, minimum and maximum ranges, degree of coverage, resolution, accuracy.
(d) Rated outputs, such as: wattages, voltages, horsepower, gallons per minute.
(e) Environmental characteristics, such as: ambient temperatures, heat dissipation per unit, humidity limits.

3.5.3.5 Equipment, accessories and documents supplied. A tabular listing of all equipment and documents supplied shall be included. List the equipment and its units, and accessories (special tools, test equipment, miscellaneous parts, and Government furnished items) which form a part of, or are supplied with the equipment. The table shall include the following:

(a) Column 1, Quantity. This column shall contain the quantity of each unit and accessories supplied with the equipment.
(b) Column 2, Item name or nomenclature. This column shall contain the official name (e.g., pump, winch) or nomenclature (name and designation) of each component, unit, or accessory.
(c) Column 3, CID number, RIC number, or unit number. This column shall contain the CID, RIC, or unit number of each equipment unit or accessory,
(d) Column 4, Overall dimensions. This column shall contain the crated (if available) and uncrated height, width, and depth in inches (or inches and centimeters) of each unit or accessory.
(e) Column 5, Weight and volume. This column shall contain the crated (if available) and uncrated weight and volume in cubic feet of each unit and accessory.

Type I	Type II
3.5.3.5.1 - Equipments and systems (such as controls, foundations mounts, couplings, and appurtenances) and related publications, which are not furnished with the basic equipment covered by this manual, but which attach or relate importantly to the basic equipment, shall be described in sufficient detail to establish their correlation with respect to physical and functional interfaces. If applicable, a reference should be made to the shipbuilder's power plant technical manual which contains additional information. TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.4	3.5.3.6 Equipment and publications required but not supplied. A tabular listing of all test equipment and publications required but not supplied with the equipment or system shall be included. Unless otherwise specified, NAVSEAS 0969-LP-019-7000 shall be used as a guide for the selection and application of test equipment required for shipboard use. If a measurement requirement is established which cannot be satisfied by the SCAT (subcategory) codes listed in this guide, or alternate test procedures cannot be developed utilizing the listed SCAT codes, a request for assistance should be directed to: Commander, Naval Ship Engineering Center Washington, D.C. 20362 ATTN: SEC 6181D

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Type I (Continued)	Type II (Continued)
TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.4	The listing shall contain: (a) Column 1, SCAT code. When applicable, this column shall contain the four digit group of numbers, which is included in NAVSEA 0969-LP-019-7000, used to identify a measurement requirement. (b) Column 2, Test equipment category. This column shall contain the item name of each item of equipment required. (c) Column 3, Representative test equipment model number. This column shall contain the model number of the Standard or Substitute Standard General Purpose Electronic Test Equipment (GPETE), listed in MIL-STD-1364, which can be used to satisfy the measurement parameters. (d) Column 4, Equipment test parameters. This column shall specify the range of test parameters which must be satisfied by this test equipment item (it shall not define the range(s) of the test equipment model). (e) Column 5, Application. This column shall indicate the intended use of the test equipment (e.g., scheduled maintenance, troubleshooting, corrective maintenance, or installation). 3.5.3.7 Field changes and factory changes. A table of field and factory changes shall be included as follows: (a) Column 1, Change number. This column shall list the change number for each field and factory change considered and included in the preparation of the manual. (b) Column 2, Nomenclature. This column shall list the equipment nomenclature and serial numbers of equipments affected by the change. (c) Column 3, Description. This column shall contain a brief statement identifying the change and its purpose.

3.5.4 Chapter 2, Operation. Operating instructions shall include all the procedures necessary to enable operating personnel to efficiently and effectively use the equipment in accomplishing its designated task. These operating instructions shall be in sufficient detail to allow operators, not trained on this equipment, of independently and safely operating the equipment without additional training or explanation.

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3.5.4.1 Introduction. The introduction shall describe each operator's relationship to the equipment and each other, and shall identify those units having controls and indicators which he is expected to use in the performance of his duties. The introduction shall be supported by illustrations which identify and locate all operator controls and indicators. The intended function and application of the equipment shall be fully explained so that the operators will know exactly what he/they should expect to accomplish with the equipment.

3.5.4.2 Controls and indicators. A description of all operator controls, indicators, protective devices and jacks shall include the following:

(a) Names of panel designations as marked on the equipment.
(b) Positions and operating functions for each control, and the normal operating condition of each indicator in each of the operating functions.
(c) The text shall be supported by detailed illustrations (see figure 6).
(d) When more than one operator is required to operate the equipment, his designated position and function, and the relationship to his controls and indicators shall be specified.

3.5.4.3 Operating procedures. Operating procedures shall include the following:

(a) Operator turn-on. All steps necessary to bring the equipment from OFF through STANDBY condition to full operation.
(b) Modes of operation. Procedures for each mode of operation, e.g., manual, automatic, local, remote, etc. The use and relative advantage of each mode shall also be described.
(c) Operation under interfering conditions. Describe the equipment anti-jamming and interference reduction features, the advantages of each feature, and the operating procedures to be followed in all possible situations. Supporting illustrations (such as indicator displays, wave forms, etc.) shall be included which provide typical observations of jamming and interference for evaluation by the operator.
(d) Operator turn-off. This procedure shall include all steps necessary to bring the equipment from full operation through STANDBY to OFF condition.
(e) Battle-short or emergency operation. This procedure shall cover operating the equipment during emergency conditions (control failure, air failure, lube oil failure, loss of cooling water, etc.). Emergency operator instructions shall be included. Provide a warning or caution to return the equipment to proper operation when the emergency is over.
(f) Emergency turn-off. This procedure shall cover turning the equipment off during an emergency (fire, water, smoke, hazard to personnel, loss of coolant, normal power, etc.).

3.5.4.3.1 Method of presentation. Operating procedures shall be presented in tabular form and shall be in concise, simply-worded, step-by-step procedures and shall include the following:

(a) A short explanation of the operation to be performed.
(b) Initial safety requirements (actions, inspections, and reference to emergency turn-off procedures).
(c) Connection of any accessory equipment not permanently connected.
(d) Instructions for obtaining or confirming the presence of all critical inputs such as power, coolant, air, signal, air conditioning, etc.
(e) Procedures for setting controls and making adjustments which must be accomplished by the operator prior to equipment turn-on.
(f) Procedures for determining operational readiness and the acceptable indications expected from built-in indicators such as meters, lamps, gages, cathode ray tubes, and recorder readouts.
(g) Milestones in the operational status of the equipment shall be identified and included by brief statements such as "the generator is now in STANDBY."
(h) Visual or aural observations which occur as a result of an operator action, such as boom lowering, sweep rotation, blower motor running, etc.
(i) Procedures that can be hazardous to personnel or equipment shall be emphasized by WARNINGS or CAUTIONS placed immediately prior to the specific step involving the possible hazard. "Notes" shall not be used (see 3.3.3.1).
(j) Illustrative material supporting the procedures shall identify and locate all operating controls and indicating devices as well as normal in-use positions or indications.
(k) Operator's checks and adjustments in proper sequence.
(l) Operator's maintenance actions and schedules.

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3.5.4.4 Operators maintenance instructions and schedules. These instructions shall define maintenance tasks and schedules to be performed by the operator. The maintenance tasks shall be limited in scope such that they will not be in conflict with his operational commitment; will not be beyond his technical training; will not be dangerous; will not be the responsibility of the maintenance technician; and will not potentially compromise the operation of the equipment. Normally these tasks will be restricted to minor adjustments, cleaning, filter replacement, servicing and lubrication.

3.5.4.4.1 Method of presentation. Maintenance procedures shall be presented in tabular form and shall be in concise, simply-worded, step-by-step procedures and shall include the following:

(a) A short explanation of the task to be performed.
(b) Initial safety requirements (actions, inspections, and reference to emergency turn-off procedures).
(c) Separately identified and defined steps for each task and major sub-routine.
(d) A limit of 10 steps to each identified task or sub-task.
(e) Procedure that can be hazardous to personnel or equipment shall be emphasized by WARNINGS and CAUTIONS placed immediately prior to the specific step involving the possible hazard. "Notes" shall not be used (see 3.3.3.1).
(f) Illustrative material supporting the procedures shall identify and locate all maintenance points with clear pictorials showing essential details.
(g) Reference to the standard log sheet.

3.5.4.5 Standard log forms (model set). Unless otherwise specified by the contract or order, a model, or model set, standardized log form will be developed and included at the end of chapter 2 for equipment requiring the periodic recording of data for monitoring its performance and accomplishing trend analysis.

3.5.4.5.1 Applicability. Operating machinery and equipment which are integrated into functioning ships/facility systems similar, but not limited to, the following types of equipment, shall include the model log form:

(a) High pressure air compressors (HPAC).
(b) Diesel engines.
(c) Air conditioning plants.
(d) Carbon dioxide scubbers.
(e) Carbon monoxide - hydrogen burners.
(f) Steam/vapor distilling plants.
(g) Turbine generators.
(h) Electronic equipment cooling water and dry air systems.
(i) Boilers.
(j) Propulsion plant.

3.5.4.5.1.1 Waiver of requirement. Guidance regarding standard log form requirements should be obtained from the procuring activity; sample forms are contained on figures 7 and 8 and listed in NAVSEA S9086-CZ-STM-000/CH 90. When it is deemed that the development of such a form would be of no value to the Government for the equipment being covered, the preparer shall provide justification and rationale and request, in writing, a waiver for standard log form requirements at least three months prior to submittal of the review manuscript.

3.5.4.5.2 Criteria. The "model" standardized log form shall serve as an operating log which will be a guide to all ships of all classes. It is not intended that this model will in any way inhibit an individual ship from accommodating its specific needs through the addition or deletion of readings as deemed necessary on a case basis. The basic criteria to be utilized in developing the model shall be:

(a) Analyzing the various parameters associated with monitoring the performance of the equipment to ascertain those that provide trend determinant information.
(b) Synthesizing the results of the above analysis to ensure that only "key" (i.e., necessary) parameters are included: the goal being to keep to an absolute minimum the total number of log readings required in running the plant.

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(c) Arranging the order of monitoring on the appropriate log forms in such a manner that the log keeper and the reviewing officer can quickly tell when problems, either short-term or long-term in nature, are beginning to develop (i.e., by function as opposed to order of physical arrangement).
(d) Duplicate readings shall be eliminated where feasible - i.e., where the parameter is generally associated with a single function and where duplicated readings are not normally compared with one another.

3.5.4.5.3 Format. The following considerations should be given in development of the log form:

(a) Log forms shall be planned in columnar format for simple use, with adequate space for identification, data and check off. The time line should normally be on an hourly or daily basis and may be arranged on a vertical (left side) or horizontal axis (top), depending on the number of monitoring points.
(b) Sheets shall have one dimension of 8-1/2 inches. Sheets should limit the second dimension to 11 inches when practical, by using the reverse side for continuation.
(c) Monitoring points should indicate the normal values for each monitoring point, and should indicate the maximum (MAX) and minimum (MIN) tolerance for which normal operation can be expected.
(d) Values, that if were to fall below the minimum or above the maximum, that would present danger to personnel, or would cause damage to the equipment, should be indicated in "red" or by bold face type or other method to be conspicuous.
(e) Instructions for use of the form shall be included on the form or with the set, when necessary. Codes, such as H = High; L = Low; B = Bubbles that are used should be described in the instructions or in a legend.
(f) Action procedures shall be indicated for danger readings (see (d)) such as shutdown; cutback; open valve, etc. in the instructions or by a separate related column.
(g) Space should also be provided to indicate the make, size, model, application, and location of the equipment, or other relevent information.
(h) Space for signing the watch, normally on the reverse side.
(i) Space should also be provided, if required, for certification by the watch officer, engineering officer, etc., normally on the reverse side.

3.5.4.5.4 Training summary. A training summary for use of the form shall also be included in chapter 2. It shall indicate how to use the chart and an explanation of the significance of the monitoring points and readings.

3.5.4.5.5 Request for number. The contractor should request a log form stock number from the procuring activity so that copies of the form can be reprinted and placed in stock.

3.5.5 Chapter 3, Functional description.

Type I	Type II
3.5.5.1 Chapter 3 shall include a description of how the equipment operates. The description shall be simplified technical language and shall be supported by simple line illustrations, preferably on the same page. A building block technique shall be used to functionally describe the operation of the equipment as follows: (a) Major subassemblies of the equipment shall be described and illustrated as shown on figure 9. (b) Interaction of major subassemblies shall be described and illustrated as shown on figure 10.	3.5.5.2 Chapter 3 shall include a detailed analysis of the principles of operations of the overall equipment and its functions. The development of the equipment output in every mode of operation shall be described. The structure and organization of this chapter shall parallel the organization of chapter 5, troubleshooting. The text shall refer to and support the chapter 5 functional troubleshooting diagrams. The descriptions shall be presented in successive levels of increasing detailed as follows: (a) Overall level (level 1). The text shall support and refer to the overall functional block diagram (see 3.5.5.2.1). All major functions such as transmit, receive, control, display, power distribution, cooling, etc. shall be described at this level.

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Type I (Continued)	Type II (Continued)
(c) Detailed mechanical and electrical functional description shall be described and illustrated as shown on figure 11. (d) A description of how it works or operates shall be illustrated as shown on figure 12. (e) Simplified diagrams in accordance with 3.5.5.2.2 through 3.5.5.2.5.	(b) Major function level (level 2). Describe the development of each major function and equipment output at the level of detail shown on its related signal flow diagram, logic diagram, or other diagram included in Chapter 5. The text shall support and refer directly to the diagrams. (c) Level 3 (circuit level). Briefly describe conventional electronic circuits found in NAVSEA 0967-LP-000-0120; refer to the maintenance schematic diagrams in chapter 5. Describe in detail circuits not covered in NAVSEA 0967-LP-000-0120; support these circuits by simplified schematic diagrams (see 3.5.5.2.2). Describe alternating current (a.c.) and direct current (d.c.) power distribution in detail; support the descriptions by reference to the power distribution diagrams in chapter 5. Describe mechanical devices, cooling systems, etc.; support the descriptions by reference to diagrams specified in 3.5.5.2.4 and 3.5.5.2.5.
TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.5.2.2	3.5.5.2.1 Overall functional block diagram. The overall functional block diagram shall show the major functions of the equipment correlated in a logical manner to show outputs, inputs, cooling, built-in-test-equipment, air pressurization, power distribution, etc. (see figure 13). Hardware packaging shall be subordinated to the functional arrangement (see figure 14). The following shall apply: (a) For multifunction equipments, whether single or multiunit, each major function shall be represented by a block and shall show the functional generation of outputs, cooling, air pressurization, power distribution, etc. All functions covered in Chapter 5 shall be shown on this diagram. (b) The blocks shall be corrected by lines and arrowheads showing the direction of the flow. (c) Each block shall be identified by the functional name only. (d) Each equipment input and output shall be identified by title. Waveforms shall be included as applicable. (e) Modes of operation shall be identified by title or symbols, as applicable. (f) Ancillary equipment shall be shown by blocks when the ancillary equipment is associated with major functions. Ancillary blocks shall be identified by nomenclature and shall be identified as "(Ancillary)".

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3.5.5.2.2 Simplified schematic diagrams. These diagrams shall show, by means of graphic symbols, the electrical connections and functions of a specific circuit arrangement. These diagrams shall be arranged functionally to show the operation of the circuits in the same manner as illustrated in NAVSEA 0967-LP-000-0120.

3.5.5.2.2.1 Integrated circuits and micro-miniature capsules. Simplified schematics in accordance with NAVSEA 0967-LP-000-0120 or logic diagrams showing input and output connections, truth tables, and waveforms as applicable, supported by a brief description of overall operation, shall be included for circuits within different types of nonrepairable integrated circuits and micro-miniature capsules. Manufacturer's data sheet information may also be included if it contains the technical contents of this requirement and meets the format and legibility requirements of this specification.

3.5.5.2.3 Simplified piping diagrams. These diagrams (hydraulic, pneumatic, or fluid) shall show the interconnection of components by piping, tubing, or hose, and sequential flow in the system. Pumps, heat exchangers, valves, gages, etc., shall be clearly identified.

3.5.5.2.4 Mechanical schematic diagrams. These diagrams shall show sufficient detail to explain the operational sequence and arrangement of a mechanical device including the electrical control circuits (see figure 15). Nomenclature, symbols, part identifying numbers, and necessary descriptive data shall be shown as required. Gears, shafts, clutches, levers, mechanically-driven switches, motors, synchros, etc., shall be shown in functional arrangement. Gear ratios or number of teeth and direction of rotations, etc., shall be given. Symbols used on these diagrams shall conform to MIL-STD-17-1 and MIL-STD-17-2.

3.5.5.2.5 Digital equipment. The functional description of digital equipment shall be similar to that for conventional equipment. However, the functional and hardware makeup of digital equipment requires a variation in the method of presentation and descriptive requirements. Special considerations relative to the functional description of digital equipment are specified herein.

3.5.5.2.5.1 Method of presentation. An overall description of the functional relationship of the logic sections, units, and assemblies comprising the equipment shall be provided. The following shall be included:

(a) An introduction to, and description of the signal characteristics:

(1) Signal levels or bi-stable states utilized by the equipment; i.e., true (1) and false (0), and their relative voltage levels.
(2) Interpretation of the word-code bit structure; i.e., address, instruction, or data bit codes.
(3) Signals, identified by their operational accomplishment.

(b) Overall and functional block diagrams and descriptions as required by 3.5.5.2.1.

(c) Logic principles shall be described beginning with an introduction to the basic digital logic symbology used in the manual. Each logic function shall be described and supported by Boolean equations, truth tables, simplified logic diagrams, and timing diagrams.

(d) Functional description of power distribution, power supplies, and regulators shall conform to 3.5.5.2(c).

3.5.6 Chapter 4, Scheduled maintenance. It is intended that the engineering effort required to develop preventive maintenance data be expended only once and that the data, where applicable be used both in this chapter and in Maintenance Engineering Analysis (MEA), Logistic Support Analysis (LSA) and/or maintenance requirements cards (MRC) where one or more of the latter three data items are required by the contract. The preparation of this chapter of the manual should therefore be delayed until completion of the MEA, LSA and/or MRC; submissions of the manual manuscript for review should be marked "TO BE SUPPLIED UPON COMPLETION OF LSA". When the approved MEA, LSA or MRC data is available, it shall be included in the manual in the identical technical content and, wherever practicable, in the identical format. When included in the manual, such MEA, LSA or MRC data shall be integrated with other technical data required by this specification to be in this chapter. The manual as delivered shall reflect the detailed maintenance requirements of the completed Maintenance Requirements Cards. This chapter shall contain preventive maintenance procedures and performance test instructions to be accomplished on a scheduled or condition monitoring basis. NAVSEA S9086-00-STM-000/NSTM and MIL-P-24534 may be used as baselines for developing the procedures. When a maintenance requirement card or a reference standards book is available or prepared under the same contract, this information may be duplicated in the technical manual without change in format. These procedures shall be correlated with the installation

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checkout requirements of 3.5.10.9 (types I and II) such that any maintenance or performance test procedure also required for checkout may be properly referenced from chapter 8, Installation. The Installation Standards Summary Sheet (see 3.5.10.9.3.2) shall also include space for any maintenance or performance standard test result that should be recorded by the installer. The recorded information will provide a reference standard to the Navy technician when troubleshooting or when needed for installation acceptance certification or for certifying proper overhaul.

3.5.6.1 Introduction. The introduction shall be an explanation of the purpose, scope, and arrangement of the scheduled maintenance data. When a preventive maintenance procedure is critical to the operation of the equipment and the schedule for servicing is absolute (not just recommended), this information shall be conspicuously written as a caution. The following applicability statement shall be included: "The scheduled maintenance instructions in this manual are intended to duplicate those furnished in the Planned Maintenance System (PMS). In case of conflicts, the PMS documentation takes precedence. Such conflicts should be reported immediately on the user comment sheet in accordance with the maintenance procedures for this manual."

Type I	Type II
3.5.6.5 Maintenance plan. A comprehensive, systematic, step-by-step maintenance plan, providing a complete description of the requirements and tasks to be accomplished for achieving, restoring, and maintaining the operational capability of the equipment shall be included. The maintenance plan shall be a time-phased presentation defining and detailing the most effective and efficient procedures for performing examinations, tests, replacements, preventive maintenance tasks, and overhauls. 3.5.6.5.1 Representative considerations. Representative considerations to be evaluated in developing the maintenance plan shall include the following: (a) Reliability, maintainability, simplicity, and ruggedness. Tasks shall be straightforward and allow some margin for error. Comparative overall benefit or nonrepairable versus repairable parts shall be considered. (b) Possibility of maintenance induced faults. (c) Transient and static operating conditions (maximum and minimum seawater, air, and other ambient), shock, vibration, wear, corrosion, erosion, fatigue, operating temperature, moisture, roll, pitch, air-blant, and other environmental forces, as applicable, which will affect the continued satisfactory performance of each part. (d) Accessibility, ease of removal and replacement of parts requiring frequent removal.	3.5.6.2 Scheduled maintenance action index. This index shall include all scheduled performance tests and preventive maintenance procedures. The index shall be tabulated as follows: (a) Column 1, Periodicity. This column shall contain an alphanumeric list of all maintenance actions contained in the chapter. The following periodicity symbols, as appropriate, shall be used in the order of increasing periodicity as listed below: Interval — Symbols Daily — D Weekly — W Monthly — M Quarterly (3 months) — Q Semiannually (6 months) — S Annually — A Overhaul cycle — C As specified (explain circumstances) — R 1/ 1/ An R periodicity will be preceded by a recommended calendar periodicity (e.g., DR, WR, MR, etc.). (b) Column 2, Maintenance action. This column shall list the maintenance action which corresponds to the periodicity number in column 1. (c) Column 3, Reference. This column shall state the paragraph number that contains the procedure listed in column 2.

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Type I (Continued)

(e) Safety.
(f) Methods of calibration and adjustment.
(g) Rotatables (replacement of assembly for subsequent overhaul and reissue versus spot overhaul or repair).

3.5.6.5.2 Detailed requirements. This plan shall be developed and time-phased to allow major items of work to be conducted during the ship overhaul period. (Overhaul need not return the item to new condition, but to the condition necessary to perform acceptably on a lowest overall life cycle cost basis until the next overhaul.) The procedures shall provide independence from shore-based activities except for major repairs and major overhauls.

3.5.6.5.3 Schedule. The plan for maintenance shall set forth a time schedule of all maintenance actions (examinations, tests, adjustments, reconditioning, overhaul, and acceptance criteria) which should be taken, all parts which should be replaced, and all other parts which should be available to allow replacement if examination so indicates over the life cycle of the ship. The procedures shall provide detailed maintenance and overhaul procedures at all maintenance levels (organizational, intermediate, or depot) to provide for the reconditioning or replacement of each part subject to continuing degradation, before the degradation results in major failure. It shall specify in detail the examination measurements, tolerances, work, tests, and quality assurance operations required to provide satisfactory maintenance and overhaul. Each task for the continuing shipboard preventive maintenance schedule shall be in the following format:

(a) Subcomponent (such as "motor-controller").
(b) Frequency (such as "semi-annually" (see 3.5.6.2)).
(c) Related maintenance (such as "take insulation resistance readings").
(d) Maintenance task description (such as "clean and examine controller").
(e) Safety precautions (such as "WARNING
Controller operates at high voltages - to prevent a personnel casualty, assure that the controller remains de-energized throughout task").

Type II (Continued)

3.5.6.3 Preventive maintenance procedures. These procedures shall include information required to examine, clean, and lubricate the equipment, and shall contain:

(a) Safety precautions.
(b) Tools, parts, materials, and test equipment required.
(c) Cleaning solvents. Solvents shall be identified by Government specification numbers and National stock numbers.
(d) Instructions for properly maintaining all safety devices and interlocks, with warnings, and cautions.
(e) Instructions for lubrication at shipboard operating temperatures; also types and quantities of lubricants to be applied. Lubricants shall be identified by Government specification numbers and National stock numbers. Specific lubricants for arctic or tropic environments shall be included. When a proprietary lubricant is approved, a Government specification lubricant shall be listed as an emergency substitute.
(f) The minimum rating of the technician who can be expected to perform the task.
(g) Procedures of obtaining access to subassemblies of subcomponents.
(h) Instructions for in-place balancing and noise reduction.
(i) Inspection procedures for parts which deteriorate due to cycles of use, age, or climatic conditions.
(j) Illustrations to identify lubrication points and other pertinent data.
(k) Other information pertinent to these procedures.

3.5.6.4 Scheduled performance tests. These tests shall contain step-by-step procedures necessary to verify that the equipment is operating within standards in all modes of operation and shall contain the following:

(a) Safety precautions.
(b) A list of tools and test equipment identified by type, manufacturer, and model number.
(c) The title of the test to be performed.
(d) The minimum rating of the technician expected to perform the task.
(e) Preliminary setup data required

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Type I (Continued)	Type II (Continued)
(f) Tools, parts, materials, test equipment (such as "lint-free rag", "six inch screwdriver" and "vacuum cleaner with non-metallic nozzle"). (g) Procedure - List a detailed step-by-step procedure such as: (1) Open the access door to controller. (2) Test with voltage tester to ensure power supplies are de-energized. (3) Remove arc chutes, where applicable. Examine for carbonizing. Scrape clean or renew as necessary. (4) Examine all contacts. If main line contacts are rough, burned, or pitted, dress with sandpaper or contact dressing file. (5) Vacuum inside of panel; use dusting brush to loosen dirt, wipe components with a rag. (6) Examine components and wiring for evidence of overheating. Refer to wiring harness replacement procedures in Chapter 6, see 6.5.2 if wiring has been overheated. (Corrective maintenance required as a result of any preventive maintenance inspection shall be included in chapter 6 and referred herein - Corrective maintenance (see 3.5.8)). 3.5.6.5.4 Reconditioning procedures. The plan shall include reconditioning procedures on replaced parts to allow for reuse. 3.5.6.5.5 Parts availability. This plan shall establish the range, depth, and schedule for availability of each preventive and potentially corrective maintenance part resulting from normal operation and potential failures throughout the service life of the ship. The procedures shall also specify any required examination, storing, and testing requirements on stocked parts (and periodic replacements where shelf life is a factor). 3.5.6.5.6 Overhaul maintenance and repair standard. These standards shall provide a schedule and procedures in correlation with Chapter 6, Section 2, repairs for the detailed overhaul maintenance and repair including: (a) Pre-overhaul performance, hydrostatic, or other evaluation tests.	perform the test. (f) Detailed procedures for accomplishing the test. Procedures requiring lengthy and identical setup data may be presented in detail in one procedure and referenced in succeeding procedures. (g) Values or conditions, with tolerances, indicative of normal operation. (h) References to troubleshooting or corrective actions to be used if the test values are not within tolerances. (i) Illustrations to support the test. TYPE II MANUAL REQUIREMENTS, CONTINUE WITH 3.5.7

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Type I (Continued)	Type II (Continued)
(b) Pre-overhaul and maintenance inspections--GO/NO-GO accept and rejection criteria for wear, dimensions, clearances, surface finishes, electrical resistances, alinements, backlash, play, leakage, vibrations, noise, aging, corrosion, erosion (or other standards of acceptance, as appropriate) for each applicable part. Criteria shall be predicated on the need for satisfactory performance with no more than routine maintenance until the next similar inspection. Calibration requirements for special tools and instruments. (c) List of items which should be replaced and those which should be available for replacement if inspection so indicates. (Provide guidance on comparative suitability of worn-in parts versus new parts, wherever appropriate.) (d) Detailed disassembly procedures. (e) Procedures for reconditioning reusable parts and subassemblies before reassembly. (f) Critical inspection and review procedures for new parts (including comparison with parts being replaced to establish equivalence before installation). (g) Critical operations and checks during reassembly. (h) Quality assurance and post-overhaul (or maintenance) tests and examinations to assure suitability (including method of testing and run-in operations where required).	TYPE II MANUAL REQUIREMENTS, CONTINUE WITH 3.5.7

3.5.7 Chapter 5, Troubleshooting. Troubleshooting procedures and data shall contain all information necessary for a technician to locate a malfunction in the equipment. Troubleshooting shall be presented based on locating potential faults in a unit, assembly, subassembly module, or piece part, depending on the maintenance concept for the level of maintenance being performed. When the troubleshooting concept limits the location of a fault to a repairable item (such as the unit, assembly, subassembly or module) for organizational level maintenance, troubleshooting procedures and data shall also be presented that will permit a repair of the item at the intermediate- or depot-level maintenance facility, except when this information is included in another technical manual and the procedures can be directly referenced. When more than one level of maintenance must be provided for, organizational- and intermediate-level troubleshooting procedures shall be separated such that the organizational level troubleshooting procedures are identified as "Organizational level troubleshooting and are presented first. Intermediate level troubleshooting procedures shall follow organizational level procedures and shall be identified "Intermediate level procedures". Immediately following this title the following statement shall be included: "These procedures are authorized for Immediate-level (or higher) activities (tender/shore facility/depot)." Depot-level troubleshooting procedures may be included as a separate appendix or shall follow the Intermediate level procedures and shall be identified as "Depot-level troubleshooting procedures." Immediately following this title, the following statement shall be included: "These procedures are authorized for accomplishment by Depot-level activities only."

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3.5.7.1 This chapter shall contain all instructions and information necessary to locate troubles and conduct tests on each component, assembly, or subassembly of the equipment as follows:

(a) Troubleshooting guides providing step-by-step procedures for logical isolation of faults. This information shall direct the technician to observe meters, fuses, circuit breakers, valves, and other available indicators which would indicate the presence of trouble.
(b) Complete instructions on signal tracing for electric circuits including the use of special test instruments and unusual servicing techniques.
(c) Where appropriate because of equipment complexity, troubleshooting diagrams including schematics giving details of mechanical and electrical assemblies and relationships as follows:
(1) Mechanical schematic diagrams in accordance with 3.5.5.2.4.
(2) Maintenance schematic diagrams in accordance with 3.5.7.2.4.
(3) Piping diagrams in accordance with 3.5.7.2.3.2.
(4) Control diagrams in accordance with 3.5.7.2.3.3.
(5) Power distribution diagrams in accordance with 3.5.7.2.3.4.
(6) Signal flow diagrams in accordance with 3.5.7.2.3.1.

Type I	Type II
3.5.7.1.1 This chapter shall also provide guidance regarding diagnostic analysis of possible trouble situations wherein malfunction, fault or failure of this equipment or related equipment could render this equipment inoperative or unable to fully perform its intended function. This information shall direct the operator to observe lights, gages, meters, fuses, circuit breakers, valves, and other available indicators which would indicate the presence of trouble. These instructions shall also cover tracing out mechanical system flows and signals in electric circuits, including the use of special test instruments and unusual servicing techniques. Kinds of troubles to be dealt with include, but are not limited to, the following: (a) Low lube oil pressure on loss of oil supply. (b) Off-design voltage, power or current (input or output). (c) Off-design temperatures (input or output). (d) Off-design vacuum (input or output). (e) Off-design regulation of (or inability to adequately control) speed, load, voltage, current or fluid flow. (f) Fluid leaks which endanger personnel or functional adequacy of equipment. (g) Valves or fittings stuck in off-design or non-operative positions. (h) Inability to initially start or initially operate equipment or accessories. (i) Inability to keep equipment or accessories running or operating. (j) Inability to stop, arrest or de-energize equipment or accessories.	3.5.7.2 This chapter shall contain all the information required to enable the technician to locate malfunctions in the equipment. 3.5.7.2.1 Introduction. The introduction shall explain the approach and logic of the troubleshooting principles presented in the manual. Describe the troubleshooting data and show how they relate to one another. 3.5.7.2.1.1 Troubleshooting index. The troubleshooting index shall be presented in tabular form. The index shall list all equipment, major and supporting functions in alphabetical order, provide references to the technician to the appropriate procedures and diagrams that are to be used to troubleshoot a specific function (see figure 16). 3.5.7.2.1.2 Relay, lamp indexes. These indexes shall be prepared in tabular form for all relay coils and indicator lamps. The relay and lamp indexes shall include the item reference designation, the functional name, energizing voltage, and a reference to the troubleshooting diagram(s) (see figures 17 and 18). 3.5.7.2.1.3 Protective device index. This index shall list all protective devices, such as fuses, circuit breakers, etc. The index shall include the item reference designation, front panel marking of the device, trip-out value of the circuit breaker and rating of fuses, name of the circuit protected and a reference to troubleshooting diagram(s) (see figures 17 and 18). 3.5.7.2.1.4 Maintenance turn-on procedure. Include a maintenance turn-on procedure to energize the equipment from the fully de-energized condition to full operation (see figure 19). This procedure shall enable the technician to determine which

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Type I (Continued)	Type II (Continued)
(k) Inability to restart equipment or accessories once stopped or de-energized. (l) Inability to develop design or rated power or output. (m) Circuit faults (open or closed circuits). (n) Faulty or off-design performance of circuit elements, readout equipment or instrumentation. (o) Malfunction or non-function of safety devices. (p) Vibration. (q) Bent or bowed rotating elements. (r) Damaged or broken gear teeth or mechanical or electrical interfaces.	major function or supporting function is malfunctioning. Each step of the procedure shall include the action to be taken (STEP), the observation to be made (OBSERVE), and shall resume that normal conditions have been observed in previous steps. Reference shall be made to the procedure for troubleshooting or corrective action to be used (REFERENCE) if the observation is out of tolerance. Built-in monitors such as meters, dials, lamps, etc., shall be used when possible for making observations, as opposed to the use of external test equipment. The procedure is complete when the equipment is fully energized and all switches and controls are positioned for proper operation. The maintenance turn-on procedure may be presented in a troubleshooting dependency diagram (see 3.5.7.2.12).

3.5.7.2.2 Troubleshooting procedures. Troubleshooting procedures shall be directly related to, and support the troubleshooting diagrams (see 3.5.7.2.3). Troubleshooting information shall be developed to guide a technician in the logical order of isolating a fault. This information shall direct the technician to observe meters, fuses, circuit breakers, valves, built-in test equipment, and other available indications showing the presence of trouble. The analysis of normal indications in relation to faulty indications shall be stressed. This information may be in tabular, illustrative, or narrative format, whichever lends to faster troubleshooting. Test programs shall be developed for digital devices when the method is the most practical method of troubleshooting logic.

Type I	Type II
3.5.7.2.2.1 Troubleshooting guides in the form of charts shall be included to enable the technician to quickly identify (in the manual) the trouble or symptoms along with the immediate action to take, the probable cause, and the concurrent follow-on corrective action or remedy. Trouble analysis or troubleshooting charts should resemble figure 18 herein and be included as tables in this chapter. Troubleshooting analysis charts shall be by a list of same. When diagrams included elsewhere in the manual are useful troubleshooting tools or aids, they shall be cross-referenced in the pertinent column of the applicable troubleshooting charts; to this end, the following shall be cross-referenced: (a) Mechanical schematic diagram (see figure 15). (b) Wiring or signal tracing diagrams (see figure 20). (c) Piping diagrams (see figure 21 and 22). (d) Control diagrams (see figure 23). (e) Power distribution diagrams (see figure 24). (f) Maintenance schematic diagram (see figure 25).	3.5.7.2.3 Troubleshooting diagrams. Troubleshooting diagrams shall consist of signal flow diagrams, piping diagrams, control diagrams, power distribution diagrams, logic diagrams, maintenance schematic diagrams, and dependency diagrams, as required. 3.5.7.2.3.1 Signal flow diagrams. Signal flow diagrams shall consist of detailed block diagrams illustrating the functional development of each major function from its origin to its measurable output (see figure 20). The flow path shall begin with one or more initial input (or appropriate interface conditions) and proceed through each unit, assembly, and subassembly influencing the signal flow. Each hardware level block shall reference a schematic diagram to isolate the faulty part. All items shown on the signal flow diagram shall be identified by their reference designations. The following shall apply: (a) Titles of diagrams shall correspond to the signal flow described.

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Type I (Continued)	Type II (Continued)
TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.7.2.3.2	(b) Diagrams shall depict such signal flow as receive, transmit, RHI display, PPI display, bearing data, antenna rotation, elevation data, etc. (c) Show all test points necessary to isolate the trouble to the lowest level of hardware block (e.g., subassembly). Include test parameters required to define satisfactory operation. Where signal flow diagrams depict signal flow in more than one mode of operation, that data shall be presented on the apron for all modes. Apron notes shall also include test data for test equipment setup. All inputs and outputs shall have signal description information. (d) References shall be made to the functional description, troubleshooting procedures, corrective actions, etc., as appropriate by paragraph number. Normally these references shall be included with the notes. (e) The display of more than one function or mode of operation on one diagram shall be allowed only when clarity is not sacrificed and the functions are relatively simple. (f) Screwdriver adjustments, dial adjustments, and adjustable controls shall be shown. (g) The reference designations (e.g., 1A1A2) shall be placed in each hardware block. Reference to the figure number of the schematic diagram shall be placed adjacent to the reference designation. (h) All input/output signals and connectors and terminals in the signal path shall be shown. Identify the signal, and show all lead numbers, connector numbers, and terminal identifiers. (i) All built-in controls and monitoring devices shall be shown. Do not show external test equipment, unless it is a permanent part of the equipment. (j) Hull grounds, chassis grounds, signal grounds, and power grounds shall be shown. (k) All leads of motors, generators, synchros, etc., shall be identified.

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Type I (Continued)	Type II (Continued)
TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.7.2.3.2	(l) All relay coils that are energized by the signal shall be shown. (m) All relay contacts and relay terminals in the flow path shall be shown and identified. All relay contacts shall be depicted in operational mode. References to control diagrams on which the relay coils appear shall be shown adjacent to the relay contacts. (n) All switches which affect signal flow shall be shown and identified. Switch terminals and panel markings corresponding to the switch positions shall also be shown. (o) Mechanical couplings of all controls, switches, potentiometers, synchros, etc., shall be shown. (p) Signal paths shall be identified by weighted lines and arrowheads. (q) Test instruction procedures, and test data shall be shown on the apron pages of the diagrams. This data may satisfy the requirement of 3.5.7.2.2.

3.5.7.2.3.2 Piping diagrams. Piping diagrams shall be prepared for fluid cooling, air, gas, steam, oil and hydraulic systems. Fluid symbols shall be in accordance with ANSI Y32.10. These diagrams shall show, when significant, flow rate, temperature, pressure, and all devices which measure, control, or modify the flow (see figure 22 or 26). Also, a test data table shall be included on the apron of figure 22 reference shall be made to appropriate corrective actions and functional descriptions.

3.5.7.2.3.3 Control diagrams. Control diagrams shall be included for all control circuits. Control circuits shall be grouped according to energizing voltage, control function, mode of operation, or physical limits of cabinet or assembly, as applicable (see figure 23. Supporting information required to clarify the use of the diagram shall be provided in the general notes. The functional name and reference designation for each relay, switch, lamp, etc., shown shall be included. All relay energizing circuits shall be shown with all tie points and terminals and with switches and relay contacts in their operating positions. All terminal connections, switches, interlocks, contacts, or other relays in series with the energizing path, plus lamps or indicators (electrically connected in the energizing or indicating status of contact closures), shall be shown. The following note shall appear on all control diagrams: "All switches and relay circuits are shown in operating positions." In cases of multiple operating positions, switch and relay positions shall be explained by a specific note on the diagram.

3.5.7.2.3.4 Power distribution diagrams. Power distribution diagrams shall depict the distribution of primary alternating current (a.c.) power, secondary a.c. power, and direct current (d.c.) power from the terminal board, breaker, or fuse box to the various assemblies, subassemblies or modules of the equipment (see figure 24). Normally, a separate diagram shall be prepared for each voltage level used within the equipment. The following rules apply in the preparation of power distribution diagrams:

(a) Show and identify motors, transformers, regulators, power supplies, assemblies, subassemblies, and modules.
(b) Show and identify all power line devices such as fuses, circuit breakers, switches, and relay contacts.
(c) Show and identify all connections including plugs, jacks, and terminal boards in the distribution path.

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(d) Use dot and dash lines to set off hardware boundaries such as units, assemblies, and subassemblies. Identify each unit, assembly, and subassembly by reference designation. Include a figure reference to the schematic diagram covering the unit, assembly, and subassembly.
(e) Reference all relay contacts to the appropriate control diagrams. All relay contacts shall be shown in the operating condition.
(f) Include voltages and tolerances, as required.
(g) Show and identify all metering circuits and indicators.
(h) Show all grounds, commons, neutrals, and return lines.
(i) Whenever practicable, power path is to be displayed from left to right and from top to bottom.
(j) The functional names of all "main line" switches and circuit breakers shall be conspicuously marked on the diagram. In addition, set off any power control markings engraved or stencilled on the equipment in a rectangular box, for example, [MAIN POWER] .
(k) Show all relay coils in series with the main power distribution path. Relay control circuits shown on control diagrams need not be repeated on distribution diagrams.
(l) The following note shall appear on all control diagrams: "All switches and relay circuits are shown in operating postions."

3.5.7.2.4 Maintenance schematic diagrams. Maintenance schematic diagrams shall include unit-to-unit interconnection diagrams, intra-unit interconnection diagrams, and unit, assembly, and subassembly schematic diagrams. Complete coverage of the equipment shall be provided by these diagrams (see figure 25). Maintenance schematic diagrams shall be prepared in accordance with the following:

(a) The schematic diagram for each unit shall be drawn so that, together with the interconnecting diagrams, all circuit elements are included and all circuits can be traced from unit to unit. .
(b) Schematic diagrams shall be zoned by alpha-numeric coordinates in accordance with 3.7.4.5. The location of all circuit elements by zones shall be included in a table located on the apron of diagrams containing more than [illegible] parts. When a part such as a relay or a twin tube is drawn in sections at different locations, list as many coordinates as necessary to locate all sections.
(c) Represent major and minor signal paths by different line weights. The heavier line weight shall show the major signal path. Whenever possible, signal flow shall be from left to right and from top to bottom. Arrowheads denoting the direction of signal flow shall be placed on the signal flow lines.
(d) The use of ground and voltage buses is discouraged except in the power supply. However, voltage bus connections can be shown by broken lines directly beneath the connection. As a substitute for ground buses, individual grounds should be used and appropriate notes shall be included to indicate sources. If separate a.c., d.c., and signal grounds are actually used in the equipment, they shall be shown by keyed symbols.
(e) Show all significant voltages at buses, tube pins, transistor elements, etc., except when this data can be presented best in a voltage chart (see 3.5.7.2.4(j)). Indicate whether the voltage is a.c. or rf; d.c. voltages shall be shown by polarity. Where critical voltages occur within the equipment, tolerances for those voltages shall be shown in the illustrations.
(f) The functional names of all operating controls and adjustments shall be conspicuously marked on the schematic. For example: [VERT CENT], [BIAS ADJ], etc. In addition any operating front panel markings on the equipment shall be set off in a rectangular box. For example: [RF GAIN] , [AGC ADJ] . The functional name of all stages (tubes, transistors, etc.) also shall be included.
(g) The function, source, and destination of all input and output circuits shall be identified and indicated by figure number.
(h) Power and signal frequencies shall be designated in hertz (Hz). Resistance values, if more than one ohm, shall be noted for all wire-wound devices such as motors, relay coils, and transformers.
(i) Rated current and voltage values of primary and secondary windings of power transformers shall be indicated.

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(j) A resistance and voltage chart for a schematic diagram shall be provided on the apron. This chart shall give the normal resistance and voltage to ground (or other points of significance) for each tube socket pin. In addition, list all conditions which affect the resistance or voltage values given, such as control settings, equipment connections, tubes removed from sockets, etc. If semiconductors (transistors, diodes, etc.) are employed in circuits, adequate caution notices must be included to prevent damage to these devices when making resistance measurements in the circuit. No intra-element resistance measurements (i.e., between emitter, base, and collector) are required to be made on transistors. Also, resistance of power supply buses and other points of significance shall be indicated.

(k) Each schematic diagram shall be identified by the reference designation number, located in the lower right-hand corner of the image area.

(l) Schematic diagrams shall be presented in alpha-numeric order corresponding to the referenced designation of units, assemblies, subassemblies, etc. When two or more identical assemblies or modules are used, redundant schematic diagrams need not be repeated. However, a table which cross references the reference designation to the figure number of the common schematic diagram shall be provided immediately preceding the schematic diagram. Schematic diagrams covering more than one unit, assembly, or module shall include in the lower right-hand corner of the illustration (see figure 25), for identification purposes, all the reference designations of the unit, assemblies, and modules to which they refer. (For example, 1, 1A1, 2A7, 3A19.)

(m) Circuit elements shall be grouped functionally and arranged to make signal flow obvious from left to right and top to bottom. Circuit elements shall not be arranged to fill up white space or to maintain tube or transistor alinement. Circuit elements shall be arranged in textbook form for the convenience of the user. Layout shall not be distorted to achieve fit.

(n) Use breaks in lines as frequently as possible to avoid cluttered diagrams. Add necessary notes or text to explain how to use break symbols, where to find mating ends of broken lines on drawing, etc.

3.5.7.2.5 Logic diagrams. Logic diagrams shall be provided for digital devices and digital circuitry of conventional analog equipment in accordance with ANSI Y32.14. Distinctive shapes shall be utilized. Internal and external data shall be included as shown on figure 27. Logic diagrams shall cover digital functions such as Input-Output Control, Memory Control, Data Transfer, Clock-pulse generation and distribution, etc. Emphasis shall be placed on functional development and presentation rather than on hardware groupings.

3.5.7.2.6 Integrated circuitry packages and non-repairable assemblies. When integrated circuit packaging techniques, such as Dual In-Line packages, are utilized, the level of detail on the logic diagrams may be restricted to that necessary to isolate problems to either a specific integrated circuit package or if the maintenance concept so dictates, to a specific non-repairable assembly, providing there is sufficient detail to understand the process of troubleshooting.

3.5.7.2.7 Timing circuit diagrams. Timing circuit diagrams shall be provided for all significant timing relationships (see figure 28). These diagrams shall show the exact timing relationships and the origins of all timing signals (conventional and digital).

3.5.7.2.8 Single-function diagrams. Single-function diagrams for nonprogrammable devices which result in a unique output function may be prepared to the requirements of signal flow diagrams (see 3.5.7.2.3.1).

3.5.7.2.9 Assemblies/subassemblies, modules, standard electronic modules, designated non-repairable. Logic diagrams, functional information, hardware and test data shall be supplied to the extent that it is required to understand the operation of the equipment, concept of troubleshooting, and to serve the basis for affecting a critical repair.

3.5.7.2.10 Flow charts. Flow charts for digital devices shall be provided to support the explanation of machine instructions and test programs, and shall conform to ANSI X3.5.

3.5.7.2.11 Coding instruction sheets. Coding instruction sheets shall be provided for all programs. The listing shall contain all coding and address data as well as an adequate notes section to ensure understanding (see figure 29).

3.5.7.2.11.1 Test programs. Test programs with coding instruction sheets shall be developed and included to support troubleshooting procedures.

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3.5.7.2.12 Troubleshooting dependency diagrams. One of the following types of troubleshooting diagrams shall be provided to augment the troubleshooting procedures. Supporting information required to explain the use of the diagrams shall be provided in the text or general notes.

3.5.7.2.12.1 Fault logic diagrams. Fault logic diagrams shall be based on a fault indication observed during troubleshooting (see figure 30). The diagrams shall comprise a branching series of questions pertaining to fault isolation. Each question shall pertain to a further observation or measurement, and shall result in a "yes" or "no" answer, thereby progressively narrowing the possible functional area of the fault. Tolerance values shall be presented in those instances where a definitive "yes" of "no" is not obtained. This progression and elimination shall isolate the functional area of the equipment containing the fault and then refer the user to the portion of the manual containing that information needed to complete the fault isolation and repair. Each diagram shall include or make reference to information necessary to establish the test or operating conditions required for starting the fault isolation procedure. Only three types of blocks shall be used. Shaded blocks (right and bottom border lines weighted) shall contain questions which may be answered from observation, without changing test setup and without special equipment. Single-line blocks shall contain questions requiring measurement by special setup of external test equipment. Double-line boxes (conclusion boxes) shall list the functional area within an equipment that is the probable source of malfunction and shall reference a procedure or another diagram for further isolation or correction of a fault.

3.5.7.2.12.2 Troubleshooting-maintenance dependency-matrix chart. Matrix charts shall show the functional dependency of output signals or indications upon circuit elements, circuits, modules, etc. (see figure 31). These charts shall be developed in accordance with MIL-M-24100 and the following:

(a) Each vertical column is annotated to represent a circuit element, circuit, assembly, etc.
(b) The horizontal rows are annotated to represent a procedural step which results in an observable output or indication.
(c) Symbols shall be used, in the body of the grid, to show the relationship between circuit elements, circuit, etc., and observable output or indication.
(d) All circuits, assemblies, modules, etc., shall be exercised in a manner to permit logical diagnosis.
(e) All outputs shall be clearly defined and performance specifications given.
(f) All symbols shall be defined.
(g) Use of chart shall be fully explained.

3.5.8 Chapter 6, Corrective maintenance. This chapter shall contain instructions required to adjust and aline the equipment, remove, repair, reinstall, and aline all repairable parts, modules, subassemblies, and assemblies. The instructions shall identify the action to be accomplished; safety precautions to be observed; tools, parts, materials, and test equipment required; preliminary control settings; test equipment setup instructions; and step-by-step instructions, with supporting illustrations, to accomplish the maintenance task. Corrective maintenance instructions shall be provided for all items designated repairable irrespective of the maintenance concept unless this information is included in another technical manual and can be referenced.

3.5.8.1 Introduction. The introduction shall contain an explanation of the purpose, scope, and arrangement of the corrective maintenance data.

3.5.8.2 Section I, Adjustments and alinement. This section shall contain all information and procedures required to perform all necessary adjustments and alinements as follows:

(a) Non-operator type adjustments.
(b) Alinements requiring external jigs, test equipment, or bench setups.
(c) Alinements that are accomplished after a repair or replacement of a part or module.
(d) Test equipment setup and other illustrations necessary to support the procedures.

3.5.8.3 Section II, Repair. The repair section shall contain all procedures required in the repair of assemblies and repairable parts. Repair procedures shall include but not be limited to the following:

(a) Removal, disassembly, and inspection.
(b) Repair or replacement of piece parts.

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(c) Cleaning, reassembly, adjustment, installation, calibration, and checkout.
(d) Exploded views, sectional views, wiring diagrams, and photographs necessary to support the procedures.
(e) Obvious repair actions such as soldering, use of multimeters, hand tools, etc. shall not be included except where these actions involve hazards to personnel or equipment.
(f) Repair procedures shall be arranged in numeric-alpha unit designation order of the equipment.

3.5.8.4 Supporting illustrations and data. The use of clear, sharp illustrations to supplement description and maintenance coverage is required. Exploded views are required for parts location illustration. As a minimum, illustrations or sketches of the following shall, when applicable, be included:

(a) Typical bearings for rotating or moving equipment.
(b) Method of taking clearance measurements where required.
(c) Typical mechanism for absorbing thrust where applicable.
(d) Locking devices when applicable.
(e) Typical seal assembly (pressure and/or vacuum seals or controlled leakage between rotor and casing).
(f) Typical assembly of blading to rotor, with lock devices, where applicable.
(g) Typical assembly of field poles, laminated core iron, electrical windings, commutator, slip rings, and brush rigging where applicable.
(h) Equipment assembly with upper casing partially removed.
(i) Series of illustrations showing installation of supervisory instruments (such as RTE's in bearings).

3.5.9 Chapter 7, Parts list. Unless a separate illustrated parts breakdown is procured under a separate line item of the contract for the system/equipment in accordance with MIL-M-8910 or MIL-M-38807, the parts list shall conform to all requirements for chapter 7 herein. When an illustrated parts breakdown is so procured, the requirements for chapter 7 shall be modified as required to assure compatibility between the documents and harmony of the technical manual requirements. The parts list shall list and identify and shall reference or include an illustration that will show the location of all repair parts, including attaching hardware. All mechanical parts subject to replacement, although not provisioned as a spare part, such as handles, slides, plates, covers, etc. included in the Provisioning Documentation shall be included in the parts list and identified in the respective parts-location diagram. This chapter shall include the following:

(a) Introduction.
(b) List of major components or list of major units.
(c) Parts list.
(d) List of common item descriptions.
(e) List of attaching hardware.
(f) List of manufacturers.
(g) Parts location illustrations.

3.5.9.1 Introduction. The introduction shall contain an explanation of the scope and arrangement of the parts list. The following type of information shall be included:

(a) Models of equipments and, where all equipment of a specific model do not have interchangeable parts, serial number of equipments covered.
(b) Explanation of any special notes.
(c) Explanation and instructions for using the list of common item descriptions.
(d) Explanation and instructions for using the list of attaching hardware.
(e) Explanation and instructions for using the parts list.
(f) Explanation and use of the list of manufacturers.

3.5.9.2 List of major components or major units. A list of major components or a list of major units in tabular form shall be included as follows:

(a) Column 1, CID, RIC, or unit number. This column shall list the CID or unit number for each major component, if available.

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(b) Column 2, Quantity. This column shall list the quantity of components.
(c) Column 3, Name or nomenclature. This column shall contain the official name or nomenclature of the major components and shall be subdivided as follows:
(1) Name of unit. This column shall list the official name of the unit.
(2) Designation. This column shall list the type number of the unit.
(d) Column 4, Page number. This column shall list the number of first page of the parts listing for the major unit.

Type I	Type II
3.5.9.? Parts list requirements. Except when reference designators are assigned to parts in accordance with ANSI Y32.16, the parts list shall be prepared in accordance with 3.5.9.3.1. When equipment part designators are assigned in accordance with ANSI Y32.16, the parts list shall conform to the requirements of 3.5.9.3.2. 3.5.9.3.1 Parts list. Parts listing shall be prepared in tabular form similar to the following (see figure 32): (a) Column 1, Figure and index number. This column shall contain the figure number and index number which shows the location of the part. (b) Column 2, Name and description. This column shall contain the designation name of the part and descriptive data to identify the parts of the equipment. Descriptive information shall include the military identification number of the part whenever available. Those parts which do not have a military identification number shall include physical characteristics (material, grade, series, dimensions, specification, and any other information necessary to order replacement parts from the original supplier of the part without going to the prime contractor). (c) Column 3, Quantity. This column shall identify the quantity of parts required. (d) Column 4, Federal supply code number. This column shall contain the original item manufacturer's Federal supply code identification number. Identification numbers are not required for common hardware items (screws, washers, "O"-rings, etc.) that are available from many sources. However they shall be identified by military type number.	3.5.9 3.2 Parts list. The parts list shall be divided and arranged by major units in numerical sequence (e.g., unit 1 with its parts, etc., will precede unit 2 with its parts). All parts attached to the unit shall be listed first in alpha-numerical order, followed by unit assemblies with parts and subassemblies with parts, also listed in alpha-numerical order, and shall be arranged as follows: Unit (Cabinet parts): 1, 1AT1, 1B1, 1C1, 1CR1, 1R1, etc. Assembly (Assembly parts): 1A1, 1A1AT1, 1A1B1, 1A1C1, 1A1CR1, 1A1R1, etc. Subassembly (Subassembly parts): 1A1A1, 1A1A1AT1, 1A1A1B1, 1A1A1C1, 1A1A1CR1, 1A1A1R1, etc. Unit: 2, etc. 3.5.9 3...1 Parts listings shall be prepared in tabular form with columns headed as follows (see figure 33): (a) Column 1, reference designation. This column shall contain the reference designations of all parts listed in sequential order. The unit numbering method of assigning reference designations, as specified in ANSI Y32.16, shall be used to identify units, assemblies, subassemblies, and parts. MP numbers shall be assigned to mechanical parts subject to replacement, such as handles, slides etc., that are included in the APL but not assigned MP

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(e) Column 5, Original manufacturer's part number. This column shall be the part number assigned by the original manufacturer of the part. Part numbers are not required for common hardware items that are available from many sources.	numbers in the engineering drawings. With the exceptions of screws, nuts, bolts, and other attaching hardware, every functioning part in the equipment shall have a reference designator. The parts list shall be divided and arranged by major units in numerical sequence (e.g., unit 1 with its parts will precede unit 2 with its parts, etc.). When reference designations have been cancelled for more than two consecutive items, only the first and last of the designations are to be listed, separated by the word "through". For example: 3A1R69 through 3A1R100 not used. (b) Column 2, notes. This column shall contain equipment reference information such as serial number, model number, configuration data, etc. (c) Column 3, name and description. This column shall include descriptive data to identify the parts of the equipment and aid in determining substitutes. Such information shall consist of the name, electrical or mechanical characteristics, and military identification number of the item, and when applicable, attaching hardware. Common parts (e.g. washers, springs, nuts, bolts, etc.) shall be identified only by the military identification numbers. Those parts not having a military identification number shall also include physical characteristics (material and sufficient dimensions) to identify the parts within the set the manufacturer's part number and FSCM or the equipment contractor's part number and FSCM, Federal supply code number or NSN, and drawing number. Replaceable mechanical parts that are assigned as "MP" numbers in accordance with (a) preceding, shall include the manufacturer's part number or engineering drawing number along with the name and description of the item. The statement "Same as ....," or equivalent, shall not be used for describing identical parts. Identical parts that are used more than two times, but less than five times, shall include the complete description or conform

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TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.9.7	to the requirements for identical parts used more than five times. For identical parts that are used more than five times in the equipment, the complete item description shall be given in the list of common item descriptions and reference made thereto by the item number. Attaching hardware, with quantity required, shall be identified by the assigned letter code (see 3.5.9.5). For example, C(4) would be the third listed piece of attaching hardware in which four pieces are used. When non-standard parts have been approved and there are multiple sources, each source shall be cited in the description. If selected values for critical parts are required, sufficient information, such as criteria for selection and range of values, shall be provided to permit the repair activity to make selection. For each part, the part number of the actual item manufacturer shall be used, unless the part is physically modified by the equipment contractor. (d) Column 4, figure and item number. This column shall reference the parts location illustration (see 3.5.9.7) by figure number and item number enclosed in parenthesis (e.g., 6-119(17)). 3.5.9.4 List of common item descriptions. This list shall include the description of all multiple use parts (over five applications). The description shall contain the same information specified in 3.5.9.3.2.1. Like parts should be grouped and arranged in alphabetical order. Item numbers shall be assigned consecutively, for example: Item number — Description 1 — CAPACITOR, FIXED, CERAMIC: DIELECTRIC 3PF, 500 VDCW; MIL type CCZ1UJ03OC. 2 — CAPACITOR, FIXED, GLASS: DIELECTRIC 5100 PF, ± 1% 300 VDCW, mfr 86969, dwg 231B743H15 3 — RESISTOR, FIXED, COMPOSITION: 3000 ohms ± 5% 1/2w; mfr 42384 dwg 4469D69 4 — RESISTOR, FIXED, COMPOSITION: 4000 ohms ± 5% 1/2w; MIL type RC20GF402J 3.5.9.5 List of attaching hardware. A

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TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.5.9.7	list of standard attaching hardware shall be included which shows items of attaching hardware. For example: Letter code — Name and description A — SCREW, CAP, HEX HEAD, DRILLED HEAD, CRES: MS51100-8 1/4-28, UNF-2A, 1 in lg. B — WASHER, FLAT, STEEL, ROUND, 0.750 in. ID, 0.312 in. OD, 0.066 in. thickness. Items used fewer than five times need not be included in the list of Attaching Hardware providing the item is completely identified in the parts list, following the part identification that is being attached. 3.5.9.6 List of manufacturers. The list of manufacturers shall contain the names, addresses, and code symbol of all manufacturers supplying items for the equipment as referenced in the parts list. The list shall be presented in numerical sequence by code number. Code numbers shall be in accordance with Handbooks H4-1 and H4-2.

3.5.9.7 Parts location illustrations. Illustrations shall be included to provide positive and rapid location of parts. Each part used in the equipment shall have its location identified in an illustration. Types of parts location illustrations include exploded views (see figure 34), engineering drawings (see figure 33), when command approved, photographs (see figure 36), and sectional views, as applicable. (Suitable parts location illustrations located in other chapters of the manual may be referenced.)

3.5.9.7.1 Criteria for illustrations. Items shall be called out on illustrations by call-out leader, index number, or grid coordinates. Standard attaching hardware items (such as nuts, bolts, washers, screws) need not be called out or illustrated, except when they are on exploded views or are referenced in a procedure. Exploded views shall be supplied for mechanical units, subassemblies, and components designated Navy repairable but shall normally be included in Chapter 6, Repair and replacement, for parts-location purposes.

3.5.9.7.2 Exploded views. Parts in an exploded view shall be arranged in correct relative disassembled position and shall be shown in proportional size (see figure 34). The spacing of parts shall achieve maximum clarity and effective use of space. The relationship of parts shall be shown by the use of assembly lines where the main line of exploded parts has been broken into two or more groups for convenience of layout on the page. Leader lines and index numbers shall be used to assist in locating parts.

3.5.9.7.3 Line drawings and photographs. Unless otherwise permitted in the contract or order, line drawings shall be used in lieu of photographs. Engineering drawings are acceptable if they meet the format and content requirements of this specification and the legibility requirements of MIL-M-38784. Figure 35 is a typical line drawing and figure 36 is a typical photograph.

3.5.9.7.4 Printed-circuit board. Printed-circuit boards shall be illustrated foil side up. When printed wirings appear on both sides of the board, both sides shall be illustrated. All parts mounted on the board shall be outlined in black solid (front) or dashed line (rear) (even though mounted on the reverse side of the board) and their connections to the printed wiring clearly illustrated. If insufficient room exists, separate illustrations, top and bottom views, shall be provided. Each part shall be labeled with the applicable reference designation. To facilitate parts location, a locating grid and corresponding guide chart shall be provided when more than 30 items are mounted on a board.

3.5.10 Chapter 8, Installation. Drawings and information concerning installation shall be provided in this chapter. The following type of information shall be included: site selection, or installation location guide lines (such as moisture precautions and maximum temperature allowed as appropriate), special tools and materials requirements, unpacking, and handling (if unusual procedures or precautions are required), preparation of foundations,

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mechanical assembly procedures, mounting instructions, bolting diagrams, safety precautions, grounding and bonding, clearances for access, ventilation, fluid cooling requirements, clearances for motion under shock, and recommendations for reduction of electrical and electromagnetic interference, and other interface requirements, as applicable. In addition, this chapter shall contain tests and test procedures required to demonstrate that the equipment after installation is capable of satisfying operational requirements.

3.5.10.1 Installation drawings. Legible reproduction of appropriate installation control drawings in accordance with MIL-D-23140, except that zoning and title blocks shall be omitted, shall consist of the following, as applicable:

(a) Drawing list.
(b) Block diagram.
(c) Outline and installation drawing.
(d) Auxiliary cooling diagram.
(e) Auxiliary dry gas diagram.
(f) Cable running sheets.
(g) Summary list of installation material.
(h) R.f. transmission line diagram.
(i) Hydraulic fluid piping diagram.

3.5.10.2 Site or installation location information. Data shall contain information supplemental to the installation drawings. If all site information is contained on the installation drawings, reference shall be made to the applicable drawing(s) by figure number.

3.5.10.3 Reference publications. Reference shall be made to Military-issue general publications required to complete the installation, when it is impractical to include such information.

3.5.10.4 Tools and materials required for installation. Include information supplemental to the summary list of installation material. If no supplemental information is required, reference the drawing by figure number.

3.5.10.5 Unpacking and repacking. Include information supplemental to the installation drawings regarding unpacking and repacking. Include step-by-step procedures to prevent damage to the equipment or injury to personnel. Supporting illustrations shall be provided to clarify procedures. When packing for reshipment is required, step-by-step procedures for packing shall be included and illustrated. When packing is simply the reverse of unpacking, this fact need only be stated. Any special environmental conditions required for storage shall be provided. Instructions shall be included for items in the following categories:

(a) Depreservation procedures required at time of installation.
(b) Represervation packaging required prior to repacking for storage or shipment.
(c) Intricate mounting, blocking, or bracing.
(d) Special cushion inserts.
(e) Repairable items.
(f) Sensitive or fragile components.
(g) Items held in special cradles.
(h) Items furnished in reusable containers.
(i) Special environmental conditions required for storage.
(j) Special handling procedures required.
(k) Container storage or disposition instructions, as applicable.

The instructions shall be detailed to prevent handling damage to the equipment or injury to personnel. In addition, step-by-step procedural illustrations as shown in MIL-M-38784 may be used to supplement the packing and handling instructions.

3.5.10.6 Preparation of foundations. Include information supplemental to the installation drawings. If all foundation preparation information is contained on the installation drawing, refer to the drawing(s) by figure number.

3.5.10.7 Input requirements. A summary of the input data contained on the installation drawings shall be included. Parameters with tolerances should be included with each of the inputs listed. Examples of inputs are as follows:

(a) Power.
(b) Ventilation.
(c) Dry air (waveguide pressure).
(d) Ship gyro information.
(e) Fluid cooling.
(f) Steam.
(g) Freon.

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3.5.10.8 Installation procedures. The following types of supplemental information which are not provided on the installation drawings shall be included:

(a) Instructions required to assemble units.
(b) Instructions required to mount units. Include bolting and bracing diagrams and data on shock mounts.
(c) Instructions for making electrical, plumbing, transmission line, and all other interface connections (external) to the equipment.
(d) Instructions for interconnecting units comprising the equipment.
(e) Servicing procedures, such as initial lubrication.
(f) Instructions for bonding and grounding.

3.5.10.9 Installation checkout. Provide step-by-step procedures to demonstrate that the equipment operates correctly and within tolerances. These procedures shall provide for equipment checkout in three test phases as follows:

(a) Phase 1 - Installation inspection and pre-energizing procedures.
(b) Phase 2 - Turn-on and preliminary tests.
(c) Phase 3 - Installation verification test.

3.5.10.9.1 Phase 1 - Installation inspection and pre-energizing procedures. Provide inspection procedures in the form of check lists to verify the following:

(a) That all units of the equipment and required auxiliary equipments have been installed and that their location and orientation is proper; that all cables, filters, antennas, waveguides, transmission lines, dehydrators, coolant lines, piping, etc., have been installed in accordance with plans and specifications; that continuity exists in all interconnections.
(b) That the test equipment listed in Chapter 1 is onboard, operating satisfactorily, and has been calibrated.
(c) That the Allowance Parts List (APL) is onboard, that the Coordinated Shipboard Allowance List (COSAL) includes the equipment data.
(d) That all field changes, shipalts, and mandatory retrofits have been accomplished.
(e) That all installation and reference standards are entered on the Installation Standards Summary Sheet.
(f) That all rotating devices are free from obstruction.
(g) That there is access to the equipment for maintenance.
(h) That all pre-energizing servicing procedures, including lubrication have been accomplished.
(i) That it is safe to turn-on the equipment.

3.5.10.9.2 Phase 2 -Initial turn-on and preliminary test. Procedures shall be included for energizing the equipment for the first time. This may be accomplished by reference to the applicable portions of Chapter 2. Step-by-step procedures shall be included for testing the equipment electrical supply circuits including distribution panels, switches, breakers, relief valves, and interlocks. Procedures shall be included for testing piping, electrical cables, wire rope, stays, for proper installation of transmission lines and waveguides, hangar spacing, torqueing of connectors, pressure testing, flow rates, standing wave ratio and attenuation checks, etc.

3.5.10.9.3 Phase 3 - Installation verification test. Complete instructions shall be included for testing the equipment in all modes of operation. Where applicable, refer to the scheduled performance tests in Chapter 4. Procedures shall cover checking gages, meters, alarms, and other sensing devices for proper operation and calibration. The tests shall verify that all inputs are in tolerance. Where applicable, include voltage standing wave ratio (VSWR) and insertion loss tests to verify the proper installation of antenna-to-equipment waveguide runs; transducer impedance and source level checks to verify proper installation of transducers, domes, and cables. Preliminary setup data shall be included in each procedure. When it is required that an alinement be accomplished prior to performing a test, the alinement shall be included or referenced in the procedure.

3.5.10.9.3.1 Test procedure. Testing procedures shall be presented in a logical order as follows:

(a) Energize the equipment.
(b) Test the first units (normally power supplies) which must be operating properly. When test results are within the required tolerance, include reference to the next logical test. Include a reference to the corrective maintenance or troubleshooting data to be used if test results are not within tolerances.

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(c) Enter installation and reference standards in Installation Summary Sheet (see 3.5.10.9.3.2). Include instructions for installer to enter actual test and reference standard values, and certify proper installation on the Installation Summary Sheet.

Type I	Type II
	3.5.10.9.3.2 Installation standards summary sheet. The installation standards summary sheet shall provide spaces for recording the results of all installation verification tests and significant reference standard tests that should be made a matter of record for reference by the technician during troubleshooting and for a standard to be re-established after overhaul (see figure 37). Each space shall be identified by the step or paragraph number which provides the instructions for accomplishment. Each space shall contain the respective unit of measurement (e.g., amp., dB., or V.). When applicable, waveforms shall be included to show the points on the pattern where the measurement is to be taken. In addition, critical installation data shall be included, such as the length of the transmission line. 3.5.11 Type IIS and type IIX manuals. 3.5.11.1 Type IIS manuals for service test equipment. Type IIS manuals for service test equipment shall conform to the production requirements of 3.8.6 and to the content requirements for type II manuals except that the following data need not be supplied: (a) Descriptions of circuits in NAVSEA 0967-LP-000-0120. (see 3.5.5.2(c)) (b) Scheduled maintenance action index. (see 3.5.6.2) (c) Reference to troubleshooting actions. (see 3.5.6.4(h)) (d) Troubleshooting index. (see 3.5.7.2.1.1) (e) Relay, lamp indexes. (see 3.5.7.2.1.2) (f) Control diagrams. (see 3.5.7.2.3.3) (g) Troubleshooting dependency diagrams. (see 3.5.7.2.12) (h) Attaching hardware data. (see 3.5.9.3.2.1(c) and 3.5.9.5) (i) Phase 1 - installation examination and pre-energizing procedures. (see 3.5.10.9.1)
TYPE I MANUAL REQUIREMENTS, CONTINUE WITH 3.7	3.5.11.2 Type IIX manuals for experimental equipment. Type IIX manuals for experimental equipment shall conform to the production requirements of 3.8.6 and the content requirements for type II manuals except that the following data need not be supplied: (a) Descriptions of circuits in NAVSEA 0967-LP-000-0120. (see 3.5.5.2(c))

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Type I (Continued)	Type II (Continued)
TYPE I MANUALS REQUIREMENTS, CONTINUE WITH 3.7	(b) Scheduled maintenance action index. (see 3.5.6.2) (c) Reference to troubleshooting actions. (see 3.5.6.4(h)) (d) Troubleshooting index. (see 3.5.7.2.1.1) (e) Relay, lamp indexes. (see 3.5.7.2.1.2) (f) Protection device index. (see 3.5.7.2.1.3) (g) Maintenance turn-on procedure. (see 3.5.7.2.1.4) (h) Troubleshooting procedures. (see 3.5.7.2.2) (i) Control diagrams. (see 3.5.7.2.3.3) (j) Power distribution diagrams. (see 3.5.7.2.3.4) (k) Flow charts. (see 3.5.7.2.10) (l) Troubleshooting dependency diagrams. (see 3.5.7.2.12) (m) Section II, repair. (see 3.5.8.3) (n) Attaching hardware data. (see 3.5.9.3.2 and 3.5.9.5) (o) Phase 1 - Installation, examination and pre-energizing procedures. (see 3.5.10.9.1)

3.6 Technical contents for systems manuals (type III).

3.6.1 Spec. _ quirements. Type III manuals shall provide system and subsystem-oriented instructions (see 6.5.18) for operation, maintenance, installation, and test data. Detailed equipment data should be provided by reference to the equipment manuals. However, technical data shall be developed at the equipment level and be included for all equipment system and subsystem and interface components not covered by an equipment technical manual. The contents shall be arranged in chapters according to the following:

Chapter 1 - General information.
Chapter 2 - Safety precautions.
Chapter 3 - Conditions of readiness.
Chapter 4 - Operation.
Chapter 5 - Functional description.
Chapter 6 - Scheduled maintenance.
Chapter 7 - Fault isolation.
Chapter 8 - Alinement procedures.
Chapter 9 - Installation data.

3.6.2 Chapter 1, General information. Chapter 1 shall describe the system in general physical and functional terms as specified in 3.6.2.1 through 3.6.2.5.

3.6.2.1 Introduction. The introduction shall define the system and its relationship with other systems. The mission of the system shall be stated. The text shall be supported by a diagram(s) showing the interrelationships of the system equipments (see figure 38). The major functional relationship and inputs and outputs to related systems shall be indicated.

3.6.2.2 Physical arrangement. System areas and compartments shall be described and the system equipment and units contained in the areas shall be listed. The physical arrangement description shall be supported by the following illustrations:

(a) An inboard profile drawing of the ship or stations showing compartment locations and identifying topside equipment and equipment units comprising the system (see figure 39).

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(b) Separate illustrations of each compartment and area, identifying the listed system equipment (see figure 40). Other equipment, which is installed in the subject system compartments and areas, need not be listed in the text or called out in the illustrations if they do not directly affect the operation or maintenance of the subject system.

3.6.2.3 System equipment. Each of the equipments comprising the system shall be identified and described. Descriptions of operator-attended equipment shall include general statements as to the nature and purpose of units and indicators. The text shall be supported by illustrations. All equipments shall be shown, whenever possible, in relative-scale proportion. An equipment may be separately illustrated with significant features called out, if such details are necessary for proper support of the text.

3.6.2.4 Associated-system equipment. When required, descriptions and illustrations of associated-system equipment shall be limited to the major units thereof. The descriptions shall be more condensed than those of subject system equipment; otherwise, the same requirements are applicable. In the descriptions, emphasis shall be placed on those associated systems equipments that constitute operational or functional interfaces with the subject system. Such units shall be included in the system illustrations.

3.6.2.5 Reference data. Reference data shall include a list of the equipment comprising the system, and its official designations. A list of common names and abbreviated or informal nomenclature, and system characteristics together with a list of referenced publications shall be included as follows:

(a) Capabilities. A summary of system capabilities shall be provided. The summary shall include data such as gallons per minute, transfers per hour, boom capacity, rated ranges, resolution, accuracy, data handling capability, number of channels, etc. Such data shall be presented in tabular form.

(b) Reference publications. A list of the manuals that pertain to system and system equipments, and other documents of interest, such as training manuals and manuals for associated systems equipment shall be included. The list of publications shall include the title and publication number of the referenced publications.

(c) Tools and test equipment. A list of all special tools and test equipment for system-level maintenance shall be included. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)). An illustration and description of special items required shall be provided for identification. Information shall be presented in tabular form. For each item the description or table shall include:
(1) The official name or nomenclature.
(2) Identifying number.
(3) A brief description of the use of the item with a reference to the procedure(s) requiring its use.

3.6.3 Chapter 2, Safety precautions. Chapter 2 shall describe the hazards associated with system operation and maintenance. To permit wide and unrestricted use onboard ship, chapter 2 should contain only unclassified information.

3.6.3.1 Introduction. This description shall orient system supervisory personnel, and shall include the following:

(a) Purpose, scope, and organization of the system safety instructions.
(b) Basic safety concepts.
(c) Basic responsibilities for safety.

3.6.3.2 Electromagnetic radiation hazards and precautions. If applicable, describe the radiation hazards to topside personnel and the precautions to be taken. The hazards of radiation to flammable or explosive materials also shall be described. The description shall include discussions of the following:

(a) Locations of topside and inboard radiation hazardous areas.
(b) Minimum safe distances on the axis of beam radiation.
(c) Precautions to be taken when entering areas of radiation hazard (such as the wearing of copper-screen goggles).

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(d) The effect of radiation on flammable or explosive material such as induction of rf currents in metals, sparking, and the consequent possibility of igniting flammables or detonating explosives. The text shall be supported by one or more illustrations identifying the areas of radiation hazards and the location of antennas, etc.

3.6.3.3 Hazards to divers. When applicable, the description of hazards to divers shall include but shall not necessarily be limited to the following:

(a) The nature and intensity of sound energy in water as related to divers.
(b) Precautions to be taken when working near operating equipment.
(c) Illustrations identifying the hazardous areas and locations of sonar domes.

3.6.3.4 System hazards and precautions. Descriptions of system hazards and precautions shall be included, addressed to system personnel and referenced to particular system equipments. The descriptions shall be organized to be consistent with the operation of the system. The descriptions shall supplement and extend equipment safety instructions to the system level, by warning of potential hazards that can be caused during operation or maintenance.

3.6.3.4.1 Operational safety summary. A summary shall be included which emphasizes the proper use of equipment controls, describes the hazards to operators, or as applicable, the hazards to persons in areas remote from the operation, and recommends precautions. An emergency operational routine shall be included which emphasizes the controls that permit immediate braking or deenergizing of the system.

3.6.3.4.2 Maintenance safety summary. A maintenance safety summary shall emphasize the proper use of controls, describe the hazards to maintenance personnel, potential damage to the equipment, and recommend precautions.

3.6.3.4.3 Hazardous components. Identify and briefly describe the hazardous components including radioactive devices and elements used with the system and summarize the general handling precautions for such components. The description of a hazardous component shall include brief statements as to the purpose, manner of functioning, nature of built-in safety devices, and nature of the hazardous element; it shall also indicate the relative sensitivity of the component to mechanical shock, vibration, electromagnetic and radioactive radiation, and electrostatic discharges.

3.6.4 Chapter 3, Conditions of readiness. This chapter shall be provided for guidance purposes and shall represent engineering considerations. Chapter 3 shall list conditions of readiness requirements prior to the operation of the system. Activities or conditions within compartments or areas outside the system shall be described as necessary. When it is impractical or illogical to separate the conditions of readiness information from operational information, request shall be made with justification, to combine the requirements of chapter 3 (see 3.6.4) with the requirements of chapter 4 (see 3.6.5).

3.6.4.1 Conditions coverage. An orientation of system personnel shall be included. The condition of readiness shall be included as follows:

(a) Watch condition.
(b) General quarters condition.
(c) Emergency condition.

Coverage shall include but not be limited to the following:

(a) A block diagram showing signal paths of the conditions of readiness (see figure 41).
(b) Block diagrams and descriptions of interior communications links between compartments and areas, including sound-powered telephone circuits, announcing systems, and closed-circuit television.
(c) Presumptions relative to system status at specified times. (For example, from watch condition to general quarters condition, or from general quarters condition to emergency condition).

3.6.4.1.1 Watch condition. The condition of all systems equipment and the personnel situation (how many of what class/type of men to be stationed where) for watch condition shall be described. The operating console and panel positions shall be illustrated.

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3.6.4.1.2 General quarters condition. The condition of all systems equipment and personnel situation shall be described for general quarters condition. The operating control and panel positions shall be illustrated.

3.6.4.1.3 Emergency condition. The condition of all systems equipment and the personnel situation for emergency condition shall be described. The operating console and panel positions shall be illustrated.

3.6.5 Chapter 4, Operation. Chapter 4 shall describe system operating situations, modes, and procedures. The descriptions shall be detailed to the level required for an understanding of the operational interfaces of the system equipments and associated systems. Illustrations shall be included when necessary for clarity (see figure 42). The various operating modes shall first be described to acquaint the operator with all equipment combinations that can be employed to effect a given mode of operation.

3.6.5.1 Preoperational conditions and setup. Specific preoperational conditions presumed to be in effect prior to system operation shall be established. A system readiness checkoff list of significant switch positions and indicator status shall be tabulated. For indicators such as dials, where a band of readings are possible, upper and lower limits shall be delineated. The initial conditions of associated-system equipment that directly affect system operation shall be treated in a similar manner.

3.6.5.2 Operating modes. The primary operating mode shall be discussed in detail, and alternate modes shall be treated as modifications of the primary mode. Operating procedures common to all modes shall be detailed under the primary mode and referred to under the alternate modes, with such modifications of procedure as may be necessary. Each mode shall be described in the logical sequence of major phases, events, options, supervisory commands and responsive actions, and the following:

(a) Only those equipment operational controls and indicators having system significance shall be explained in the description. When controls must be actuated and indicators observed in a sequence to achieve system operation, the descriptions will cite each control and indicator with a number to indicate the position in the sequence.
(b) Emphasis shall be placed, by the use of warnings on the safe operation of controls, which, if operated improperly, could result in hazards to personnel or damage to the equipment. Each control shall be followed by a brief description of its effects (equipment actuation or display indication, or both) at the operator station and at remote stations. The primary mode description shall be supported by both general and detailed illustrations.
(c) Operational phases which involve operator judgment shall be illustrated by operational logic diagrams. The diagrams shall indicate the conditions that must be favorable prior to an operator action, or if unfavorable, indicate the alternate action. Illustrations showing dials, gages, status lights, etc., which indicate the favorable or unfavorable conditions shall be included. Special procedures to be followed when an equipment failure may be bypassed (as separate from emergency procedures) shall also be described.

3.6.5.2.1 Normal operation. The duties of system operators shall be described in terms of general responsibility and specific step-by-step procedures for operating the system in all of the primary modes. Descriptive words (such as switch, button, dial, or indicator) may be added to clarify the type of control involved, for example: "Press ACCESS button and observe channel spot." All system controls and indicators provided for the use of operators shall be covered. Controls and indicators provided only for maintenance and nonsystem application shall not be called out.

3.6.5.2.2 Emergency operation. Step-by-step procedures shall be provided for emergency operation of the system. If specially designated controls have been provided for emergencies, a short statement shall be included describing how they modify or otherwise affect normal system operation. Emergency procedures shall be supported by illustrations.

3.6.5.2.3 Special operation. Special operations such as test checkout, training, or evaluation exercises shall be described. Illustration support shall include block diagrams and pictorial diagrams.

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3.6.5.3 Standard log form (model set). Standard log forms and selected data shall be developed and prepared in accordance with 3.5.4.5 and included in this chapter for all ship systems requiring monitoring of operational functions such as chemicals, data, temperatures, etc. required for determining safe and reliable ship's operations. Such systems shall be, but not limited to the following:

(a) Propulsion.
(b) Boiler.
(c) Electrical.
(d) Battery.
(e) Distilling.
(f) Fuel and fueling.
(g) Refrigeration.
(h) Environmental.

3.6.6 Chapter 5, Functional description. Chapter 5 shall describe how the components or equipments comprising the system jointly perform major operations and functions, and how associated systems contribute to the performance of these major functions. Equipment or associated system interfaces shall be described only as necessary to identify the sources or destinations of system inputs and outputs (see figure 43). Descriptions shall not repeat the functional description provided in the equipment manuals.

3.6.6.1 Method of presentation. The presentation shall first define how the system major functions meet the purpose of the system as described in 3.6.2.1. Each major function shall then be discussed separately at progressively increasing levels of detail. The description of the entire system shall be supported by a basic block diagram of the system (see figure 44). Where information can be presented better pictorially than by text, additional diagrams and other illustrations shall be used. Reference may be made to diagrams appearing in chapter 7.

3.6.6.2 Introduction. The introduction shall describe the general approach that is used in the functional description. The introduction shall also describe briefly the interrelationship between the system and associated systems.

3.6.6.3 System function directory. A system function directory shall tabulate operation control functions and the signal data described in the detailed level of functional analysis (see 3.6.6.6). The tabulation shall include the following information, as applicable:

(a) Official name of the function, colloquial name, and symbol.
(b) Type of control or signal (a.c. frequency and voltage, d.c. polarity and voltage, hydraulic pressure, mechanical motion, synchro, etc.).
(c) The origin and termination of the control or signal.
(d) Identify of equipments (relay transmitters, coordinate converters, distribution boxes, switches, and the like) between the origin and termination of the output control or signal.
(e) Figure numbers of illustrations on which the function is illustrated, including the fault isolation diagrams in chapter 7.

3.6.6.4 Functional description. Each function shall be described by at least two levels of detail (see 3.6.6.5 and 3.6.6.6).

3.6.6.5 First level of functional description. The first level of the description shall be confined to data such as origin of the function at an equipment control or sensor; transmission of the signal via intermediate equipment such as switchboards, relay transmitters, coordinate converters, and distribution boxes; and presentation of the function at terminal equipment. Control functions essential to the development of a signal shall be introduced and briefly described in their relationship to the signal. The description shall be supported by a block diagram (see figure 45). The diagrams for electrical/mechanical systems may be illustrated as shown on figure 46. Functions involving computations may be explained in mathematical terms, but at a level no higher than high school mathematics. The second level of analysis shall be supported by one or more diagrams.

3.6.6.6 Detailed (second) level of functional description. The detailed-level description shall explain the system functions in terms directly related to the diagrams in chapter 7. Data and control functions shall be described down to the level of an equipment or an equipment group. Functional descriptions of important parts of electrical/mechanical systems may be supported by illustrations similar to figure 47.

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3.6.7 Chapter 6, Scheduled maintenance. Chapter 6 shall contain all system scheduled test procedures, together with necessary explanations and illustrations. It is intended that the engineering effort required to develop preventive maintenance data be expended only once and that the data, where applicable be used both in this chapter and in Maintenance Engineering Analysis (MEA), Logistic Support Analysis (LSA) and/or Maintenance Requirements Cards (MRC) where one or more of the latter three data items are required by the contract. The preparation of this chapter of the manual should therefore be delayed until completion of the MEA, LSA and/or MRC; submissions of the manual manuscript for review should be marked "TO BE SUPPLIED UPON COMPLETION OF LSA". When the approved MEA, LSA or MRC data is available, it shall be included in the manual in the identical technical content and, wherever practicable, in the identical format. When included in the manual, such MEA, LSA or MRC data shall be integrated with other technical data required by this specification to be in this chapter. These procedures shall be correlated with the installation checkout requirements of 3.6.10.8 (type III), such that any maintenance or performance test procedure also required for checkout may be properly referenced from chapter 9, Installation Data. The Installation Standards Summary Sheet (see 3.6.10.11.2) shall also include space for any maintenance or performance test result that should be recorded by the installer. The recorded information will provide a reference to the Navy technician when troubleshooting, or when needed for installation acceptance certification. Chapter 6 shall comprise the following:

(a) Introduction.
(b) Scheduled maintenance action index.
(c) Scheduled test procedures.

3.6.7.1 Introduction. The introduction shall be an explanation of the purpose, scope, and arrangement of the scheduled maintenance material. When a preventive maintenance procedure is critical to the operation of the system and the schedule for servicing is absolute (not recommended), this information shall be conspicuously written as a CAUTION. The following statement shall be included: "The scheduled maintenance instructions in this manual are intended to duplicate those furnished in the Planned Maintenance System (PMS)." In case of conflicts, the PMS documentation takes precedence. Such conflicts should be reported immediately on the user comment sheet in accordance with the maintenance procedures for this manual.

3.6.7.2 Scheduled maintenance action index. This index shall include all required scheduled performance tests. The index shall be tabulated as follows:

(a) Column 1, Periodicity. This column shall contain a list of all scheduled actions contained in the chapter. The following periodicity symbols, as appropriate, shall be used:

Interval	Symbol
Daily	D
Weekly	W
Monthly	M
Quarterly (3 months)	Q
Semiannually (6 months)	S
Annually (12 months)	A
Overhaul cycle	C
As specified (explain circumstances)	R

(b) Column 2, Maintenance action. This column shall list the title of the maintenance action which corresponds to the periodicity number in column 1.
(c) Column 3, Reference. This column shall state the paragraph or table number of the maintenance procedure that corresponds to the maintenance action listed in column 2.

3.6.7.3 Scheduled test procedures. Include the detailed procedures for setting up and performing complete system tests. Each procedure shall be numbered and titled to clearly define the test action and the output to be tested. The procedures shall contain the data specified in 3.5.6.4.

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3.6.8 Chapter 7, Fault isolation. Chapter 7 shall contain fault isolation procedures, illustrations, and an explanation of the use of the information presented. The major objective of the system fault isolation procedures contained in chapter 7 shall be described. In addition, this shall contain a brief description of each type of maintenance diagram. A fault directory shall be included which relates fault symptoms found during operation to the fault-isolation procedures. Fault-isolation procedures, fault logic diagrams, control function diagrams, and data function-diagrams shall be included as follows:

(a) Operation-based symptom fault directory.
(b) Fault-isolation procedures.
(c) System fault logic and troubleshooting maintenance dependency matrix diagrams.
(d) System control function diagrams.
(e) System data function diagrams.

3.6.8.1 Operation-based symptom fault directory. The directory shall relate system faults observed during operation described in Chapter 4 to fault isolation diagrams. The directory shall be presented in tabular form by operational modes (see figure 48). The table(s) shall include references to system fault isolation diagrams, and where applicable, directly to equipment troubleshooting diagrams. The content of the table shall include, as applicable, the following information:

(a) Table headings. Table headings shall be identified by the operational mode to which the table relates.
(b) Column headings. The fault directory shall contain the following columns.
(1) Column 1, Operating procedure step. This column shall list the step of the operation procedure in Chapter 4 for which a fault symptom can be observed. (For example, if step 1 is an action step "set XYZ switch to ON" with no operational response, step 1 would not appear in the column. However, if step 1 stated "Set XYZ switch to ON and check to see that XYZ lamp lights," then step 1 would be included in the column.)
(2) Column 2, Functional description. This column shall include the reference to the paragraph number of the functional description in Chapter 5. In addition, when an operation can be identified with an output, this column shall also contain the name or symbol of the output.
(3) Column 3, Fault isolation procedure. This column shall reference by paragraph number the fault isolation procedure.
(4) Column 4, Alinement procedure. This column shall reference by paragraph number the alinement procedure in Chapter 8.
(5) Column 5, Fault isolation diagram. This column shall reference the fault isolation diagram by figure and sheet number.
(6) Column 6, Equipment document. This column shall reference the equipment technical manual when the fault can be isolated to the specific equipment causing the fault.

3.6.8.2 Fault isolation procedures. Procedures shall be included for isolation of a trouble to a single equipment or functional area of an equipment. The procedures shall provide for the analysis of switching combinations and observable indications (dials, gage lamps, and meters). The use of any required test equipment shall be described. The procedures shall support the fault-logic, control-function, and data function diagrams. The supporting diagrams shall be referenced by figure number. Prerequisite control settings and conditions shall precede each procedure.

3.6.8.3 System fault logic and troubleshooting-maintenance dependency-matrix diagrams. System fault logic diagrams shall be prepared for fault indications observed during either scheduled tests or operation. Fault logic diagrams shall be constructed in accordance with 3.5.7.2.12.1. These diagrams shall isolate the functional area of the equipment at fault and then refer the user to the equipment technical manual containing the information needed to complete the fault isolation and repair. Each diagram shall include or make reference to information necessary to establish the system test or operating conditions required for starting the fault-isolation procedure. The conclusion boxes shall list the equipment or functional area within an equipment that is the probable source of malfunction and the technical manual reference or references for further isolation and repair of the fault. Troubleshooting-matrix diagrams in accordance with 3.5.7.2.12.2 may be substituted for or augment fault logic diagrams.

3.6.8.4 Control function diagrams. Control function diagrams shall be provided for all system control circuits. The control function diagrams shall be at the system level but shall be constructed in accordance with 3.5.7.2.3.3. The diagrams shall show essential fault isolation test points or comparable indicators, and shall include appropriate references to equipment manuals.

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3.6.3.5 Data function diagrams. Data function diagrams shall show in detail the system information needed to isolate faults within signal or data flow paths. Data function diagrams shall include tolerance values and shall contain references to equipment publications where necessary. All inputs required to develop the output shall be shown. The data function diagrams shall be constructed in accordance with 3.5.7.2.3.1(a), (c), (d), (e), (f), (i), (j), (k), (l), (m), and (p).

3.6.9 Chapter 8, Alinement procedures. Chapter 8 shall present the corrective adjustment procedures and support information necessary to restore electrical and mechanical alinement between the various system equipments. All values and tolerances shall be included. The alinement shall be cross-referenced to respective fault isolation procedures and diagrams in chapter 7. Alinement procedures shall include references to equipment publications where further procedures are required at the equipment level. The alinement procedures shall be presented in step-by-step form.

3.6.10 Chapter 9, Installation data. Installation drawings and information not contained in the equipment technical manuals as necessary to install and checkout the system. System installation data shall include the following:

(a) Summary of all utilities required by the system, such as air, water, power, steam, freon, etc.
(b) System interconnection diagrams.
(c) Cable run diagrams.
(d) System piping diagrams.
(e) System cable interconnection check.
(f) Active system tests.
(g) System component installation procedures.

3.6.10.1 Utilities list. A utilities list shall be included that presents in tabular form all utilities required, and the quantities of each, in each system, compartment, and area.

3.6.10.2 Interconnection diagrams. Interconnection block diagrams shall be presented with each equipment or component shown as a block. All cables running between equipments shall be identified by cable number. The number of active and spare leads in each cable shall be included. The illustrations shall also indicate all junction boxes, switchboards, etc., into which interconnection cables enter or leave.

3.6.10.3 Cable run diagrams. Isometric diagrams shall be used to indicate the location of all cable runs between compartments or areas. Each cable run diagram shall indicate by deck, compartment, and frame identification the location of all cables shown on the interconnecting diagrams.

3.6.10.4 System piping run diagrams. Isometric diagrams shall be used to indicate the location of all system piping runs between compartments and areas. Each piping run diagram shall indicate by deck, compartment, and frame identification the location of all pipes, valves, fittings, tanks, etc.

3.6.10.5 System cable interconnection check. Cold-wire check procedures shall be provided to verify the proper installation of all system cables. These checks shall be prepared to be conducted with all power off and all equipment completely shut down.

3.6.10.6 Active system tests. All active system test procedures required to verify the proper installation and operation of the system shall be included. Reference may be made to applicable tests and procedures in Chapter 6. Procedures for complete setup, testing, shut down, and data analysis shall also be provided.

3.6.10.7 System component installation procedures. Complete step-by-step instructions shall be provided for installation of system components not covered in any of the equipment manuals for the equipment comprising the system. The following types of supplemental information not provided in the equipment technical manual shall be included:

(a) Instructions required to assemble components.
(b) Instructions required to mount components. Include boring and bracing diagrams and data on shock mounts.
(c) Instructions for making electrical, waveguide, plumbing, and all other interface connections between equipments, components, and other systems.
(d) Servicing procedures, such as initial lubrication and adjustments.
(e) Instructions for bonding and grounding.

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3.6.10.7.1 Installation drawings. The following drawings shall be included:

(a) Pictorial diagrams.
(b) Outline and mounting dimension data.
(c) Interconnecting wiring and cabling diagrams.
(d) Primary power distribution.
(e) Piping diagrams.

3.6.10.8 Installation checkout. Step-by-step procedures shall be provided to demonstrate that the system operates correctly and within tolerances. These procedures shall provide for system checkout in three test phases as follows:

(a) Phase 1 - Installation inspection and pre-energizing procedures.
(b) Phase 2 - Turn-on and preliminary tests.
(c) Phase 3 - Installation verification test.

3.6.10.9 Phase 1 - Installation inspection and pre-energizing procedures. Inspection procedures shall be provided in the form of check lists to verify the following:

(a) That the system and required auxiliary equipments have been installed and that their location and orientation is proper; that all cables, antennas, waveguides, transmission lines, dehydrators, coolant lines, piping, etc., have been installed in accordance with plans and specifications; that continuity exists in all interconnections.
(b) That special system test equipment listed in Chapter 1 is onboard, operating satisfactorily, and has been calibrated.
(c) That all field changes, mandatory retrofits, or shipalts have been accomplished.
(d) That all rotating devices are free from obstruction.
(e) That there is access to the system components for maintenance.
(f) That all pre-energizing servicing procedures, including lubrication, have been accomplished.
(g) That it is safe to operate the system.

3.6.10.10 Phase 2 - Initial turn-on and preliminary test. Procedures shall be included for energizing the system for the first time. This may be accomplished by reference to the applicable portions of Chapter 4. Step-by-step procedures shall be included for testing electrical, steam, and fluid supply circuits, distribution panels, breakers, and interlocks. Procedures shall be included for testing piping, cables, transmission lines, and waveguides for proper installation, including checks for hanger spacing, torqueing of connectors, pressure testing, standing wave ratio, and attenuation checks.

3.6.10.11 Phase 3 - Installation verification test. Complete instructions shall be included for testing the system in all modes of operation. Where applicable, refer to the scheduled performance tests in Chapter 6. Procedures shall cover checking gages, meters, alarms, and other sensing devices for proper operation and calibration. Tests shall verify that all inputs are in tolerance. Where applicable, include VSWR and insertion loss tests to verify the proper installation of antenna-to-equipment waveguide runs; transducer impedance and source level checks to verify proper installation of transducers, domes, and cables; dynamic and static load tests for hoists, winches, etc. Preliminary setup data shall be included in each procedure. When it is required that an alinement be accomplished prior to performing a test, the alinement shall be included or referenced in the procedure.

3.6.10.11.1 Test procedures. Testing procedures shall be presented in a logical order as follows:

(a) Energize the system.
(b) When test results are within the required tolerance, include a reference to the next logical test. When test results are out of tolerance, include a reference to the corrective maintenance or troubleshooting data. Reference shall be made to troubleshooting diagrams except where probable causes of failure can be predicted, in which case reference may be made directly to an alinement or repair procedure.

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3.6.10.11.2 Installation standards summary sheets. The installation standards summary sheet shall provide spaces for recording the results of all installation verification and maintenance standard tests. Spaces shall also be provided for recording the results of all maintenance or performance standards tests included in Chapter 6, scheduled maintenance, that are required for reference by the Navy technician when troubleshooting or for Government acceptance certification. Each space shall be identified by the step or paragraph number which provides the instructions for accomplishment. Each space shall contain the respective unit of measurement (e.g., amp., dB., or V.). When applicable, waveforms shall be included to show the points on the pattern where the measurement is to be taken. In addition, critical installation data shall be included, such as length of transmission line.

3.7 Format.

3.7.1 General requirements. Style and format shall conform to MIL-M-38784 (except that for commercial manuals, 8-1/2 X 11-inch page size may be used) and 3.7.2 through 3.7.8. Graphic symbols and reference designations for electrical and electronic diagrams shall be in accordance with ANSI Y14.15, Y32.2, Y32.14, and Y32.16. Abbreviations shall be in accordance with MIL-STD-12. The use of foldouts, foldups, and fold-downs is discouraged (see 3.8.2(d)).

3.7.2 Volumes. A single bound volume manual shall not exceed a 3-inch thickness. When the thickness exceeds 3-inches, Chapter 2 (types I and II) or chapter 4 (type III) shall be bound separately. When three or more volumes are required to maintain the 3-inch thickness limitation per volume, one or more additional chapters shall be bound separately in the following order:

(a) Installation.
(b) Parts list.
(c) General information and safety precautions.
(d) Scheduled maintenance.
(e) Corrective maintenance.

3.7.3 Distribution statements. The applicable distribution statement shall be included on the cover and title page (see 6.2 and 6.2.1).

3.7.4 References.

3.7.4.1 Referencing between diagrams. Referencing between diagrams shall be by figure number; sheet and zone numbers shall also be included for multi-sheet and zoned diagrams.

3.7.4.2 Text references. All illustrations shall be referenced in the text. References to illustrations shall be by figure number in accordance with MIL-M-38784. If the figure is located in another volume of the manual, reference shall also be made to the number of the volume. Where multiple sheet figures are referred to, the sheet number(s) involved, shall also be included, for example: "See figure 3-7, sheet 2."

3.7.4.3 Reference placement. References shall conform to the following:

(a) When a reference applies to one item within a sentence, place the reference parenthetically immediately after the item being referenced, for example: "The test set (figure 1-1) comprises a main panel (figure 1-2) and a power supply (figure 1-3)." When a reference applies to an entire sentence, place the reference at the end of the sentence; for example: "This characteristic causes the first portion of control movement to be less effective (see figure 6-3)." or show the reference in a complete sentence: "The procedure for making the adjustment is shown on figure 6-23."
(b) When reference applies to an entire paragraph or paragraphs, place the reference after the paragraph head, for example: "6-2 Simulation Diode Monitor (figure 6-56)."
(c) When reference is made to items in figures by reference designations, the numbers shall be indicated in the following manner: "The ON-OFF switch (1S8, figure 3-6) on the center console controls the TWT cooling."

3.7.4.4 Paragraph references. For single volume manuals, reference to text matter shall be by paragraph numbers only. For multi-volume manuals, reference shall be by volume number and paragraph number.

3.7.4.5 Zoning on diagrams. Diagrams containing the symbols for more than 100 parts shall be zoned. Diagrams shall be divided into equally spaced horizontal zones (ordinates) designated A, B, etc., from bottom to top along the outside left and right borders. Diagrams

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shall be divided into equally spaced vertical zones (abscissa) designated 1, 2, 3, etc., from right to left along the outside top and bottom borders. The zone size shall be as needed to clearly locate referenced points. The location of all circuit elements by zones shall be included in a table located on the apron.

3.7.4.6 Reference to zones. In reference to zones, the numeral shall be listed first to avoid possible confusion with reference designations, for example: use "3C" instead of "C3."

3.7.5 Notes for diagrams. Notes on diagrams shall be confined to clear spaces of the image area. Notes for foldout diagrams, with the exception of installation control drawings, shall be placed on the apron. Notes shall be identified with the legends GENERAL NOTES and SPECIFIC NOTES, as applicable. General notes shall precede specific notes and shall be identified by capital letters (A, B, etc.). Specific notes shall be identified by arabic numerals (1, 2, etc.).

3.7.5.1 General notes. General notes shall apply to the entire diagram and shall appear only on the first sheet of multi-sheet diagrams. No reference shall be made to general notes from the diagram or from specific notes. Examples of general notes are: a warning that high voltage exists throughout the entire equipment, the general instructions for positioning switches, and a list of the test equipment needed to take measurements throughout the diagram.

3.7.5.2 Specific notes. Specific notes shall apply only to a specific item on the diagram. Specific notes shall be repeated on each sheet of a multiple sheet diagram to which they apply and it shall not be required to refer to a specific note on another sheet of a diagram.

3.7.5.3 Apron notes. Apron notes for foldout diagrams should be placed in a final size image area of 7 by 10 inches with a minimum letter height of 0.060 inch. Notes shall be arranged to make best use of the available space. When notes for a given diagram foldout sheet require more than the 7 by 10 inch apron, the notes shall be extended with additional columns on the diagram image area if the space permits. If the space does not permit, single pages shall precede and shall contain the additional notes. When additional note sheets are required, the notes shall start on the first additional sheet and be printed as a right hand page; note pages may be printed on both sides. Foldout sheets shall not be used for running text. Each sheet of the diagram, including the note sheets, shall contain the figure number, title, and sheet number.

3.7.5.3.1 Multiple page notes. For multiple page note pages, applicable to illustrations, each note page if used shall be consecutively numbered front and back. Multiple note pages shall not be delineated as being part of (or sheet of ___) an overall notes page, but illustrations shall be delineated as being sheet ___ of ___ an overall illustration, as applicable. Notes shall precede the illustration or appear on the illustration apron, as applicable. Notes pages shall be of the single sheet type.

3.7.6 Foldout pages. All foldout illustrations shall be located at the rear of the chapter in which they occur, provided such location does not reduce the clarity of the text.

3.7.7 Running heads and feet. Complete running heads and feet shall be included on all pages of all volumes and parts (see figure 5).

3.7.7.1 Running heads.

3.7.7.1.1 Security classification. The security classification, including unclassified pages, of classified manuals shall be at the top and bottom center of each page and 8-1/2-inch segment of foldouts. Unclassified back-up pages of classified data will be marked with the security classification of the classified data and contain the following statement at the top: "The data for this page is unclassified."

3.7.7.1.2 Technical manual identification number. The command technical manual identification number, as assigned for each volume and part shall be at the top and bottom center of each page and 8-1/2-inch segment of foldouts. This number shall not be changed on change pages.

3.7.7.1.3 Inboard edge. The inboard edge should indicate the equipment or subject to which the manual applies. Appropriate abbreviations may be used.

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3.7.7.1.4 Outboard edge. Significant information such as chapter, section or subject titles paragraph number or figure number can be used or added. Appropriate abbreviations may be used.

3.7.7.2 Running feet.

3.7.7.2.1 Page number. Page numbers shall be located at the outboard edge in accordance with MIL-M-38784.

3.7.7.2.2 Issue indicator. The word "ORIGINAL" shall be located at the inboard edge of all pages of basic manuals and superseding revisions. The change number and date shall be located at the inboard edge of change pages.

3.7.8 Paragraph numbering. Paragraph numbering for primary paragraphs shall be in accordance with MIL-M-38784, e.g. 1-1, 1-2. Decimal numbers shall be used for subparagraphs, e.g. 1-1, 1-1.1, 1-2, 1-2.1.

3.8 Production. Production of reproducible copy and the printed manual, unless otherwise specified in the contract or order, shall be prepared for conversion to microfiche with a minimum of reformatting and reorganization as follows:

(a) Foldout pages shall be divided so as to produce 8-inch final size image area segments.
(b) There shall be no drawing data, other than horizontal lines, placed closer than 1/8-inch from the image area limit and between segments.
(c) Lines shall not be vertically displaced during the transition from one image area to the next.
(d) Printer position marks appearing as bold crosses not less than 1/4-inch in height shall be located on clear space in the image area near the upper and lower borders so that they divide the printed image into a final size segment not more than 8 inches wide.
(e) Halftone artwork shall not be used without prior command approval.
(f) Each text reference to a figure, table, paragraph, etc., shall be followed by a bracketed blank of five spaces for future microfiche referencing. This requirement does not apply to references to material located on the same page.

3.8.1 Typography. Typography, unless otherwise specified (see 6.2), shall be as specified in MIL-M-38784 for double-column boldface type for both single and double-column page layout. Letters, lines, and symbols shall be of a uniform contrast throughout the technical manual. Blurred or smudged printing or dropout characters or lines shall be cause for rejection. All manuals shall meet the requirements for good reproduction to 24 X 98 frame microfiche in accordance with MIL-M-38748 and 35mm microfilming in accordance with MIL-M-9868 and MIL-M-9868/1. Typography for change pages shall conform as nearly as possible to the original manual. Preferred typography is specified in MIL-M-38784.

3.8.2 Illustrations. Illustrations shall conform to MIL-M-38784 and the following:

(a) Lettering. Final size lettering shall be a minimum of 0.060 inch.
(b) Line artwork. Line artwork shall be prepared with line weights of sufficient strength to reproduce a minimum final line weight (when reduced) of 0.01 inch. Final size line artwork shall be suitable for reduction to 24 X 98 frame microfiche in accordance with MIL-M-38748 and 35mm microfilm in accordance with MIL-M-9868 and MIL-M-9868/1.
(c) Identification of part symbols and piece part symbols. All part symbols and piece part symbols shall be identified by reference designations. In addition, all parts such as switches, controls, indicators, blowers, etc., shall be identified by name. All control panel markings relating to these parts shall have the name printed exactly as shown on the panel and enclosed in a rectangle.
(d) Illustrations exceeding one page width. Whenever possible, to reduce the number of foldouts to the essential minimum, illustrations shall be divided between facing pages and identified "figure ___ sheet 1 and sheet 2 respectively. When more than two pages are required, and except for such illustrations as schematic, wiring and logic diagrams where the use of the diagram would be adversely affected, additional sheets may be continued on immediately succeeding pages and similarly identified.

3.8.3 Paper stock for printed manuals. Paper stock for text pages and foldouts shall be in accordance with MIL-P-38790. Blank apron pages shall be limited in size to one page unit (normally 8-1/2 X 11 inches).

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3.8.4 Cover and title page stock for printed manuals. Unless otherwise specified in the contract or order, covers of manuals shall be printed on cover stock JCP R10 (891 pound type) in accordance with MIL-P-38790. Information to be imprinted on the covers shall not be stamped in gold or other metal foil. Colors for covers are as follows:

Unclassified	Blue
Confidential	Green
Secret	Yellow
Top Secret	Pink

Title pages shall be printed on JCP A-60 (100 pound type) white paper.

3.8.5 Binding. Binding of manuals shall be in accordance with MIL-P-38790. Manuals shall be prepared in loose-leaf form to facilitate the insertion of replacement pages. Unless otherwise specified in the contract or order, corrosion-resistant posts and screws or snap-ring metal prong fasteners shall be used. All metal parts shall be treated to resist corrosion. Manuals shall be punched or drilled in accordance with MIL-P-38790. Page sizes and punching or drilling for change pages shall be the same as for the original manual. Fillers shall be used to build up the binding edge to the same thickness as the right-hand edge of the manual.

3.8.6 Preliminary manuals (types I, II, III), service test equipment manuals (type IIS), and experimental equipment manuals (type IIX).

3.8.6.1 Typography. Except when it is more economical to the Government to supply a higher quality of typography, preliminary, and type IIS, and IIX manuals shall be typewritten (single-spaced). The copy shall be such that clear, readable reproductions may be obtained. Any method of duplication, as defined in NAVEXOS P-35, which will provide the necessary quantity of black and white legible copies will be acceptable. When the method of duplication permits, the manual pages shall be duplicated on both sides of the paper.

3.8.6.2 Layout. Layout shall conform to the requirements for final manuals except that horizontal folds are acceptable on foldout pages.

3.8.6.3 Illustrations. Illustrations should be prepared by the most economical method which will result in clear, legible illustrations when produced.

3.8.6.4 Paper and cover stock. Paper stock which is suitable for the intended use will be satisfactory for text and foldout pages. Cover stock satisfactory for the intended use is acceptable; however, the color shall conform to the requirements for final manuals (see 3.8.4).

3.8.6.5 Binding. Binding shall conform to the requirements for final manuals (see 3.8.4 and 3.8.5). Covers need not include any printed matter (except for the security classification which shall be either stamped or printed thereon) if suitable cut-out windows and back-up pages with identifying data are provided. In addition, the cover or the backup page under the window shall indicate "Preliminary Manual for...."

3.9 Applicability of manuals.

3.9.1 Identical manuals (types I and III). When a manual that is already available is applicable in its entirety to the equipment or system being procured, the applicability shall be extended to additional ships. However, the contractor shall issue a new approval and procurement record page (see 3.4.2.1.4). Such previously accepted technical publications when provided, shall be certified technically accurate, adequate, and current by the contractor with respect to the provisions of this contract.

3.9.2 Manual modifications. When an existing manual is applicable to an equipment being procured, except for minor differences, the contractor shall modify the manual to cover the differences by the issue of a permanent change or revision, as applicable, in accordance with 3.10. Changes to previously approved manuals shall generally be limited to those necessary to correct technical errors or omissions. The intent is that changes not be made solely to correct items of an editorial nature, however such items shall be corrected when they occur on a page or illustration which is otherwise required to be changed. When a book is changed such that it does not cover all equipment in the fleet, the manual shall not supersede the existing manual.

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3.10 Changes and revisions. The changes and revisions shall correct all inaccuracies or omissions determined during usage or contractor generated design or production changes to the equipment or system, until expiration of the contractor's responsibility for the last equipment or system delivered. Where items being supplied are not identical, all changes shall be referenced to the serial number of the lowest level for which serial numbers are required (unit, assembly, printed circuit board).

3.10.1 Responsibility for changes or revisions to final manual. Contractors shall be responsible for issuing necessary corrections and changes to technical manuals throughout the life of the contract and guarantee period. The contractor shall supply either changes or revisions. Corrections shall begin not later than six months after distribution of the preliminary manual. They shall be made at least semiannually except for those which effect personnel safety or whose delay might hazard equipment through improper operation or maintenance. In the latter case, such corrections shall be prepared and promulgated as soon as the need is discovered. In the event no changes are required within such period, the contractor shall so advise the contracting officer, at the required time, in writing.

3.10.1.1 Changes. Changes to an existing manual shall be issued in the form of replacement pages as specified in MIL-M-38784. Upon application, change numbers (NAVSEA or NAVELEX numbers) will be assigned by the command or agency concerned (see 3.10.2 and 3.10.3).

3.10.1.2 Method. A change shall consist of revised pages and detailed instructions (instruction sheet) for inserting the pages into the manual. When the equipment nomenclature has been changed because of a production or field change to the equipment, the nomenclature references in the text and illustrations need not be modified merely to include the new nomenclature, providing that information equivalent to the following is included in the introduction of Chapter 1: "All references in the manual to Radar Set AN/SPS-40 apply equally to Radar Set AN/SPS-40A, unless otherwise indicated." If an equipment change does not affect the entire equipment population, the change shall be prepared in such a manner that the manual will describe both the affected and the unaffected equipment. The change shall clearly specify and identify the serial numbers of sets, units, assemblies, and subassemblies, as applicable, affected by the equipment change. If the change applies to the entire equipment population, the change shall cover only the modified equipment and a note shall be included in the first paragraph of the instruction sheet to the effect that maintenance support activities should not dispose of the superseded pages until it can be established that all of the equipment population has been modified. Equipment holders should be instructed to update their publication when the modification has been accomplished.

3.10.1.3 Instruction sheets. A change shall include an instruction sheet (see figure 47), which provides instructions for inserting new and revised pages and disposing of superseded pages (see 3.10.2).

3.10.1.4 Issue information. Changed pages shall include the change number; backup pages not affected by the change shall include the existing issue information.

3.10.1.5 Production of changes. Changes shall be prepared in the same style, format, and specifications as the basic manual. Changes shall be inserted (by the contractor) into the manuals that are shipped with the equipment, after the changes are available.

3.10.2 Interim (temporary) changes. Information to amend, correct, or modify manuals may be supplied as an interim (temporary) change, when emergency, urgent or minor changes are necessary and there is insufficient time available to publish a permanent change prior to the need, or as specified in the contract or order. Interim changes shall normally be promulgated by Advance Change Notices (ACN). Interim changes are subject to replacement by a permanent change. An interim change shall be identified and dated; shall contain an instruction sheet (see 3.10.1.2 and figure 47). The instruction sheet shall include the following, as applicable:

(a) The publication number shall appear in the upper left-hand corner of each page. The interim change shall be identified with an ACN number (NAVSEA) or a "T" identification number (NAVELEX) (assigned by the procuring activity) and the publication number to which it applies, for example, ACN 1/4 to NAVSEA S9221-A4-MMI-010 or T4-NAVELEX 0967-0000-1234 would designate the fourth interim change.

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(b) The total number of pages shall be indicated on the first page only.
(c) The security classification shall be located on all pages of all classified interim changes and shall appear at the top and bottom center. The word "UNCLASSIFIED" shall appear on top and bottom center of all unclassified pages of classified manuals.
(d) A statement shall be included indicating when the interim change is in effect. If it supersedes an earlier interim change or a permanent change, a statement of that fact should be given.
(e) If an equipment change is involved (field or production change), a statement shall be included indicating the purpose of the change and the extent (serial numbers or conditions of application) to which it applies to the equipment population.
(f) A list of revised pages (if any) issued as part of an interim change shall be included with instructions for inserting the new or revised pages for disposition of superseded material, and for correcting the title page and list of effective pages (see figure 49).
(g) Instructions shall be terse and clear; illustrations and diagrams corrected, as applicable; and specific data to be changed shall be set off in quotation marks.
(h) The instructions shall be followed by a statement that the interim change shall be inserted in the manual immediately under the front cover.
(i) Printed matter shall be arranged on the page so that all copy is readable when bound in the manual.
(j) Quantity and distribution shall be in accordance with 3.12.2.

3.10.3 Permanent changes. A permanent change shall include all corrections issued in previous interim changes (unless they have been covered in a previous permanent change) and in addition to all equipment changes required by the contract or order. The permanent change shall include an instruction sheet (see figure 50) prepared with the same requirements as for interim changes that includes step-by-step procedures for inserting the change into the manual. Change numbers shall be requested from the procuring activity when the manuscript is submitted for approval. It shall also include the following:

(a) A new cover and binding fasteners (for each volume affected) if the change includes too many pages to fit in the original cover.
(b) A revised title page including the original approval date of the publication and the approval date of the change. The NAVSEA or NAVELEX number for the change shall be inserted below the change number and date. The basic publication number appearing in the upper left-hand corner shall remain unchanged.
(c) A revised list of effective pages (see figure 46).
(d) Revised pages, table of contents, list of illustrations, list of tables, index, and content assurance pages, as applicable.
(e) If a publication consists of more than one volume, changes shall be prepared separately for each volume but shall be assigned the same change number.
(f) Parts lists may be corrected by means of supplementary parts lists which precede the parts list section (rather than by revising each applicable page) provided the supplementary lists do not exceed one-fifth the size of the main parts list. In subsequent changes, the supplementary list may be revised until it exceeds one-fifth the size of the main list. Should this percentage be exceeded, the parts list section shall be completely revised.

3.10.4 Revisions. A revision is a second or subsequent edition of a manual or volume of a manual which normally supersedes the preceding edition. A revision shall be supplied when the number of pages required to supply corrected and additional information for the manual is more than 25 percent of the total pages of the manual. Blank pages are not to be included as part of the basis for determination of the 25 percent figure. If the manual contains more than one separately bound volume, the 25 percent provision shall apply to each volume separately. Unless otherwise specified by the Government, when a revision is issued, the following supersedure notice shall be included on the title page: "This publication supersedes NAVSEA or NAVELEX (number) date (day, month, year)." Revisions shall conform to the contract requirements, MIL-M-38784, and this specification (see 3.4.1.5 and 6.5.13).

3.11 Approval and acceptance. All publications prepared by or for NAVSEA and NAVELEX shall be approved by the respective command, its field activities, or its representatives. Contractually deliverable items listed in 3.13 shall be submitted for approval in sufficient time (or as scheduled in the contract) to allow for an adequate review by the Government. Approved deliverable items will not be accepted by the Government until the contractor has complied with all conditions of approval.

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3.11.1 Publication or change number. The contractor shall request a publication or change number from the Government approving activity when the review manuscript is submitted for approval.

3.11.2 Preliminary manuals.

Type I	Types II and III
3.11.2.1 Preliminary (review) manuals. Type I preliminary (review) manuals are required to be submitted for approval by NAVSEA or its field representatives before the final manuals are printed and distributed. If the final manuals are not available at the time the equipment is shipped, preliminary manuals, as approved, shall be shipped with the equipment to be used with the equipment until the final manuals are available (see 3.12.1).	3.11.2.2 Preliminary manuals. When there is insufficient time (less than approximately six months) to permit approval by the command or agency concerned and still provide for the preparation and printing of final manuals to be shipped with the first equipment delivered, preliminary manuals shall be supplied with the equipment at no additional cost to the Government. Permission shall be obtained from the procuring activity for shipping the preliminary manual when any required data are missing. A review copy of the preliminary manual shall be submitted to the cognizant Government inspector for approval. Supplying preliminary manuals does not relieve the contractor from any contractual requirements pertaining to delivery of complete, adequate, and accurate final manuals. 3.11.2.3 Preliminary manuals, review copy of. The preliminary manual review copy shall be submitted (prior to shipment of the first equipment) to the Government inspector simultaneously with four copies of the manual to the command or agency concerned. Publications numbers and preliminary manuals shall be requested from the approving activity. After authorization and the publication numbers are received, two copies of the preliminary manuals shall be packaged with each equipment. 3.11.2.4 Self-addressed post card. A self-addressed contractor-furnished post card containing information equivalent to the notice indicated below shall be attached to the title page of all preliminary manuals which accompany the equipment. "IMPORTANT NOTICE: This is a preliminary manual for (insert nomenclature of equipment), publications (insert number). A copy of the FINAL manual will be forwarded direct to you when printed. Return this card IMMEDIATELY, indicating ship or shore activity and mailing address." Copies of final manuals shall be forwarded by the contractor to activities that submit post cards before shipment of the final manuals to stock. When post cards are received by the contractor after shipment of the bulk stock quantity has been made, the cards shall be forwarded to the cognizant Government inspector with a request to forward the cards to the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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Type I (Continued)	Types II and III (Continued)
TYPE I MANUAL REQUIREMENTS CONTINUE WITH 3.12	3.11.3 Manuscript - basic manual, changes, or revisions. Five review manuscript copies of the basic manual, changes, or revisions shall be submitted to the Government approving activity for approval and assignment of a publication number (see 3.13.4 and 5.1.4). Unless otherwise specified in the contract or order, a minimum of six weeks shall be allowed for Government approval. 3.11.3.1 Certification check-off list. A signed and completed certification check-off list for types II, IIS, and IIX manuals shall be prepared in accordance with figure 51 for preliminary, final manuals, and revisions. It shall be submitted to the procuring activity with the review copies and shall indicate the degree to which the manuscript conforms with this specification. A similar list shall be prepared for changes and type III manuals. The certification check-off list prepared for permanent changes of Government furnished manuals shall certify the extent to which the content requirements conform to this specification.

3.12 Quantity and distribution. The quantity of manuals shall be as specified (see 6.2). The contractor shall be responsible for the distribution of all manuals procured on the contract or order. Unless otherwise specified, two copies of preliminary or final manuals shall be shipped with each equipment. Bulk quantities shall be distributed at time of shipment of first equipment, except that when preliminary manuals are supplied in accordance with 3.11.2, quantity and distribution shall be as specified in 3.12.1, and the final manuals shall be distributed within 90 days after receipt of approval or as otherwise specified by the contract or order.

3.12.1 Preliminary manuals. Unless otherwise specified in the contract or order, the quantity shall be two per equipment. For type II manuals, a bulk quantity up to the bulk required for final manuals, but no less than 25 copies shall be supplied. (The total quantity of preliminary manuals to be supplied will depend upon the need at the time and the expected delay in production of final manuals. Distribution will be as directed.) Distribution of bulk quantities shall be not later than 30 days after the shipment of the first equipment.

3.12.2 Changes and revisions. The distribution of changes and revisions shall be to all activities reviewing the original manuals, including those for bulk stock, in the same quantity or as directed by the procuring activity.

3.13 Deliverable items. Deliverable items shall include the following as specified (see 6.2):

(a) Book plan (types II, IIS, IIX, and III).
(b) Quality program.
(c) Preliminary manual (review) (type I).
(d) Manuscript (review) (types II, IIS, IIX, and III).
(e) Preliminary manual (types II, IIS, IIX, and III).
(f) Reproducible copy.
(g) Replenishment material.
(h) Printed manuals.
(i) Status reports (monthly).
(j) Photolithographic negatives.

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3.13.1 Book plan. A book plan shall be based on a plan for maintenance of the equipment and shall be submitted to the procuring activity for review and acceptance prior to development of the review manuscript. The book plan shall be in conformance with the control requirements of the contract and specification and shall not be written in a manner intended to waive requirements. However, the plan shall indicate selected approaches to preparation and specify those options permitted by the contract and specification that the contractor elects to pursue. Acceptance of the book plan by the Government shall not waive any requirement within the scope of the contract or referenced specifications. The manual outline shall conform to MIL-M-38784 and the following:

(a) A text outline, including front matter, prepared in accordance with the content requirements of this specification, to the level of paragraphing necessary to show the major physical and functional aspects of planned coverage. Each chapter and paragraph title or notation listed shall be followed by a short statement outlining the data to be presented. Describe the planned approach and logic of troubleshooting principles and the troubleshooting data presented and how they relate with the description to be presented in the manual. Each illustration applicable to a particular text statement shall be referenced by the figure number and title following the statement. An index in tabular form, listing each illustration, table by number, and title shall be included. Describe each illustration by type (exploded view, schematic, line drawing, etc.), information content, and approximate size. All nomenclature shall be consistent between text and illustrations. Where an existing manual can be used as the text outline, it can be marked up and submitted.

(b) A sample of each applicable type of drawing prepared in accordance with this specification shall be part of the book plan.

(c) A summary of the major points of the quality program requirements in accordance with 4.2.

(d) When MIL-M-24365 or equivalent is invoked in the contract, the outline of the content requirements shall be based on a plan for use as supplied by the Government, and a maintenance engineering analysis or a logistic support analysis (LSA) (see 3.5.6.5.1) of the equipment. The outline shall indicate coverage on a comprehensive and systematic basis the most effective and efficient method of performing necessary maintenance. The outline shall be correlated to the maintenance engineering logistic support analysis reports by direct referencing.

(e) As an addendum to the book plan, the contractor shall furnish a milestone schedule for development of the publications for this contract. Milestones to be identified shall include, but are not limited to the following:

(1) Book plan submission.
(2) Manuscript drafting 25, 50, 75, and 100 percent completion points.
(3) Schematic preparation.
(4) Parts listing.
(5) Editing.
(6) Composition.
(7) Manuscript submission.
(8) Validation.
(9) Reproducible preparation completion.
(10) Any impacting inter-relationship with hardware development of the technical manual data items with hardware development, maintenance or logistics support analysis and training requirements.

(f) The contractor shall delineate the program that will be used to validate the manual by actual test against hardware where a validation plan is required by the contract or order, reference shall be made to the plan and summary of the highlights of the plan.

(g) A preliminary list of test equipment, required by 3.5.3.6, shall be included as part of the book plan.

(h) A detailed account of procedures to achieve standardization in writing style, artwork, nomenclature, abbreviation, and symbols shall be included.

(i) The manner in which the level of writing (reading comprehension) requirements of 3.3 are intended to be met shall be addressed.

(j) Any problems regarding specification interpretations and applications shall be identified.

(k) If more than one manual is to be furnished or is otherwise required to cover the equipment(s) of the contract, a manual tree/block diagram shall be provided showing the relationship of each manual.

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(l) Plan shall include method of handling changes and revisions and incorporating data affecting one or multiple interrelated manuals.
(m) Plan shall include procedures for incorporating approved engineering changes in affected technical manuals.

3.13.1.1 Visibility. The outline provided under 3.13.1 (a), shall be maintained and reflect Government approved changes throughout the technical manual development period until Government approval of the review manuscript, or the preliminary manual if supplied. Proposed changes to the outline shall be submitted to the Government with the next scheduled status report required by 3.13.10.

3.13.2 Quality program. A quality program in conformance with 4.2 shall be maintained.

3.13.3 Preliminary (review) manual (type I). A preliminary (review) manual shall be submitted to the procuring activity for review and approval prior to the preparation of reproducible copy and printing of final manuals. The preliminary manual shall have all text and illustrations included necessary for installation, operation, maintenance of the equipment, and suitable for engineering review. The text may be typewritten and the illustrations shall be legible. The production method may be by the most economical method at the option of the contractor.

3.13.4 Manuscript review (types II, IIS, IIX, and III). Five copies of the review manuscript shall be submitted to the procuring activity for review and approval prior to the preparation of reproducible copy (see 3.11.3 and 5.1.4). Review manuscript shall be submitted within 60 calendar days after delivery of the first article sample but not later than six months prior to delivery of the first equipment, in sufficient time to allow for preparation of reproducible copy and printed manuals. Review manuscript shall conform to MIL-M-38784 and the following requirements:

(a) The text shall be presented on one side of the paper and shall contain the exact wording and content intended for the reproducible copy. The production method may be by the most economical method at the option of the contractor.
(b) Reproduction of pencilled illustrations are acceptable, if their technical content, clarity, correctness, and adequacy meet specification requirements. Illustrations submitted for review purposes shall be legible and shall not exceed the reproducible copy size. All illustrations shall be bound separately.

3.13.5 Preliminary manual (types II, IIS, IIX, and III). Unless preliminary manuals are required as deliverable items by the contract or order, preliminary manuals should be supplied in accordance with 3.11.2.2.

3.13.6 Reproducible copy. Final reproducible copy, including original artwork and assembly sheets, complete in all respects for photographic reproduction, shall be prepared in accordance with MIL-M-38784 and shall be delivered as directed.

3.13.7 Photolithographic negatives. Photolithographic negatives shall conform to the requirements of MIL-P-38790.

3.13.8 Printing and binding. Printing and binding shall be in accordance with MIL-P-38790.

3.13.9 Replenishment material. The replenishment material consisting of one copy of the final manual with the set of photolithographic negatives used to produce the manual shall be forwarded as directed by the procuring activity. This material when forwarded shall have a covering letter advising that the material is for replenishment purposes (see 6.5.12).

3.13.10 Status reports (types II, IIS, IIX, and III). The contractor shall prepare and deliver to the procuring activity a report each month beginning 30 days after award of contract through validation until final approval of the manual has been obtained. The report shall indicate the degree of progress, problems encountered, and any delays that may affect production of deliverable items. Proposed changes to the Book Plan Text Outline (see 3.13.1 (a)) shall be submitted with the report for Government review and acceptance. The report shall further indicate what action is being taken to correct or minimize the problem and shall highlight any action required on the part of the Government.

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3.14 Rejection criteria. Failure of contractors to meet all contractual requirements, including this and other referenced specifications, shall be cause for rejection and subject to correction. The following requirements shall also apply to rejection criteria:

(a) Book plan. Failure of the book plan and the resulting manuscript or preliminary manual to conform to the requirements of this specification as defined by the contract unless permission is obtained from the procuring activity (see 3.13.1).
(b) Technical accuracy and inconsistencies. Failure of all information to exactly represent the equipment or system being described and to be consistent with descriptive terms.
(c) Omission of data. Omission of any applicable data elements required by this specification, except as defined in the contract or order.
(d) Failure to meet the readibility grade level as established.
(e) Validation of preliminary and final manuals. Failure to provide adequate evidence of validation.
(f) Workmanship. Failure to conform with the requirements of 3.8 and MIL-M-38784. Significant typographical and grammatical errors are cause for rejection and pages required to be corrected for such errors and applicable when determining whether a change or revision is to be supplied.
(g) Reproducible copy and photolithographic negatives. Failure to include changes or corrections required as conditions of approval.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor (see 6.5.17) is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality program requirements. The contractor shall document and maintain a written quality program (see 6.3) to the extent required to assure that the manual accurately and adequately reflects the equipment or system involved; includes the necessary installation, operation, maintenance, repair, and parts support procedures necessary for satisfactory performance. Acceptance of the program (see 4.3) shall be obtained from the procuring activity or its designated representative prior to development of any part of the manuscript (types II, IIS, IIX, or III) or the preliminary (review) manual (type I). The program shall assure the comprehensive control of the development and evaluation of the manual(s).

4.2.1 Quality program (general). The quality program shall be in accordance with the following:

(a) Organization. Effective management for quality shall be clearly prescribed by the contractor. Personnel performing quality functions shall have well defined responsibility and authority and the organizational freedom to identify problems and to initiate, recommend and provide solutions. Management shall review the status and adequacy of the program on a regularly scheduled basis.
(b) Initial quality planning. The contractor during the earliest practical phase of contract performance, shall conduct a complete review of the contract requirements to identify and make timely provision for any special controls, processes, tests, skills, which may be required. This initial planning shall include the development, maintenance, and implementation of methods and procedures to assure the quality of the manual. This planning shall provide for appropriate review and actions to assure compatibility of preparation, inspection, validation, and documentation.
(c) Manual (preliminary or manuscript) inspection and validation. The quality program shall assure that there is a system for inspection, validation (see 6.5.20), and correction of manuals, concurrent with the design and manufacture of the equipment or system. Validation shall provide a measure of the overall quality of the manual and shall be performed in accordance with 4.2.2(d). When revisions or corrections are required, after any inspection, validation, or approval, there shall be reinspection, revalidation, or reapproval of any characteristic affected.

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(d) Subcontracting. Not required if subcontractor does not prepare, inspect, or validate manuals.

(1) Responsibilities. The contractor shall be responsible for assuring that all manuals procured from his subcontractors conform to the contract and specification requirements. The extent of control invoked and exercised by the contractor shall be identified in the quality program. To assure adequate and economical control, the contractor shall utilize objective quality evidence (see 6.5.7) furnished by his subcontractors. When the Government performs an inspection(s) at a subcontractor's plant, such inspection(s) shall not be used by the contractor as evidence of effective control of quality by such subcontractors. The effectiveness of subcontractor quality control shall be reviewed by the contractor at regular intervals. The contractor shall examine manuals upon receipt from the subcontractor to assure conformance with requirements. All deficiences shall be promptly corrected by the contractor.

(2) Purchase orders. The contractor's quality program shall not be deemed acceptable to the Government unless the contractor requires of his subcontractor, a quality effort achieving control of the quality of the manual(s). The contractor shall assure that all applicable requirements are properly included or referenced in all purchase orders. The purchase order shall contain a complete description of the manuals and material ordered including, by statement or reference, all applicable requirements for preparing, inspecting, validating, packaging, packing and any requirements for Government or contractor inspections, certifications, or approvals. The description of manuals shall include a requirement for contractor inspection at the subcontractor's, when necessary to assure that his quality program effectively implements his responsibility for quality assurance. The purchase order shall contain a requirement for subcontractors facility, to notify and obtain approval from the contractor for unauthorized changes in design or content of the manual. Instructions shall be provided when direct shipment from the subcontractor to Government activities is authorized.

(3) Government inspection, verification, in-process reviews, approval of preliminary manuals (type I), or manuscripts (types II, IIS, IIX, or III) at subcontractor facilities. Government inspection, verification, in-process reviews (see 6.5.21), approval of type I preliminary manuals, or review manuscripts shall not constitute acceptance; nor shall it in any way replace contractor inspection or validation; or otherwise relieve the contractor of his responsibility to furnish an acceptable manual. In-process reviews and verifications will be coordinated by the Government. In-process reviews will be conducted at the contractors facility. Verification will be held as designated by the Government. The contractor shall support the in-process reviews and verifications. When the Government requires inspection or verification at the subcontractor location, the contractor shall include in the purchasing document the following statement:

> "Government inspection/verification is required prior to shipment from your plant. Upon receipt of this order, promptly notify and provide a copy of this order to the Government representative servicing your plant so that planning for Government inspection/verification can be accomplished. In the event the Government representative cannot be identified, the prime contractor shall be notified immediately."

The contractor shall report to his Government representative any nonconformances of Government source inspected/verified manuals and shall require the subcontractor to coordinate corrective action with his Government representative. The Government may elect to perform verification at any time, provided that it does not interfere with the contractor in his performance of the contract. The contractor is responsible for issuing the necessary corrections to the manual resulting from the Government's verification in accordance with 3.10.1.

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4.2.2 Quality program (specifics). The quality program shall include, but not be limited to, the following:

(a) Designation of authority, functions and duties of those personnel responsible for preparation, inspection and validation of manuals. It shall also identify the quality assurance log books, by quality assurance function and the names of the responsible points of contact, that will be maintained by the contractor for recording significant actions. Such log books shall be available during in-process reviews and verification conferences.

(b) Coordination with equipment or system design and production activities to assure the use of the latest technical data and information (e.g., operating and maintenance procedures, drawings, illustrations, etc.) for the preparation of the manual(s) and that approved changes to the equipment or system are promptly incorporated in the manual(s).

(c) Establishment of in-process inspection points and development, maintenance and implementation of inspection criteria to control the adequacy and accuracy during the development stages of the manual(s).

(d) Validation (see 6.5.20) by comparison of performance of the manual content on the physical equipment or system that is described by the manual. Performance of the instructions or procedures within the manual shall be accomplished by contractor or subcontractor personnel of the level for which the manual is written using equipment normally available at the installation site; on the ship; or otherwise.

(1) The demonstration by actual performance of instructions and procedures in each manual relating to operation, assembly, disassembly, maintenance troubleshooting, installation, etc., shall be performed by the contractor or his subcontractor at the equipment or ship construction site, or a combination thereof, to the extent technically practical. Checks may be conducted concurrently with normal assembly, test, and disassembly inspections, when applicable. Written information and drawings in each manual shall be compared against actual equipment to ensure that the information and drawings match the actual equipment in all details. Installation drawings that have been reproduced for use in the manual and have been validated in accordance with MIL-D-23140, do not require further validation. Where appropriate, descriptive material may be checked against validated drawings.

(2) All procedures and instructions, except those which will damage the item or incur extraordinary costs shall be demonstrated by review of performance. Procedures that would damage the item shall be demonstrated by review of drawings. Examples are:

a. Destruction of material, such as disassembly of electronic components or removal of boiler tubes.
b. Boring, grinding, and other shaping repair procedures.
c. Removal of distilling plant tube nest, shafting, crankshafts, inner tubes from periscopes, or other parts when accessibility can be determined by measurement, observation, and reference to drawings.
d. Emergency or damage control operating procedures which would endanger material or personnel.
e. Operating procedures which would entail excessive costs not encountered in the normal operation of the ship. (Excessive costs can be defined to include the provision of special equipment, aircraft, submarine, naval ships, etc., for testing, unless furnished by the Government.)

(3) Validation shall include certification that all information necessary to provide satisfactory installation, operation, maintenance, troubleshooting, and repair is included in the manual.

(e) Detailed review of camera ready copy/photolithographic negatives, as applicable, for printing of the manual by the Government, to ensure that they are identical to the approved, validated, verified, corrected, and approved manuscript, and have complied with the legibility and reproducibility requirements.

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(f) Examination of the final Government printed manuals (upon receipt from the Government) to insure the proper count, that no damage exists, and through the use of an acceptable sampling plan as a minimum, that the printed manuals represent the camera ready copy or photolithographic negatives, as applicable. (NOTE: Printing by the Government shall not relieve the contractor from providing adequate and accurate content in the manual(s).)

4.2.3 Identical and modified manuals and changes. When 3.9.1, 3.9.2, or 3.10.1 applies, the quality program requirements are required to insure that the modifications, changes, and revisions, reflect the equipment as supplied.

4.3 Quality program acceptance. Acceptance of the contractor's quality program by the Government in no way relieves the contractor of the final responsibility to furnish manuals as specified. Acceptance shall not preclude additions, refinements, or changes by the contractor, to the quality program where evidence indicates that the program is not or will not meet the requirements of the contract or this specification. When additions, refinements, or changes to the program are determined necessary by the contractor, acceptance shall be *obtained from the cognizant Government activity prior to implementation.*

4.4 Content assurance page. This page in its completed form shall appear in each manual, including manuscripts and preliminary manuals (see 3.4.2.1.5 and figure 2). In separate correspondence, the contractor shall provide justification for any chapter, section, and paragraph not validated. (Insufficient time or personnel is not considered adequate justification.)

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Individual and multivolume technical manuals. Individual copies and multivolume technical manual. be packed to preclude the possibility of damage in transit. Multivolume technical manuals shall be furnished as complete sets except for manuals shipped for stock. Stock copies of identical volumes shall be packed and shipped in common containers.

5.1.2 Technical manuals shipped with equipment. When copies of the technical manual are packed with the equipment, they shall be packed within the shipping container holding the main unit of equipment. The shipping container shall be marked with the words "Technical Manuals Inside." The manuals shall be so placed that they are readily accessible prior to removing the equipment, and shall not be placed within the vapor-proof barrier material used to enclose the equipment. Technical manuals which accompany equipment shipments that are packed level A or B shall be transparent packaged in accordance with method IC of MIL-P-116. Technical manuals accompanying shipments that are packed level C shall be packaged in a sealed transparent plastic bag. The invoice, packing list, or bill of lading shall indicate the publication number and quantity of the manuals and shall indicate which container includes the manual and its approximate location therein.

5.1.3 Bulk shipment. Technical manuals when shipped for stock in bulk quantities shall not be individually wrapped (see 5.1.1). Containers shall comply with the Uniform Freight Classification Rules or their carrier regulations, as applicable to the mode of transportation.

5.1.4 Manuscripts for review. Manuscript copy shall be packaged in accordance with MIL-M-38784 (see 1.3, 3.11.3, and 3.13.4).

5.1.5 Reproducible copy and artwork. Reproducible copy and original artwork shall be packaged in accordance with MIL-M-38784. Large size inked-linen or vellum illustrations may be rolled and packed in mailing tubes.

5.1.6 Photolithographic negatives. Negatives for delivery to the printer shall be packaged in accordance with MIL-P-38790.

5.1.7 Replenishment materials. Unless otherwise specified in the contract or order, the contractor shall be responsible for obtaining one copy of the final manual and the photolithographic negatives (used for printing the manual) from the printer and packaging them in accordance with MIL-P-38790. Packages shall be forwarded to the procuring activity or as otherwise directed within 30 days after printing.

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5.2 Marking. Bulk quantity shipments, interior packages, and exterior shipping containers shall be marked with the following information for each item enclosed, except for shipment of an individual copy or an individual set of manuals:

(a) Box (number) of (number) (to be listed on multiple container shipments).
(b) Publication number.
(c) Quantity (in package).
(d) Contract or order number.

When applicable, the words "FOR STOCK" shall be marked on the package or packages destined for stock. The publication numbers shall be indicated on the shipping documents. When a contract or order requires technical manuals, revisions, or changes having different publication numbers, the stock copies of each number shall be packaged as separate items. All shipments shall be made in accordance with security requirements (see 3.2).

6. NOTES

6.1 Intended use. The technical manuals prepared to this specification are intended to be used for installation, operation, maintenance, repair, and parts support of equipments and systems for NAVSEA and NAVELEX. The technical manual will be used as a training document in the classroom and as a source for on-the-job training.

6.1.1 Type I. Type I manuals are normally required for electrical and mechanical equipment.

6.1.2 Type II, IIS, and IIX. Type II manuals are required for electronic and interior communications equipment; type IIS required for service test electronics and interior communications equipment; and type IIX for experimental electronic and interior communications equipment manuals. However, types II, IIS, and IIX manuals may be specified for electrical and mechanical equipment if the electrical/electronic circuitry and other considerations justify its use.

6.1.3 Type III. Type III manuals are required for systems, providing that the system equipments are supplied with equipment-level technical manuals.

6.2 Ordering data. Procurement documents (including the Contract Data Requirements List, Form DD 1423) should specify the following:

(a) Title, number, and date of this specification.
(b) Type of manual required (see 1.2).
(c) Deliverable data items (see 1.3).
(d) Reading grade level (see 3.3).
(e) Requirements for appendixes, as applicable (see 3.4.4).
(f) Warranty/guarantee information (see 3.5.3.1).
(g) Any special requirements for a standard log form (see 3.5.4.5 and 3.6.5.3).
(h) Requirements for the parts list (see 3.5.9).
(i) Requirements for installation data (see 3.5.10).
(j) Applicable distribution statement (see 3.7.3 and 6.2.1).
(k) If typography other than that specified in 3.8.1 is required.
(l) Original manual, permanent change[1/] (see 3.10.3), interim change (see 3.10.2), revision[1/] (see 3.4.1.5, 3.10, 3.10.4, and 6.5.13.1), supplement manual (see 3.4.1.4 and 3.4.3.4).
(m) Quantity of manuals required (see 3.12).
(n) Quantity and delivery date of deliverable items (see 3.13).
(o) Type of binder.

[1/] Where the extent of change to the manual is less than 25 percent, a permanent change should be procured. When the extent of a change is over 25 percent, an updated revision (see 3.4.1.5.1) of the manual shall be procured. When the basic manual has been changed to the extent that it is wholly inadequate because of its arrangement, or other reason detrimental to the user, a complete revision (see 3.4.1.5.2) of the manual shall be procured.

6.2.1 Distribution statement. A statement used in marking a technical document to denote the conditions of its availability for distribution, release, or disclosure. This statement will be provided by the approving activity (see 6.2).

6.3 Quality program. A comprehensive quality program is necessary to assure adequate and accurate manuals for Navy use. Therefore, the quality program shall cover planning, development, production, inspection, review, validation, and implementation.

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6.4 Planning, inspections, and reviews.

6.4.1 Planning. Planning for manuals to support an equipment or system must begin during the early design phase of the related equipment or system, to assure the quality of the manual.

6.4.2 Inspections and in-process reviews. Inspections and in-process reviews conducted by the Government are held primarily to provide guidance to the contractor and to assure that manuals are being prepared in accordance with the contract and specification requirements. These inspections may be conducted at the contractor's facility or his source facility at any time during the development of the manuals. (In-process reviews are normally required at least once every three months.)

6.4.3 Review. Verification of manuals may be simultaneously performed with validation in cases where:

(a) Time and equipment or facility availability does not permit separate verification.
(b) When advantageous to the Government.

6.5 Definitions.

6.5.1 Component. A composite fabricated unit, generally complete within itself, that is designed to perform a stated service when installed in its proper position within a ship or station (e.g., boiler, motor driven centrifugal pump, etc.).

6.5.1.1 Single component. A hull, machinery, or electrical (H, M, and E) item performing only one function, relatively simple in design and operation and which is normally integrated into an operating system/subsystem such as a pump, valve, motor, etc.).

6.5.1.2 Component identification number. Component identification number (CID) is a design identifier which gives the manufacturer's name and the characteristics for a particular piece of equipment.

6.5.2 Equipment. One or more assemblies capable of performing a complete function.

6.5.3 Final manual. A final manual complies with all requirements, including production requirements of this specification and has been validated, verified, and approved to be complete and accurate.

6.5.4 Function. A group of functional devices or circuit elements, that work together to accomplish an operation of a system or an equipment.

6.5.4.1 Major function. An essential functional operation which is fundamental to operation (as opposed to a circuit) of the equipment. (e.g. transmit, receive, display, hoist, etc.).

6.5.5 Interim change. An expeditious method of promulgating an emergency, urgent or minor manual change by means of a message, change pages, addendum sheets, or published article in the Electronics Information Bulletin (EIB). (Interim changes are often referred to as temporary changes.)

6.5.5.1 Advance change notice. The formal NAVSEA method of promulgating interim changes.

6.5.6 Interior communications equipment. Equipment normally used for shipboard interior communication. These include navigation or related equipment designated Interior Communications (IC) equipment such as the following:

(a) Plotting tables.
(b) Deck reckoning indicators.
(c) Gyros and gyro amplifiers.

6.5.7 Objective quality evidence. (See MIL-STD-109.) Examples of verifiable evidence are validation reports, inspection records, certificates, and any other evidence relating to the control of quality.

6.5.8 Part. One piece, or two or more pieces joined together, which are not normally subject to disassembly without destruction.

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6.5.8.1 Part, military identification number. The number derived from a military specification such as 1439027/1-1, or type RWR74549R9FM; or a military standard such as MS 18272.

6.5.9 Planned maintenance system. Aspects of the Navy's maintenance and material management system as described in MIL-P-24534.

6.5.10 Preliminary manuals.

6.5.10.1 Preliminary manuals (type I). A preliminary (review) manual is submitted to the procuring activity for review and acceptance prior to the development of the final manual. The preliminary (review) manual serves the same purpose for a type I manual as a review manuscript does for a type II or type III manual. Preliminary manuals shall be superseded by final manuals. Preliminary (review) manuals may be packed with the hardware if the final manual has not been completed.

6.5.10.2 Preliminary manual (types II, and III). A preliminary manual is a manual which includes all data required for the final manual but has not received final approval and may not comply with the production requirements for a final manual. Preliminary manuals shall be superseded by final manuals.

6.5.11 Repair. Action required to restore a repairable part, subassembly, or assembly to a specified condition, but not necessarily to make like new in physical appearance.

6.5.11.1 Repair, troubleshooting; maintenance concept. The planned concept by which the equipment, assemblies, units, components, modules, or piece parts are to be repaired or replaced by designated maintenance level activities (organizational, intermediate or depot).

6.5.12 Replenishment material. Replenishment material is material required to provide for economical quality reprinting of technical manuals.

6.5.13 Revision. A second or subsequent edition of a manual issued complete with covers and all pages which supersedes (except as specified in 6.5.13.1) the preceding edition (see 3.4.1.5).

6.5.13.1 Updated revision. An updated revision is required when the current manual does not accurately or adequately cover the equipment or system (because of errors, production change, field changes, or shipalts made to the equipment) and when the extent of required page changes exceeds 25 percent. An updated revision can be procured when it is necessary to represent a new model or different configuration when a manual exists which can be used as the baseline and will provide an effective technical manual at less cost to the Government. This manual shall reference only the model covered and shall not supersede the basic manual used as the data baseline, and shall have a unique NAVSEA or NAVELEX number.

6.5.13.2 Complete revision. A complete revision is a new edition in full conformance to this specification and is required when the basic manual is wholly inadequate because of obsolescence, gross inaccuracies, etc.

6.5.14 Service test equipment. An equipment used for test under service condition for evaluation of suitability and performance. It closely approximates the final design, has the required form, and employs approved parts or their interchangeable equivalents.

6.5.15 Set. A unit or units and necessary assemblies, subassemblies, and basic parts connected or associated together to perform an operatonal function. Refer to ANSI Y32.16 for a detailed description.

6.5.16 Supplement manual (complementary manual). Augments a technical manual so that both manuals jointly represent the configuration of the equipment and all necessary information to install, operate, maintain, and repair a basic equipment.

6.5.17 Supplier. This is the same as "contractor" and "prime contractor".

6.5.18 System. A system includes two or more equipments (sets) or components, each having its own identity and nomenclature, arranged and interconnected to perform a specific operation, e.g.:

(a) An electronic system can be identified as an IFF system, ECM system, AEW system, ASW system, NTDS system, etc.
(b) An electromechanical system can be identified as a propulsion system, underway replenishment system, degaussing system, etc.

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6.5.19 Unit. A major building block for a set or system, consisting of a collection of basic parts, subassemblies, and assemblies packaged together as a physically independent entity.

6.5.20 Validation. The process by which the contractor assures the technical accuracy, and adequacy of the manual and that it represents the latest configuration of the equipment by actual test against the hardware.

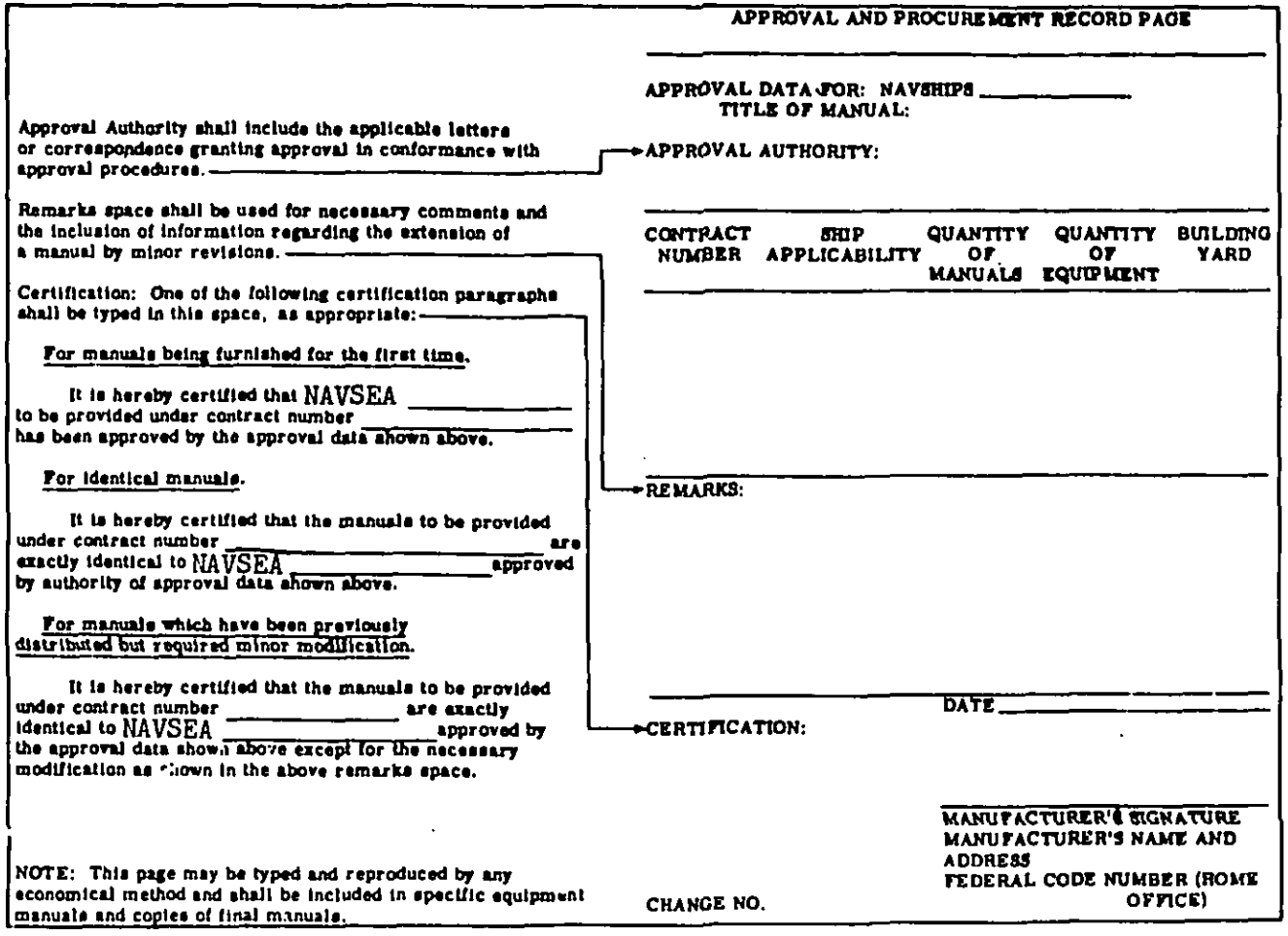
6.5.21 Verification. The process by which the Government assures the accuracy of the manual(s) by actual comparison with hardware. This includes Government action to assure proper execution of validation by the contractor.

6.6 Changes from previous issue. The symbol "†" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Review activity:
EC

Preparing activity:
Navy - SH
(Project TMSS-N096)

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APPROVAL AND PROCUREMENT RECORD PAGE

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Approval Authority shall include the applicable letters or correspondence granting approval in conformance with approval procedures. → APPROVAL AUTHORITY:

Remarks space shall be used for necessary comments and the inclusion of information regarding the extension of a manual by minor revisions.

CONTRACT NUMBER	SHIP APPLICABILITY	QUANTITY OF MANUALS	QUANTITY OF EQUIPMENT	BUILDING YARD

Certification: One of the following certification paragraphs shall be typed in this space, as appropriate:

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It is hereby certified that NAVSEA _____ to be provided under contract number _____ has been approved by the approval data shown above.

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CHANGE NO.

FIGURE 1. Approval and Procurement Record Page.

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Contractor:

Sub-Contractor (if performing validation):

Contract No(s) and Purchase Orders, if applicable

Chapter	Section	Paragraph	Date Validation Completed	Check here if not validated

Name & Authority of Validating Officer:

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FIGURE 2. Content Assurance Page.

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NAVSEA (USER) TECHNICAL MANUAL DEFICIENCY/EVALUATION REPORT (TMDER)

(Check appropriate block)

☐ UNCLASSIFIED ☐ CLASSIFIED (Specify) (If CLASSIFIED, this report must be marked and mailed in accordance with OPNAVINST 5510.1E.)

INSTRUCTIONS 1. Use this report to indicate deficiencies, user remarks and recommendations relating to the publication.
2. * To be filled in by Contractor before printing.
3. Fold on dotted line, staple and mail to Commanding Officer, Naval Ship Weapon Systems Engineering Station, NSDSA (Code 5700) Dept. of the Navy, Port Hueneme, CA 93043

*1. NAVSEA NO.	*2. VOL/PART NO.	*3. TITLE (Nomenclature)

*J. TITLE (Nomenclature) (Cont'd)	4. REV DATE (TN, CHG)	MICROFILM/MICROFICHE			
		5. CART/FICHE #	6. CART/FICHE DATE	7. FRAME NO	8. OTHER I.D. NO

9. SYSTEM/EQUIPMENT	10. IDENTIFICATION (MR/MOD/AN)	11. PRIORITY

12. USER EVALUATION (Manual)? (Check One)

☐ EXCELLENT ☐ GOOD ☐ FAIR ☐ POOR ☐ COMPLETE ☐ INCOMPLETE

RELATED REMARKS

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14. RECOMMENDED CHANGE(S) TO PUBLICATION

PAGE NO	PARA-GRAPH	LINE NO	FIGURE NO	TABLE NO	RECOMMENDED CHANGES AND REASONS

15. ORIGINATOR	16. RANK/RATE OR GRADE AND TITLE	DATE
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NAVSEA 5600/2 (REV. 6-78) (BACK)

3A. Front.

FIGURE 3. NAVSEA (User) Technical Manual Deficiency/Evaluation Report (TMDER).

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3B. Back.

FIGURE 3. NAVSEA (User) Technical Manual Deficiency/Evaluation Report (TMDER).

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USER ACTIVITY TECHNICAL MANUAL COMMENT SHEET

Title_____

NAVELEX NO._____

VOLUME NO._____

(Fold on dotted line on reverse side, staple, and mail to:

NAVAL ELECTRONIC SYSTEMS COMMAND

_____

_____

PROBLEM, QUESTION, SUGGESTION, COMMENT:

NOTE Both sides of this form shall be reproduced locally to size of manual as required.

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4A. Front.

FIGURE 4. User Activity Technical Manual Comment Sheet.

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FIGURE 4. User Activity Technical Manual Comment Sheet.

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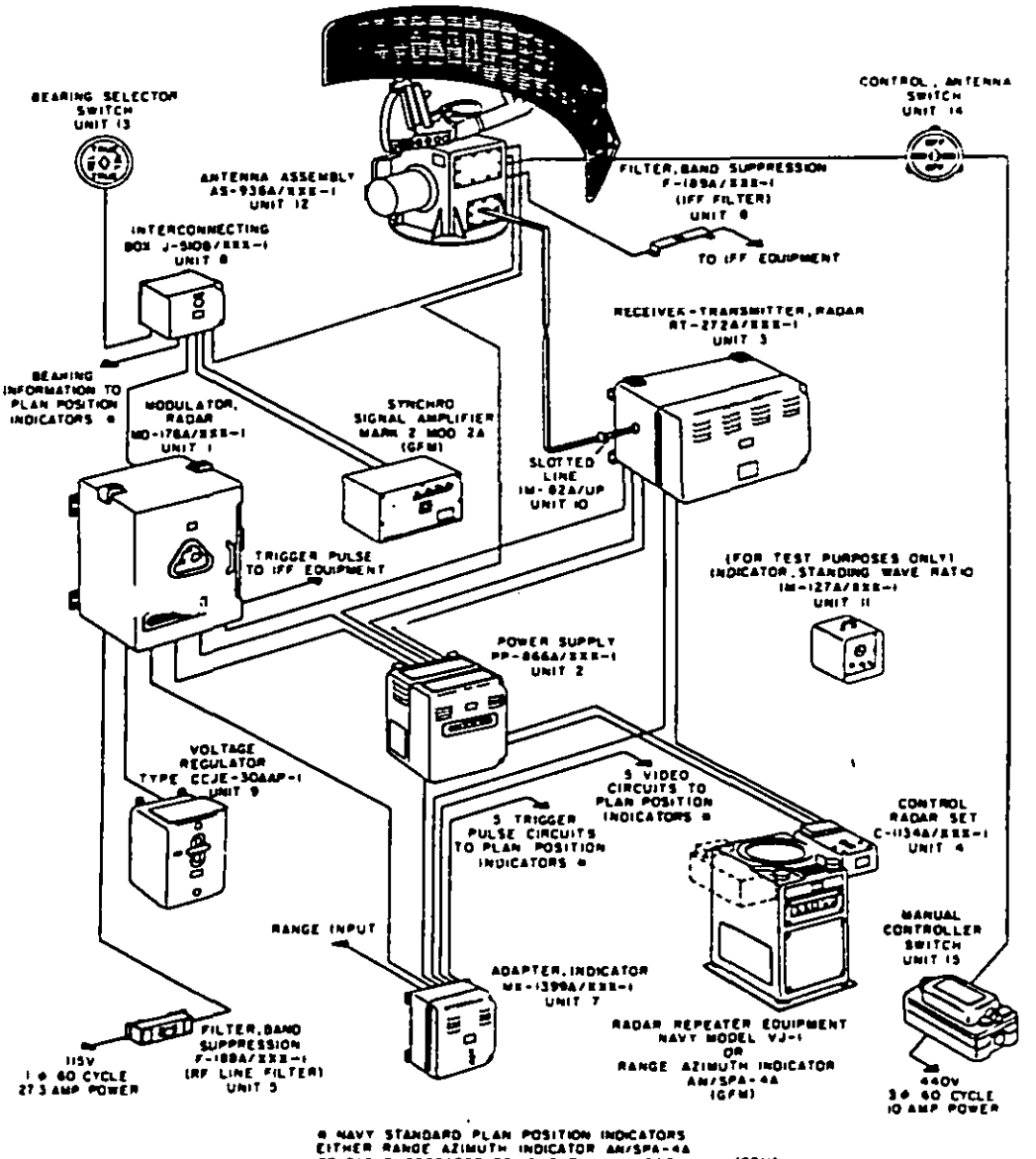
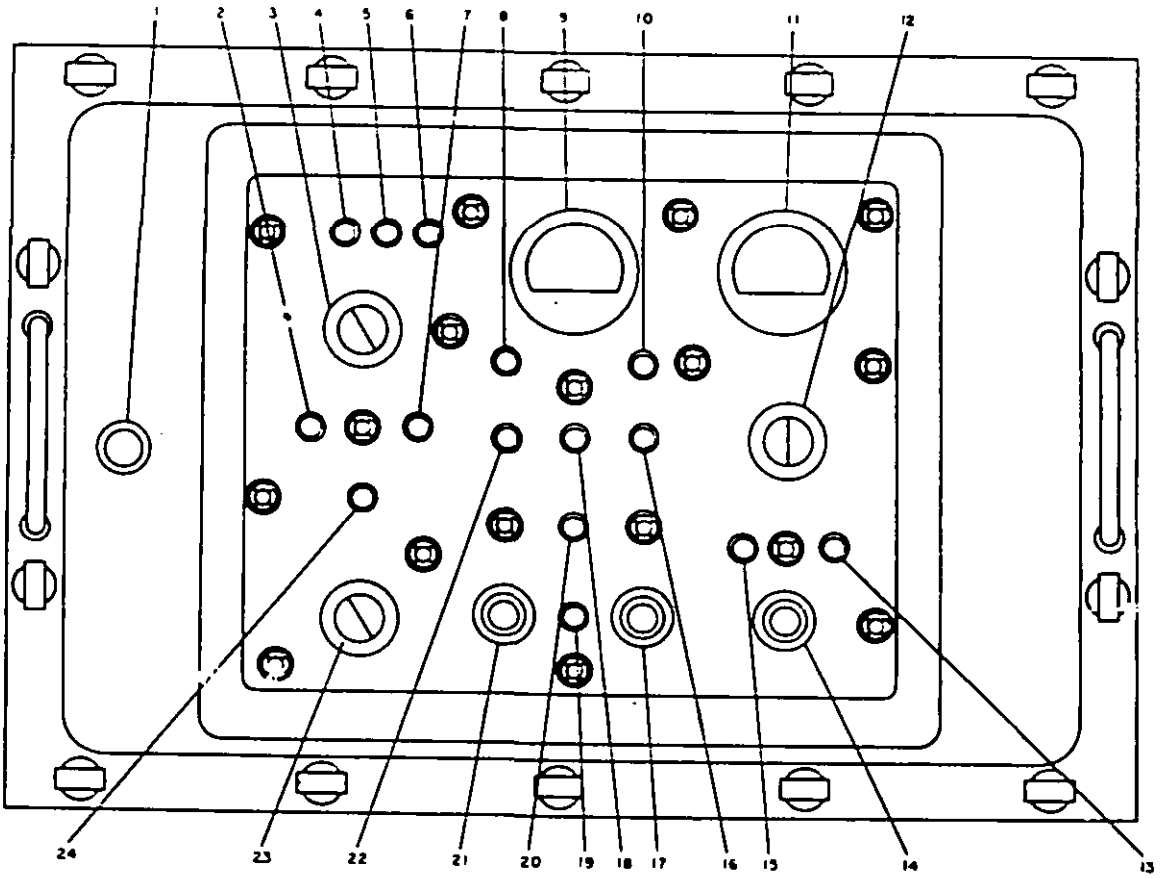


Figure 1-1. Radar Set AN/XXX-1, Relationship of Units

NOTE: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements.

FIGURE 5. Equipment Illustration Showing Relationship of All Units.

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1. PHONE BUZZ buzzer
2. POWER INTERLOCKS lamp
3. POWER switch
4. STDBY lamp
5. READY lamp
6. TEST lamp
7. CW ILLUMINATOR INTERLOCKS lamp
8. RF POWER DRIVE lamp
9. RF POWER meter
10. RF POWER RADIATED lamp
11. DEVIATION meter
12. DEVIATION switch
13. FM NOISE ALARM lamp
14. NME RESET button
15. AM NOISE ALARM lamp
16. ILLUMINATOR STATUS NO GO lamp
17. RADIATE OFF button
18. ILLUMINATOR STATUS MARGINAL lamp
19. RADIATE lamp
20. HT MODE lamp
21. RADIATE ON button
22. ILLUMINATOR STATUS FAULT lamp
23. DIRECTOR LOUVERS switch
24. DIRECTOR LOUVERS OPEN lamp

FIGURE 6. Equipment Controls and Indicators.

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Training Summary for the Ingersoll-Rand
Model NA 13.5 CFH, 4500 PSI High Pressure
Air Compressor (HPAC) Log Forms
for Use on SSBN627, Class

The following summary discusses selected values on the Air Compressor Log forms which is of concern to an operator. It will be noted on the log forms that some values outside the listed ranges need not be circled. These values are shown in Black and may be either the upper or lower value shown for that particular range. The values which are recorded on the log form which fall outside the normal range Red values are of concern and must be circled. The Red values are the significant values and values circuled on the log forms will show that a potential problem area is developing or has already developed.

TEMP CIRC WATER OUT - Values lower than the 95°F given for the normal range of the circulating water out temperature need not be circled. This value is dependent on the "circulating water" in temperatures which are recorded elsewhere on the new log form.

PRESSURE 4th STAGE - The lower value shutdown is given for this stage as it is an indication that back pressure regulating values in the air dryers are not functioning properly (see TECH MANUAL for H. P. Dehydrator and Filters - NAVSEA S6228-AB-MMA-000/NA, paragraph 1-3-4 and page 1-5). This value provides the required pressure for proper functioning of the ships air dryers. Operation of the air dryers at pressures lower than 4300 psi can cause saturation of the desiccant with subsequent breakdown desiccant carryover and permit saturated air to be sent to the air banks.

The normal range values for the 4th stage were selected in the following manner:

a) Lower range values - 4300 psi - this value is required again for proper operation of the air dryers.

b) Higher range value - 4800 psi - values this high may be required to charge the air banks to 4500 psi.

TEMP 1st STAGE SUCT - Values outside the listed normal range need not be circled as the operator has little or no control over the compartment air temperature. Values outside the normal range will not adversely affect compressor operation.

INTERSTAGE LOWER VALUE TEMP VALUES - Temperature values below the listed lower normal range for the various stages within the compressor need not be circled as these are a function of the circulating water inlet temperature and this value is recorded elsewhere on the log form. In addition these values are not normally an indication of problems developing within the compressor.

FIGURE 7. Sample Log Form Training Summary.

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HIGH PRESSURE AIR COMPRESSOR

COMPRESSOR NUMBER _____

USS _____ (SSBN~

DATE _____

	SHUTDOWN		NORM *		00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23
HPAC																												
TEMP CIRC WATER OUT	130[a]		80	110																								
PRESS 1st STAGE	85[b]	60	62	80																								
2nd STAGE	280[b]		300	320																								
3rd STAGE	1370[b]		970	1220																								
4th STAGE	ABOVE 6100[b] BELOW 6250[c]	4300	4900	4800																								
TEMP 1st STAGE SUCTION			80 110																									
1st STAGE DISCHARGE	430[d]		320	395																								
2nd STAGE SUCTION				110																								
2nd STAGE DISCHARGE	430[d]		310	345																								
3rd STAGE SUCTION			90	110																								
3rd STAGE DISCHARGE	430[d]		350	365																								
4th STAGE SUCTION			80	110																								
4th STAGE DISCHARGE	430[d]		370	410																								
FINAL DISCHARGE			80	120																								
FRAME OIL PRESS	BELOW 12[e] ABOVE 40[f]		18	30																								
LEVEL LO SUMP		AT FILLMARK																										
TEMP LO SUMP		65-120																										
LO LUBRICATORS LOWER SIGHT GLASS LEVEL		MAINTAIN ABOVE ½																										
CIRC WATER PUMP DISCH PRESS	BELOW 18[f]	20	65																									
TEMP CIRC WATER IN			75	80																								
LUBRICATOR FLOW RATE EACH GLASS	ABOVE 2[g] BELOW 1[h]	½	(DROPS/)																									

* CIRCLE LOG VALUES WHICH ARE OUTSIDE NORMAL RANGE

** SHIPS WITH ENGINE ROOM AFW COOLING SYSTEM MEASURE TEMPERATURE ON DOWN STEAM SIDE OF THERMO CONTROL VALVE IF EXISTS

a WILL SHUT DOWN AUTOMATICALLY.

b RELIEF VALVE LIFTS

c WILL CAUSE WET AIR TO BE SUPPLIED TO BANKS RESET BACKPRESSURE CONTROL VALVE

d WILL SHUT DOWN AUTOMATICALLY EXPLOSION MAY OCCUR AT HIGHER TEMPERATURE

e INADEQUATE LUBRICATION, WILL SHUT DOWN AUTOMATICALLY.

f LUBE OIL BLOCKAGE, MANUAL SHUTDOWN REQUIRED

g WILL CAUSE EXCESS CARBON

h INADEQUATE LUBRICATION

i MANUAL SHUTDOWN REQUIRED (FOR PROBLEMS SEE TECH MANUAL TABLE 4 2 1, SHEET 3)

TIME METER READINGS*

AT 24 00 _____

AT 00 00 _____

HOURS TODAY [ ]

*WHEN METER INSTALLED _____

FOR INGERSOLL-RAND MODEL NA 13.5 CFH, 4500 PSI AIR COMPRESSOR

CSS 14 HPAC LOG
REV 1

AS 34 0101 LF L34 $100

FIGURE 8. Air compressor log form.

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NAVSEA 0551-LH-1550-0010 (REV 4)

3-1. FORK ASSEMBLY. (Figure 3-1). The 48-inch long side reach forks are connected to the fork carriage. The fork carriage moves up and down by roller chains driven by a hydraulic cylinder and piston. The fork carriage is roller-mounted to the inner mast assembly. Rollers mounted on inner mast move in channels which make up the outer mast assembly. As the fork carriage moves up, the top of the hydraulic piston assembly comes in contact with the inner mast. If the hydraulic piston moves up more, it lifts the inner mast along with the fork carriage. Forks can be raised from deck level to 60 inches above the deck by moving the fork carriage.

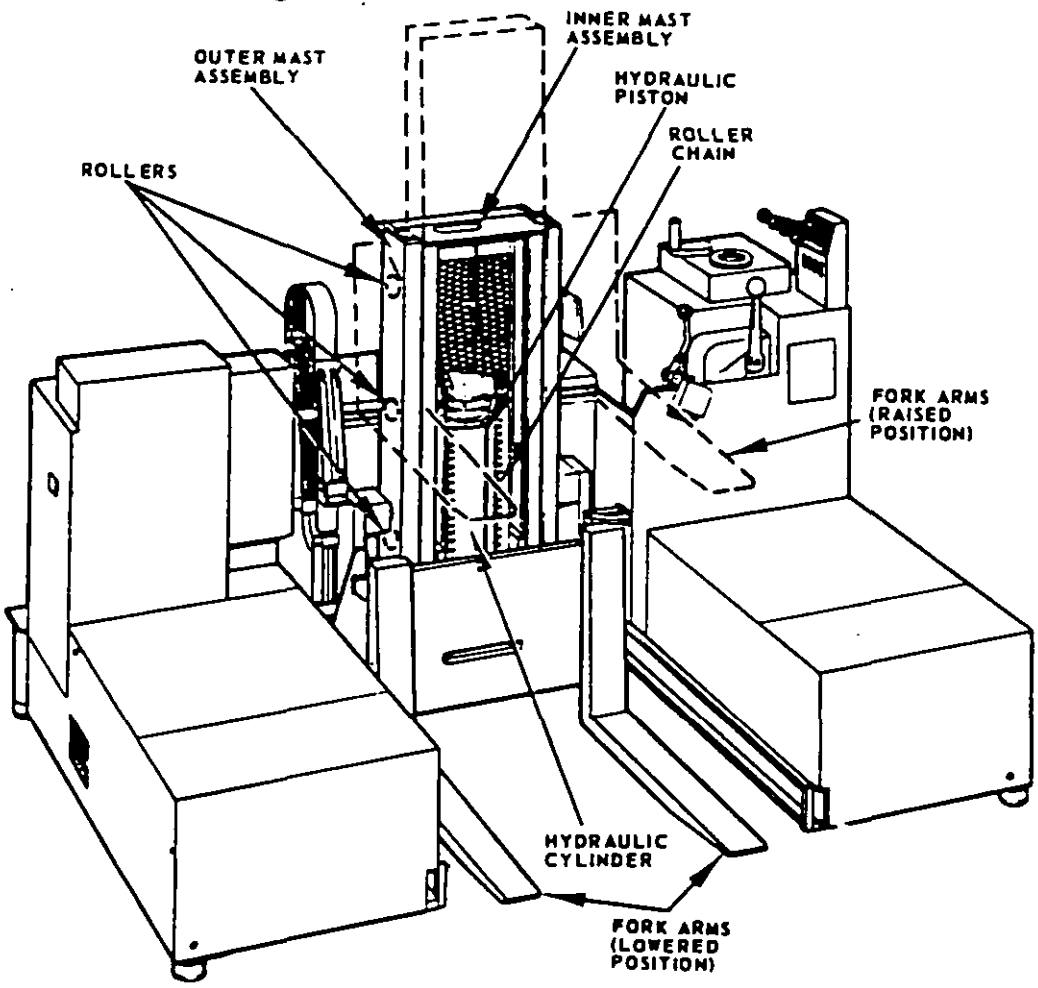


Figure 3-1. 4-D Fork Truck Fork Assembly

FIGURE 9. Major Parts of Component or Equipment.

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3-5. INTERACTION OF PARTS. (Figure 3-5). The steering mechanisms for the 4-D fork truck is used for direction changes and normal turns when the truck is in motion. The mechanisms provide steering control, an electronic system, and four steering motors connected to the wheel assemblies by chain drives. Moving the steering control causes the electronic system to guide each steering motor to move. As all four wheels are used to steer the truck, the wheels do not all turn in the same direction. The electronic steering system guides each wheel to move in the right amount and direction for the turn. The steering motors are also used to swing the wheels for 90-degree changes in direction when the 4-D fork truck is lifted on the jacks for a 90-degree turn.

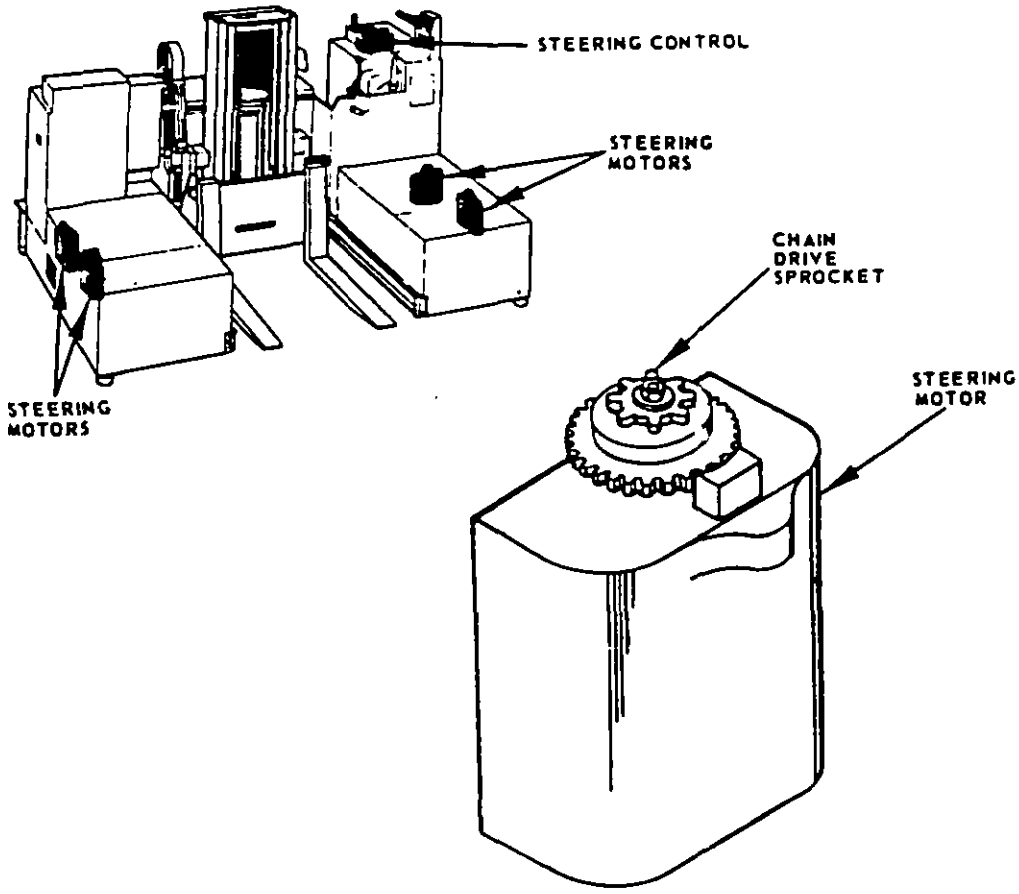


Figure 3-5. 4-D Fork Truck Steering Motor

FIGURE 10. Interaction of Major Parts.

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3-12. SERVO VALVE (See figure 3-12). The servo valve gets mechanical signals from the winch control lever in the FAST control station. If the control lever is moved to the PAY OUT position, the servo valve gets the signal and positions the pump yoke to send pressure to the hydraulic motor to turn the winch drum in the PAY OUT direction. If the servo valve gets a signal to HAUL IN, the pump yoke is positioned so that the winch drum is turned to HAUL IN.

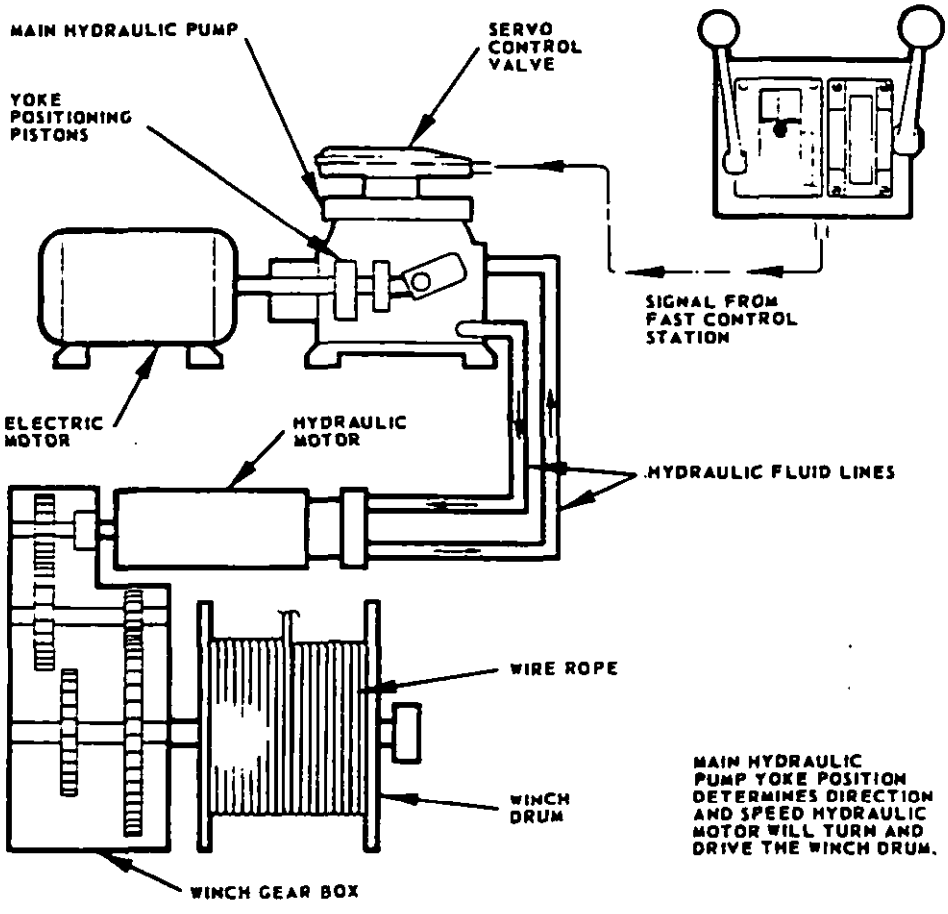
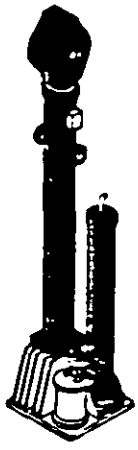


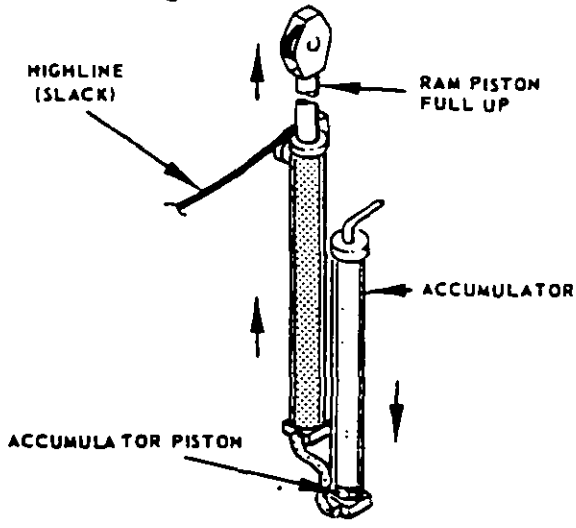
Figure 3-12. The Inhaul/Outhaul Winch Servo Valve.

FIGURE 11. Functional Description Diagram.

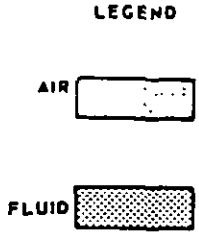
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3-10. HOW THE RAM TENSIONER WORKS (See figure 3-10). When the ram tensioner piston has traveled to the top, the accumulator piston is at the bottom of the accumulator cylinder. When this occurs, the highline is slack.





As the highline is tightened by the highline winch, the piston is forced down, forcing fluid out of the ram cylinder into the accumulator. The return of the hydraulic fluid causes the piston in the accumulator to rise, forcing the air back into the air supply source.



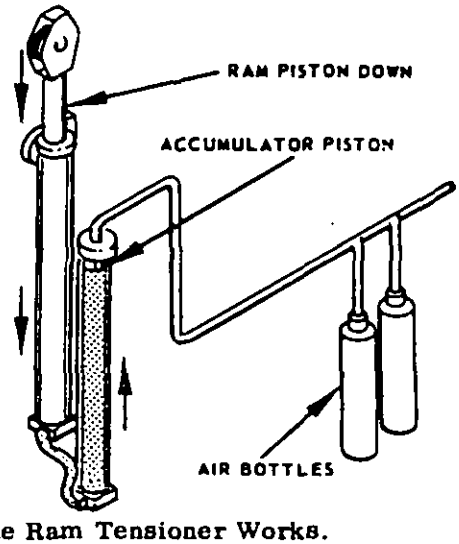


Figure 3-10. How the Ram Tensioner Works.

FIGURE 12. Equipment Inner Operation.

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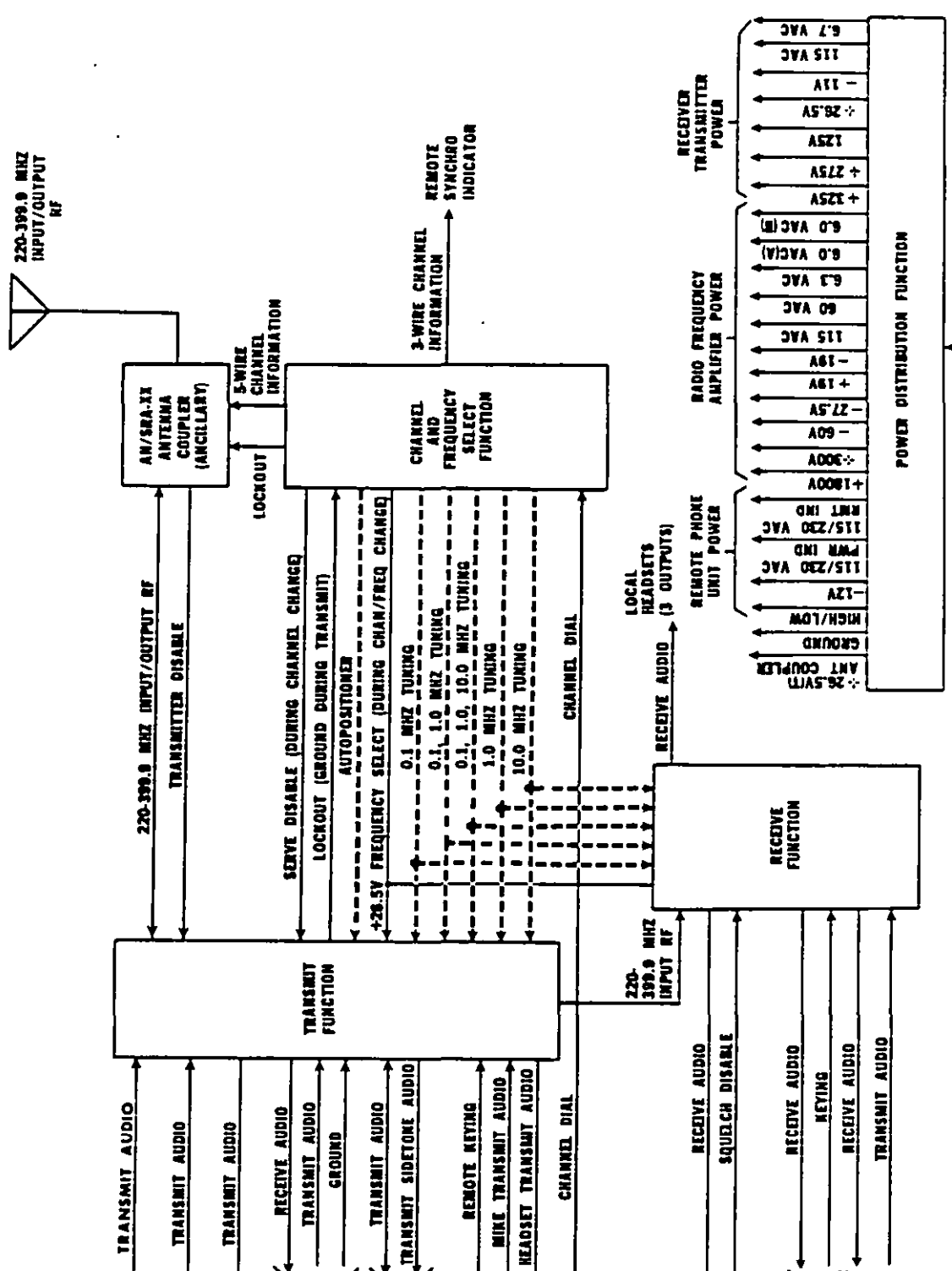


FIGURE 13. Overall functional block diagram.

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Check the source to verify that this is the current version before use.

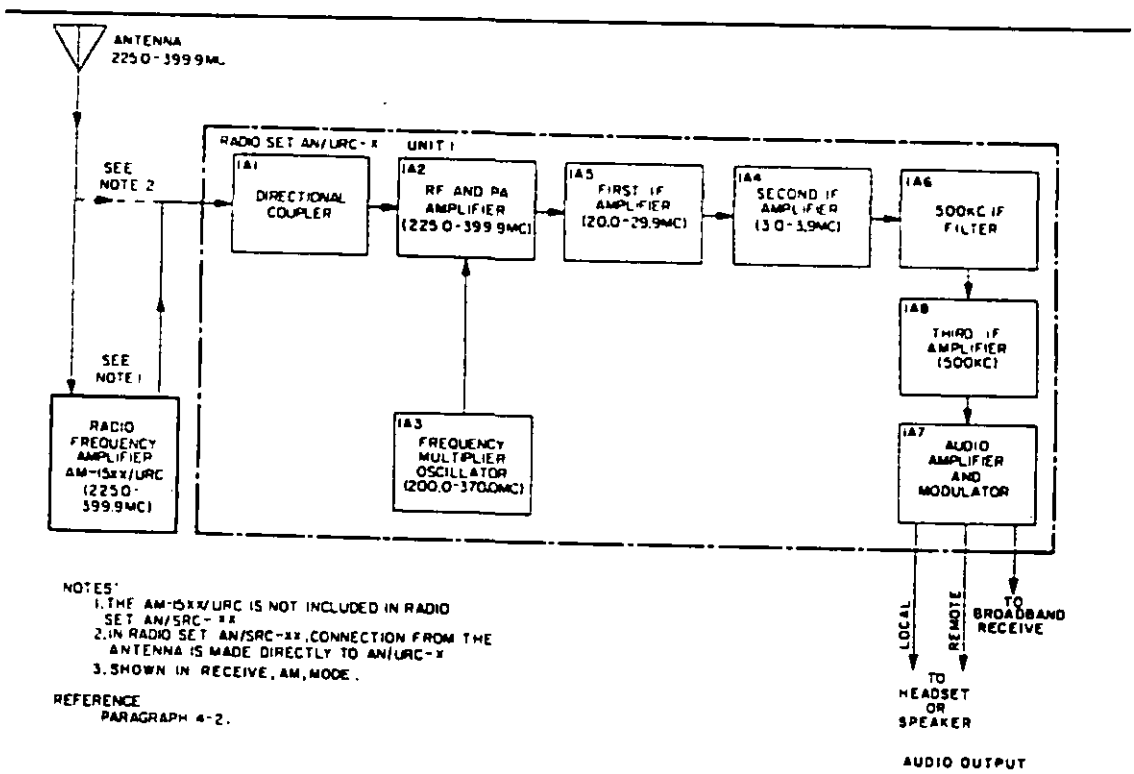
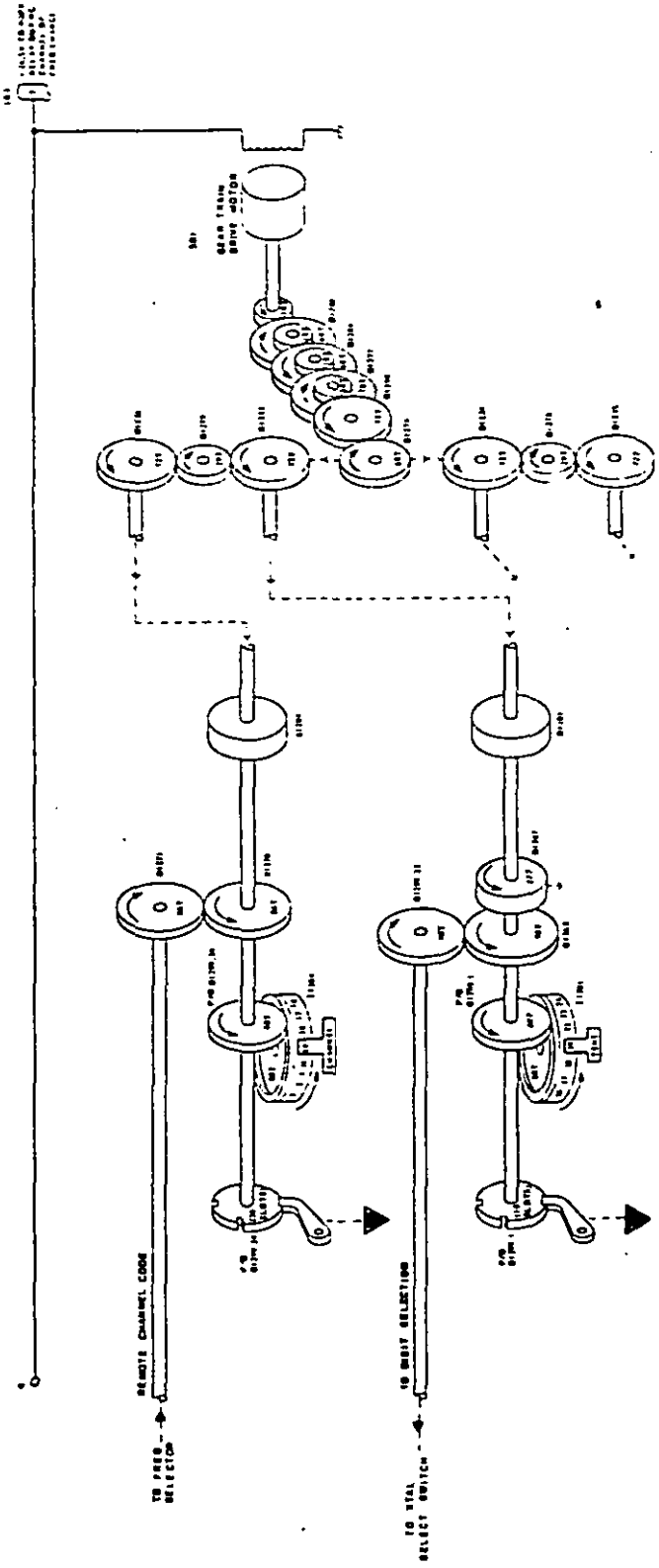


FIGURE 14. Functional Block Diagram.

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Note: Sample arrangement only. Size and legibility do no necessarily conform to minimum specification requirements.

FIGURE 15. Mechanical Schematic Diagram.

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TABLE ZZ. TROUBLESHOOTING INDEX RADIO SETS AN/SRC-XX AND AN/SRC-XX

FUNCTIONAL AREA	TROUBLE SHOOTING PARAGRAPH	TROUBLE SHOOTING DIAGRAM	FUNCTIONAL DESCRIPTION PARAGRAPH	ALIGNMENT/ ADJUST PARAGRAPH
AC Power	5-3	5-8	3-9a	6-105, 6-106
DC Power	5-4	5-19	3-9b	6-107 through 6-110, 6-127
Keying	5-5	5-24	3-13	6-22
Receive RF	5-8	5-1	3-4	6-112 through 6-115
System Channel and Frequency Selection	5-9	5-16	3-10, 3-12	6-121

FIGURE 16. Troubleshooting Index.

TABLE ZZ. RELAY INDEX

REFERENCE DESIGNATION	FUNCTIONAL NAME	ENERGIZING VOLTAGE	TROUBLE-SHOOTING DIAGRAM (FIG. NO.)
6A4K9	HV Door Interlock	115 Vac	5-21
6A4K10	Cabinet Interlock	28 Vdc	5-22
6A4K11	Buzzer Relay	28 Vdc	5-22

TABLE ZZ. INDICATOR LAMP INDEX

REFERENCE DESIGNATION	FUNCTIONAL NAME	ENERGIZING VOLTAGE	TROUBLE-SHOOTING DIAGRAM (FIG. NO.)
9A8DS15	HV INTERLOCK CONFIDENCE-VSWR TRIP-OUT	28 Vdc	5-22
9A8DS16	HV INTERLOCK CONFIDENCE-HVPS	28 Vdc	5-22

TABLE ZZ. CIRCUIT BREAKER AND FUSE INDEX

REFERENCE DESIGNATION	FRONT PANEL MARKING	RATING		CIRCUIT PROTECTED	TROUBLE-SHOOTING DIAGRAM (FIG. NO.)
		VOLTS	AMPS		
9A8F1	KLYSTRON FILAMENT FUSE ALARM 5 AMP	250	5	Klystron filament control circuit and filament transformer 9A1T106.	5-32
14A2F1	CONTROL	125	3	Voltage sensor bridge power supply consisting diodes 13A2CR1 through CR4.	5-2

FIGURE 17. Relay, Lamp and Protective Device Indexes.

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TROUBLE ANALYSIS CHART

This chart is a partial chart for a propulsion steam turbine and is considered typical. When incorrect operation is observed, it is often possible to reduce the overall servicing time by studying all the symptoms and applying the following TROUBLE ANALYSIS CHARTS:

CHART 5-1	LUBE OIL SYSTEM TROUBLES		
SYMPTOMS	IMMEDIATE ACTION	PROBABLE CAUSE	REMEDY
1. Low-lube alarm sounds. 2. Loss of lube-oil pressure.	1. Stop the shaft and engage shaft lock. 2. Start standby lube-oil pump.	1. Lube-oil pump shut down. (pump or cut-in switch failure). 2. Broken pipe. 3. Clogged strainer. 4. Insufficient oil in system. 5. Alarm setting incorrect.	1. Check the system to determine the cause, and then correct. 2. Locate and repair any leaks. 3. Shift duplex oil strainers. 4. Check oil level; replenish as necessary. 5. Reset alarm to correct value.
CHART 5-2	BEARING TROUBLES		
1. High bearing temperature/hot bearing. 2. Wiped bearing.	1. Slow down and keep turning to avoid freezing babbitt to shaft while bearing & journal cool. Then stop and replace bearing.	1. Foreign matter present, or local loss of oil.	1. Inspect turbine bearings and accessible reduction gear bearings. Take bearing-wear measurements. 2. Inspect oil strainers for babbitt particles. 3. Replace bearings as required. 4. If several bearings are wiped, do not resume operation until complete inspection and repair have been made.

FIGURE 18. Sample Trouble Analysis Chart (Type I Manuals).

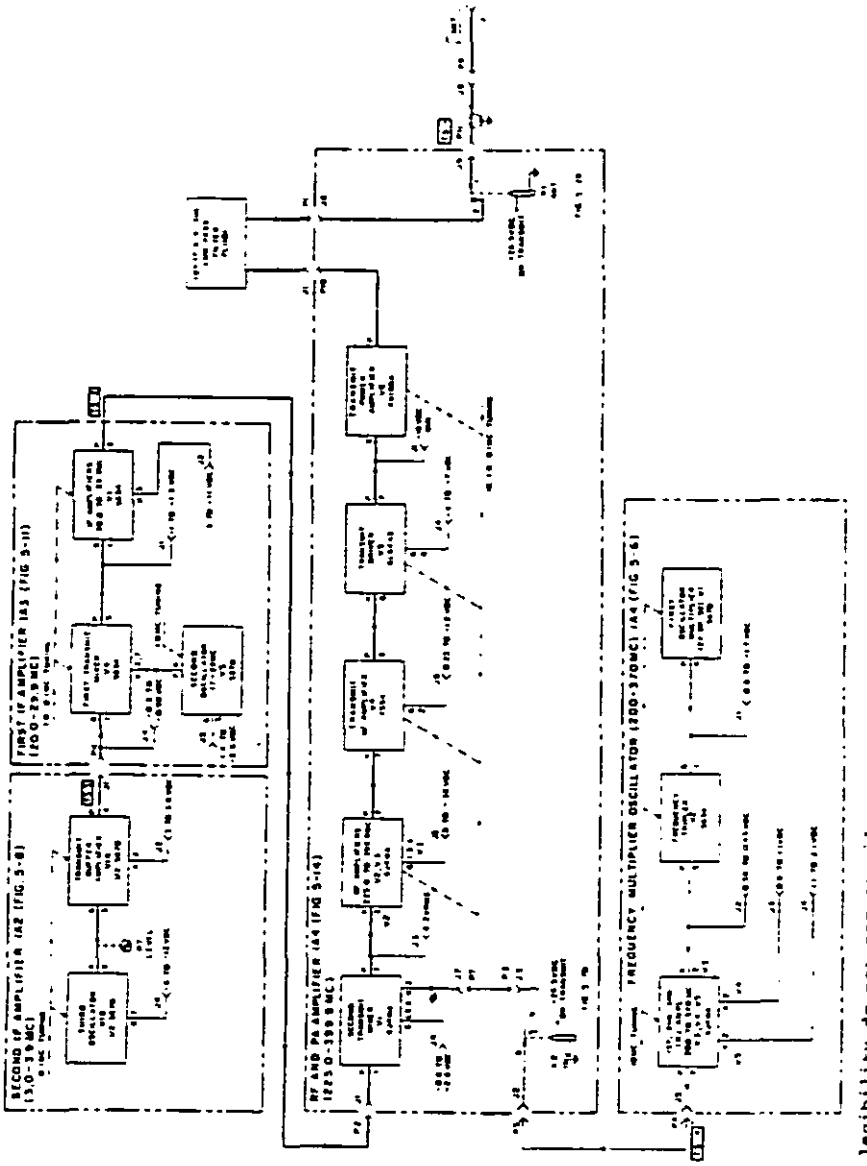
Source: https://assist.dla.mil -- Downloaded: 202640-18T05:27Z
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TABLE ZZ. MAINTENANCE TURN-ON PROCEDURE

STEP	OBSERVE	REFERENCE
1. Preliminary Procedure.		
a. Position the following switches on rear deck assembly 1A210A1 as indicated. Switch — Position POWER — OFF BATTLE SHORT — OFF STOW — BRAKES APPLY		
b. Position POWER switch on console, 1A220A20 (see figure 5-2) to OFF.		
c. Check to ensure that all chassis or subassemblies in the four compartments of electronic rack assembly, 1A70 are in the retracted position and all covers are secured.	Covers Secured	
d. Remove all obstructions from the rotational paths of the director main antenna assembly.	Director Clear	
2. Power Off.		
a. At power control panel perform the following.		
(1) Check convenience lamp indicators.	Lighted	Schematic, figure 5-233
e. At track meter panel, 1A340-02, check COOLANT FAILURES lamp.	Extinguished (Depress RESET button if lamp is lighted)	Relay diagram, figure 5-77, SH 4(4B)

FIGURE 19. Maintenance Turn-on Procedure.

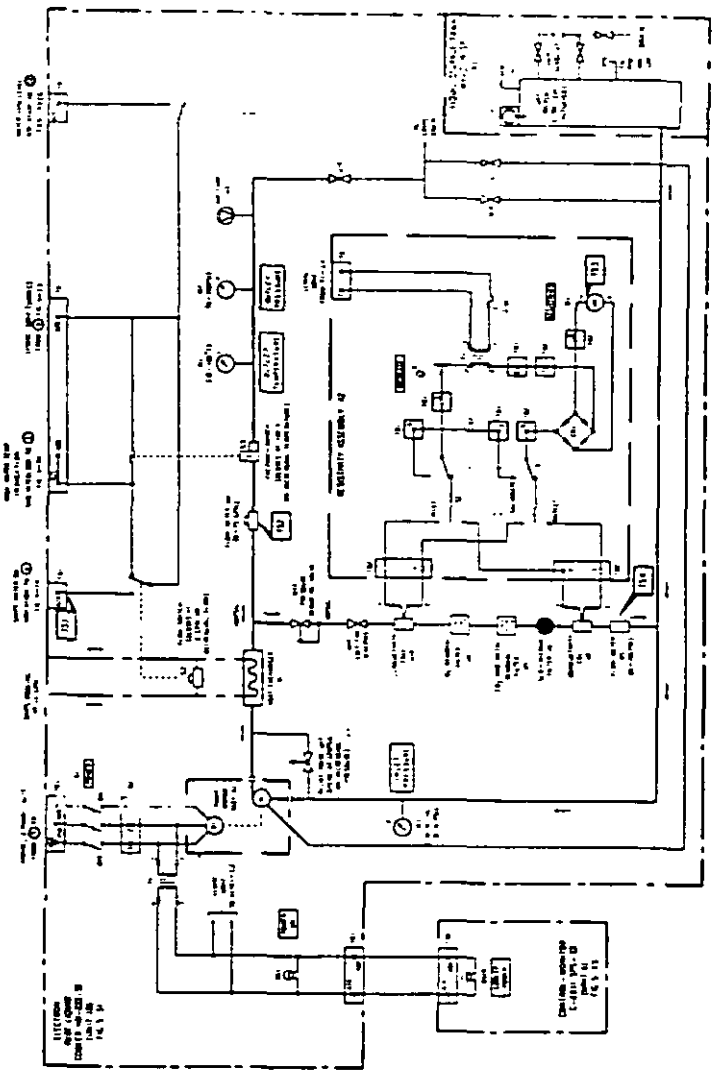
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:27Z
Check the source to verify that this is the current version before use.



Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 20. Functional Signal Flow Diagram.

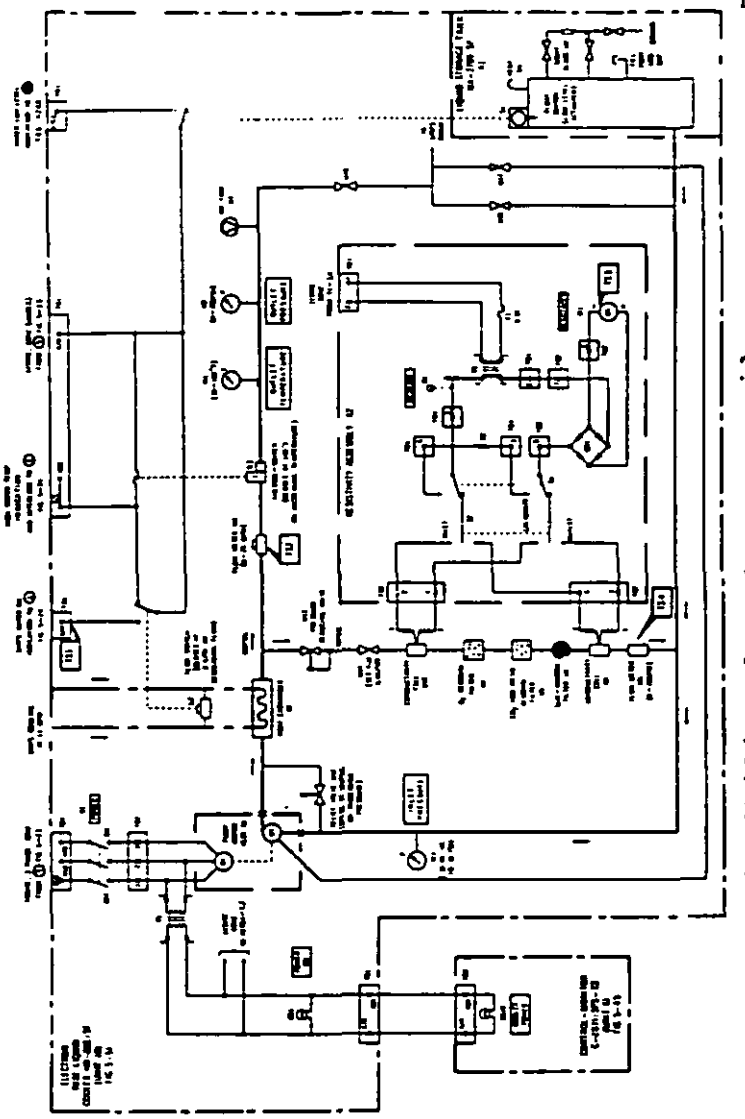
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:27Z
Check the source to verify that this is the current version before use.



Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 21. Piping Diagram.

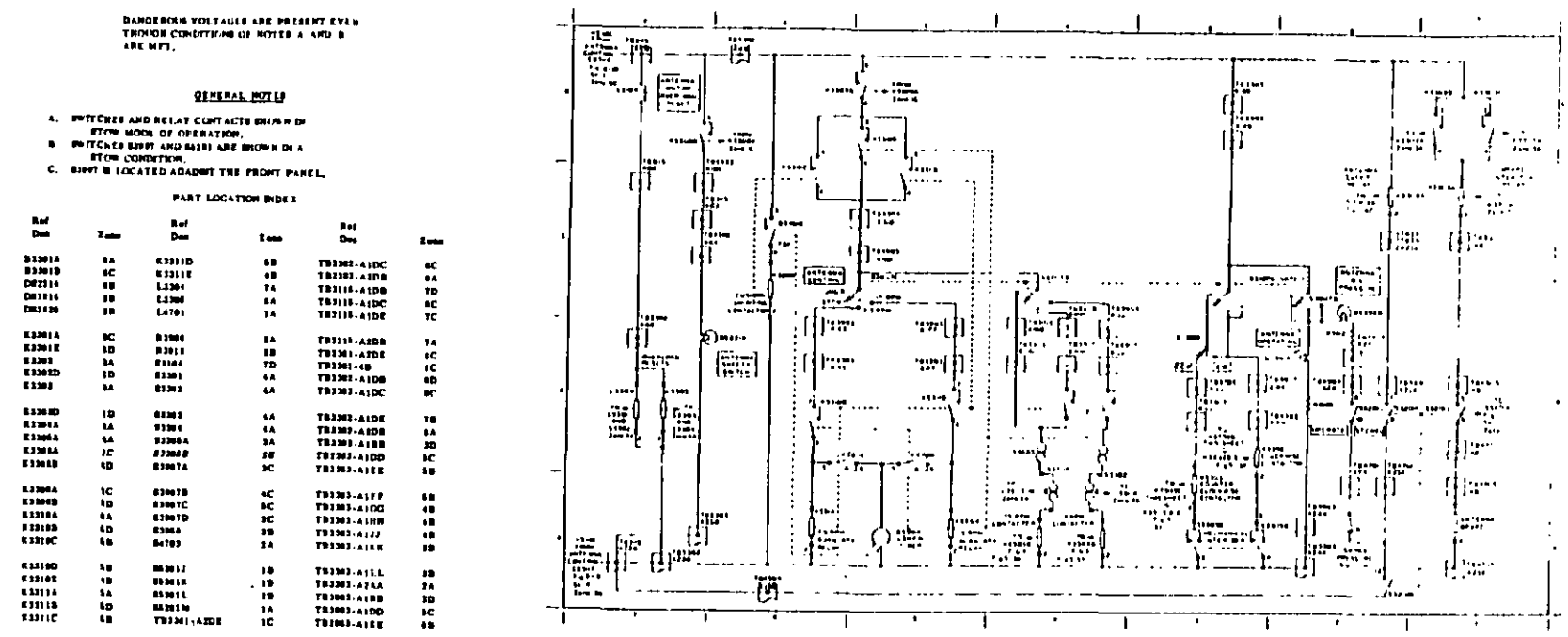
Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:27Z
Check the source to verify that this is the current version before use.



Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 22. Piping Diagram (Type I Manual).

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WARNING

DANGEROUS VOLTAGES ARE PRESENT EVEN THOUGH CONDITIONS OF NOTES A AND B ARE MET.

GENERAL NOTES

A. SWITCHES AND RELAY CONTACTS SHOWN IN STOW MODE OF OPERATION.
B. SWITCHES S3007 AND S3181 ARE SHOWN IN A STOW CONDITION.
C. S3007 IS LOCATED ADJACENT THE FRONT PANEL.

PART LOCATION INDEX

Ref Des	Zone	Ref Des	Zone	Ref Des	Zone
S3301A	6A	K3311D	6B	TB3302-A1DC	6C
S3301B	6C	K3311E	4B	TB3302-A1DB	6A
DS2314	6B	L3304	7A	TB3116-A1DB	7D
DS1314	1B	L3308	6A	TB3116-A1DC	8C
DS3426	1B	L4701	1A	TB3116-A1DE	7C
K3301A	8C	R3306	5A	TB3116-A2DB	7A
K3301E	5D	R3311	8B	TB3301-A1DE	6C
S3303	3A	R3314	7D	TB3301-4B	1C
K3302D	7D	R3301	6A	TB3302-A1DB	6B
K3302	5A	R3302	6A	TB3302-A1DC	8C
K3306D	1D	R3303	6A	TB3302-A1DE	7B
K3304A	5A	R3301	6A	TB3302-A1DB	6A
K3306A	6A	R3306A	3A	TB3303-A1BB	3D
K3304A	7C	R3302B	2E	TB3303-A1DD	5C
K3306B	6D	R3307A	3C	TB3303-A1EE	5B
K3306A	1C	R3307B	4C	TB3303-A1FF	5B
K3306B	5D	R3307C	8C	TB3303-A1DG	4B
K3310A	6A	R3307D	3C	TB3303-A1HH	4B
K3310B	6D	R3306	3B	TB3303-A1JJ	4B
K3310C	6B	R4703	2A	TB3303-A1KK	3B
K3310D	5B	R3301J	1B	TB3303-A1LL	3B
K3310E	4B	R3301K	1B	TB3303-A2AA	2A
K3311A	5A	R3301L	1B	TB3303-A1BB	3D
K3311B	6D	R3301M	1A	TB3303-A1DD	5C
K3311C	6B	TB3301-A2DE	1C	TB3303-A1EE	5B

Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

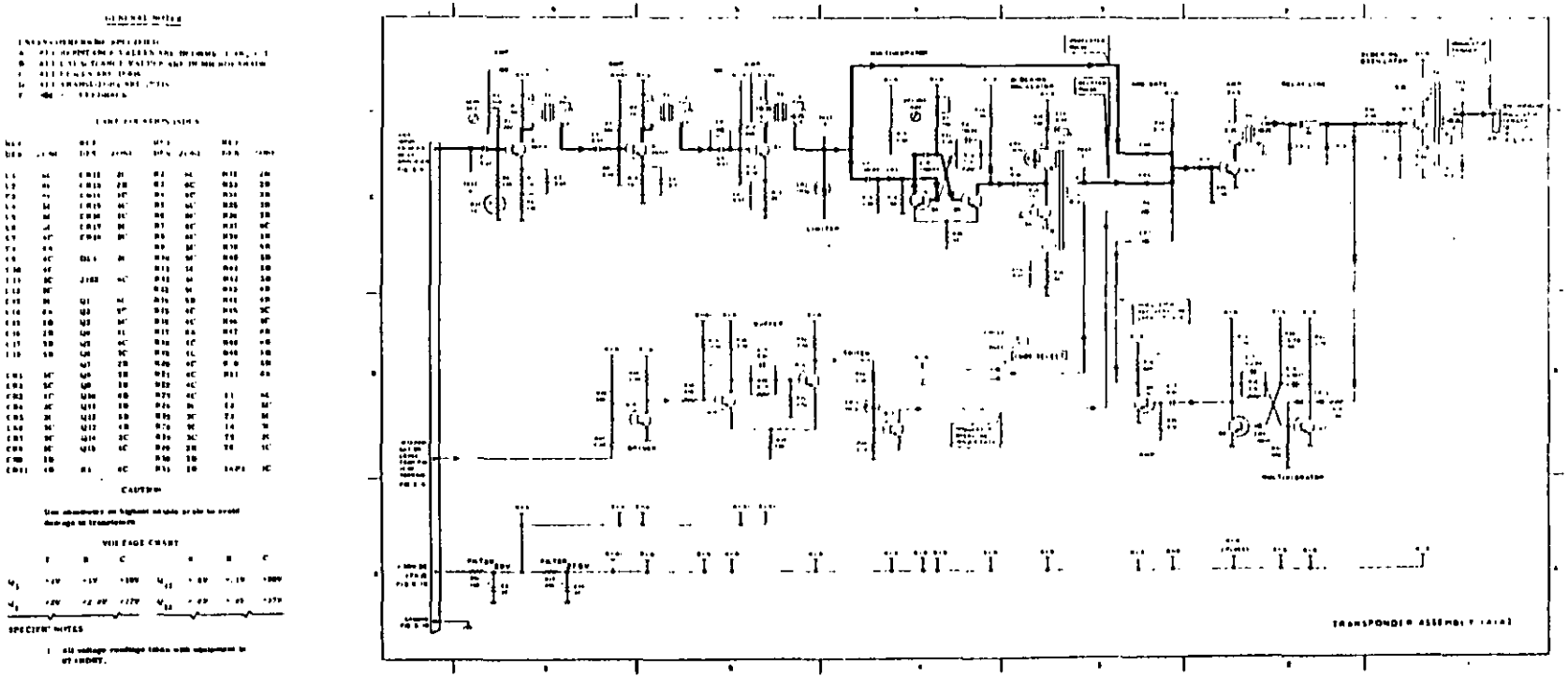
FIGURE 23. Control Diagram.

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FIGURE 24. Power Distribution Diagram.

Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

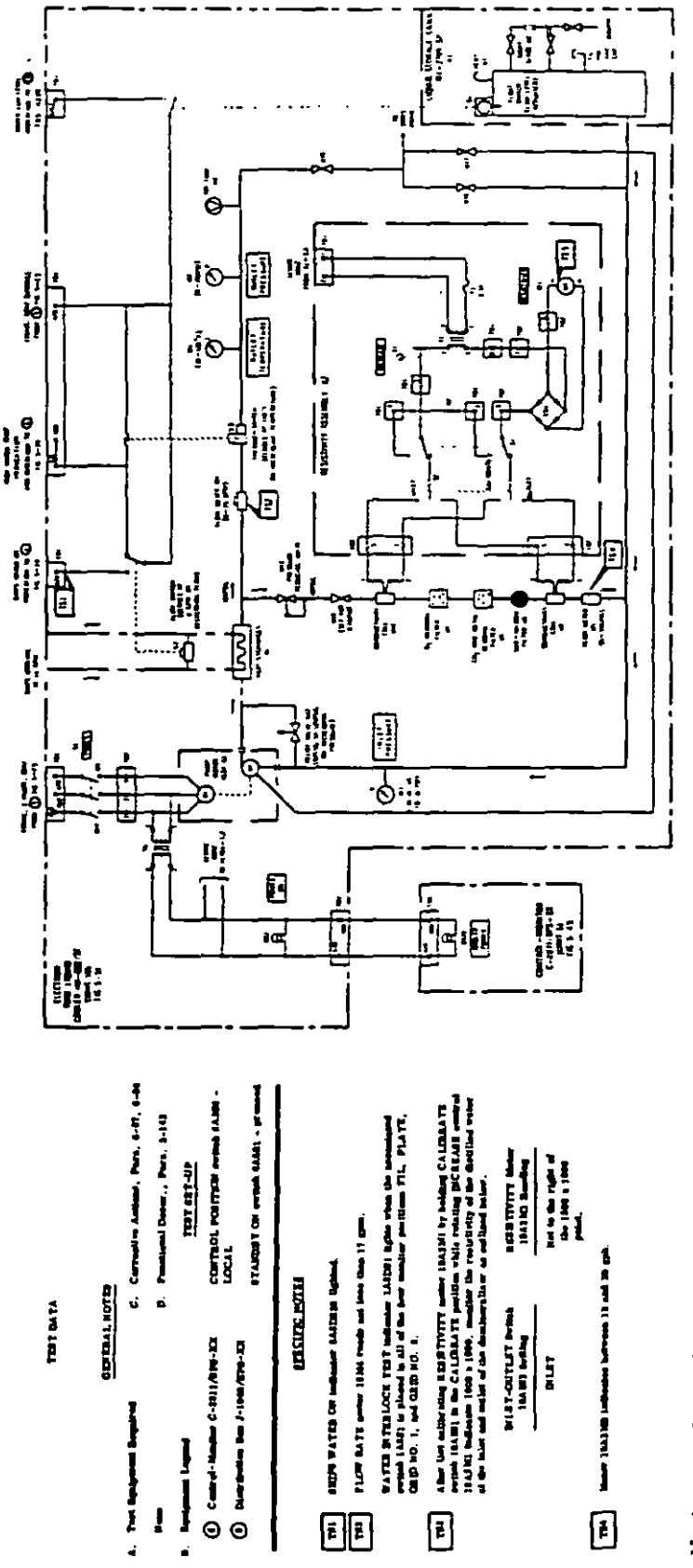
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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 25. Maintenance Schematic Diagram.

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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 26. Piping Diagram (Type II, Manual).

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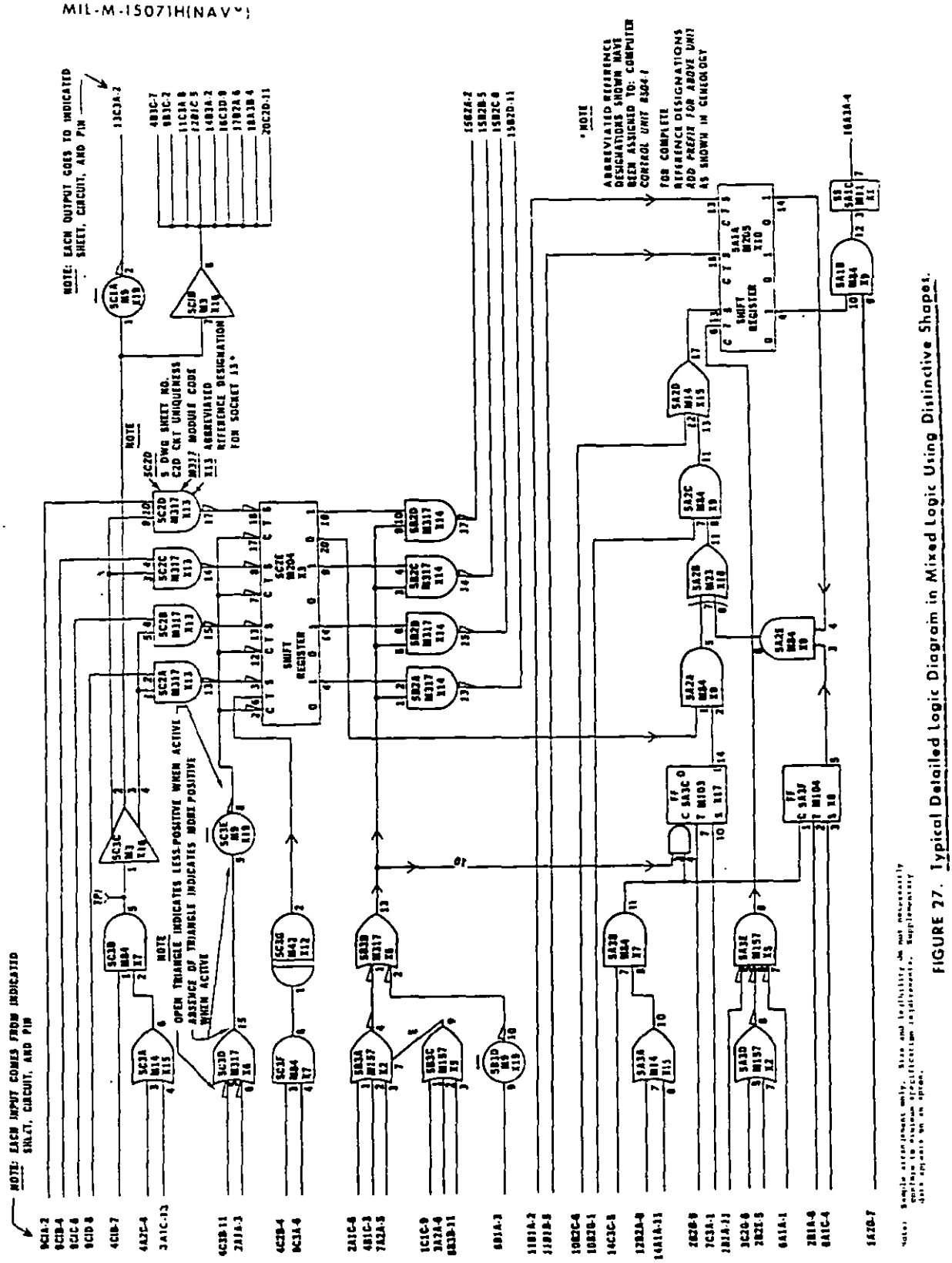


FIGURE 27. Typical Detailed Logic Diagram in Mixed Logic Using Distinctive Shapes.

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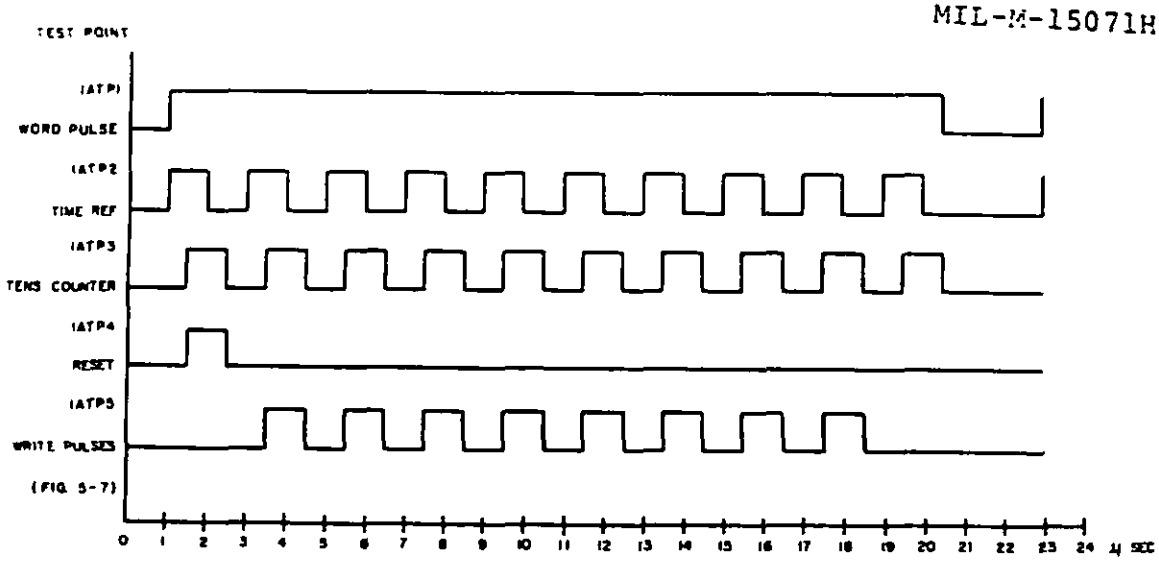


FIGURE 28. Timing Circuits Diagram.

XYZ ROUTINE - PROGRAM LISTING

LOC	MEMORY CONTENTS	LABEL	ORD	ADR	MOD	CONSTANT	SCL	NOTES
01443	000330700		FRS	30700				
01447	015101457		TMI	01457				
01453	034706277		TRA	06277				
01457	000077632		GET	37632				
01463	034706403		ORG	06403				
			ORG	06103				
06103	001037632	A1	STO	TEMPR	1			SAVE INSERTED NUMBER
06107	035106123		TM1	A3				/-/ NEGATIVE INSERT
06113	014077763	A2	GET	ONE				KEY FOR FIX RESET
06117	035035357		STO	KTYPF				TYPE OF RESET KEY
06123	000077760	A3	GET	ZERO				
06127	035037633		STO	TEMPR	2			CLEAR RESET /NRS/ KEY
06122	000077632		GET	TEMPR	1			
06137	004706147		TRA	SUB 1				GO TO RESET NUMBER SUBROUTINE
06143	000706327		TRA	A14				
06137	055437634	SUB 1	STV*	TEMPR	3			ENTRANCE - RESET NUMBER SUBRTN
06153	034313700	A4	PRS	1				
06157	035106243		IMI	RE13				CHECK LAST BCD BIT FOR SIGN
06163	034313700	A5	PRS	1				
05167	035106213		TMI	**5				CHECK 2ND BCD BIT FOR SIGN
06173	034333600	A6	FRS	2				CHECK 3RD AND 4TH BITS
06177	002206177		TMI	OUT				
06203	034077633	A7	GET	TEMPR	2			TEMPORARY RESET KEY
06207	005737634		TRA	TEMPR	3			EXIT - RESET NUMBER SUBROUTINE
06213	005077766	A8	EXT	MAXNO				
06217	034333600	FRS	FRS	2				CHECK 3RD AND 4TH BITS
06223	002206277		TMI	OUT				
06227	044041030	AO	GET			020000000		KEY TO RESET 2
06233	005477633	A10	ADD	TEMPR	2			
06237	005737344		TRA	TEMPR	3			EXIT - RESET NUMBER ROUTINE
06243	034313700	RE13	PRS					
06247	035106303		TMI	RE3				CHECK 2ND BIT FOR SIGN
06253	034077766	A11	EXT	MAXNO				
06257	034333600		FRS	2				CHECK 3RD AND 4TH BITS
06263	001106277		TMI	OUT				
06267	034077762	A12	GET	HALF				KEY TO RESET 1

FIGURE 29. Coding Instruction Sheet.

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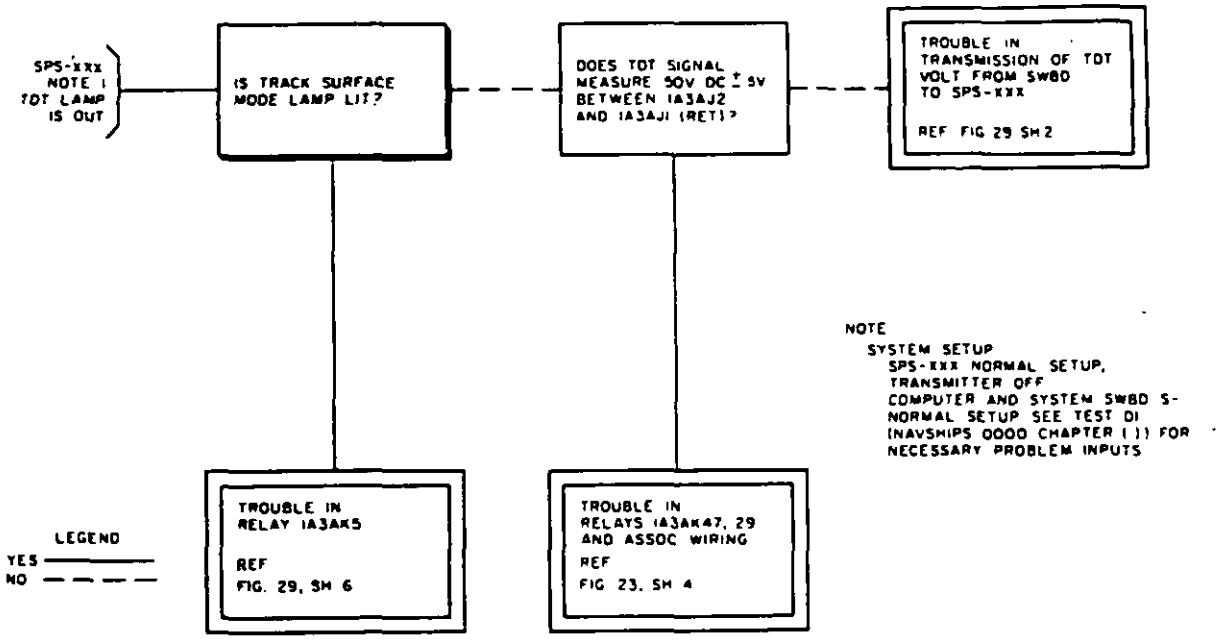
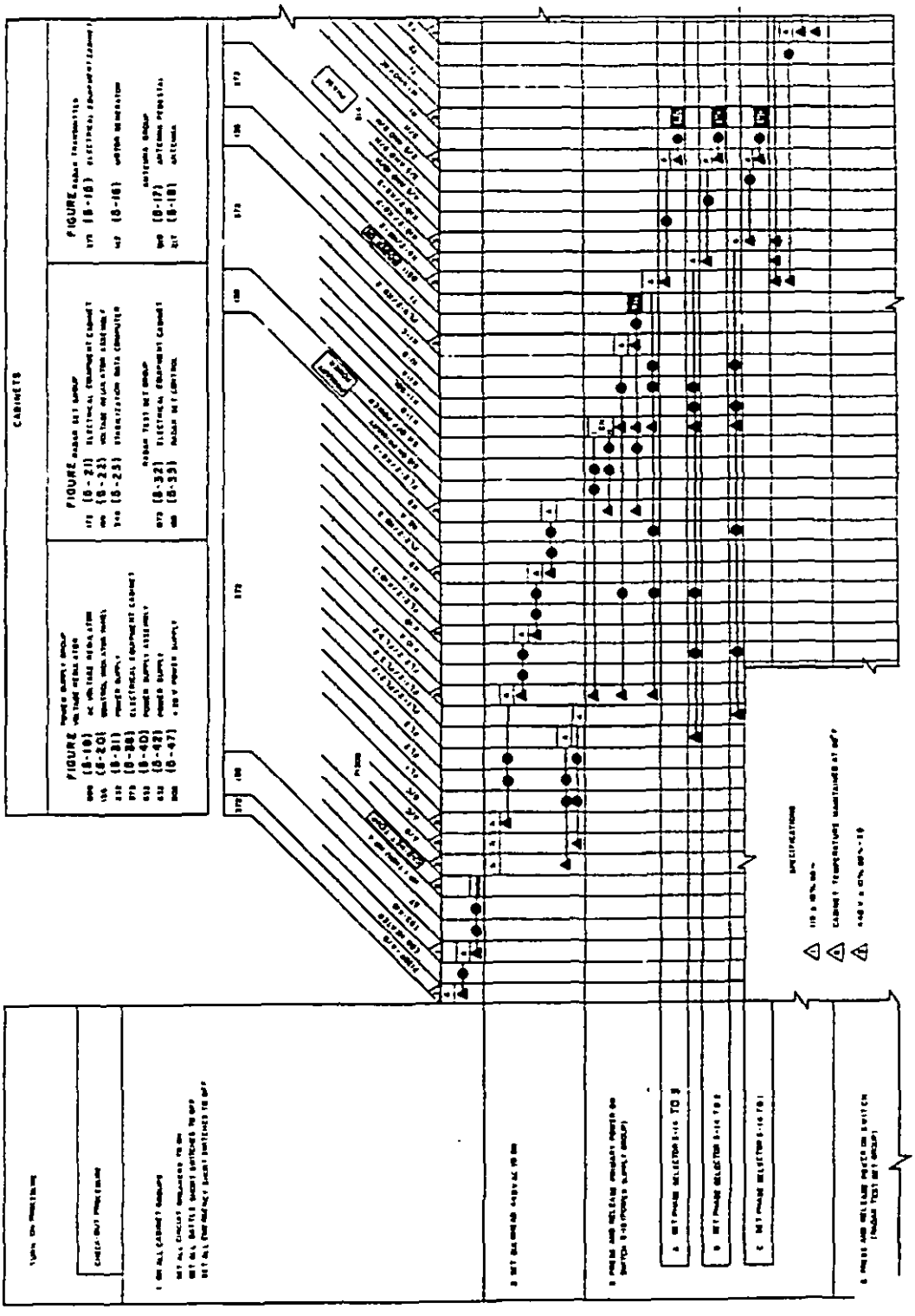


FIGURE 30. Fault Logic Diagram.

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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements.

FIGURE 31. Troubleshooting-Maintenance Dependency Matrix Chart.

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Figure and Index Number	Description	Qty.	Fed. Mfg. Code	Mfg. Part No.
6-1	Bearing - Inner	1	80648	B2212
-2	Ring, Retainer MS 9013	1	80756	RS-216
-3	Plug, Drain MS 9015	1	45681	8P5N-S
-4	Hydraulic Motor - ASD	1	16954	M15-17998
-5	"O" Ring MS-28775-234	4		
-6	Hex Hd Capscrew (Special) 1-8Nc x 3-3/4, Grade 5, MS 90725-237	4		
-7	Spring, Heavy Duty, 1-1/2 O.D. x 3/4 I.D. x 12" LG	1	15840	9-2448-26

FIGURE 32. Parts List Table (Type I Manual).

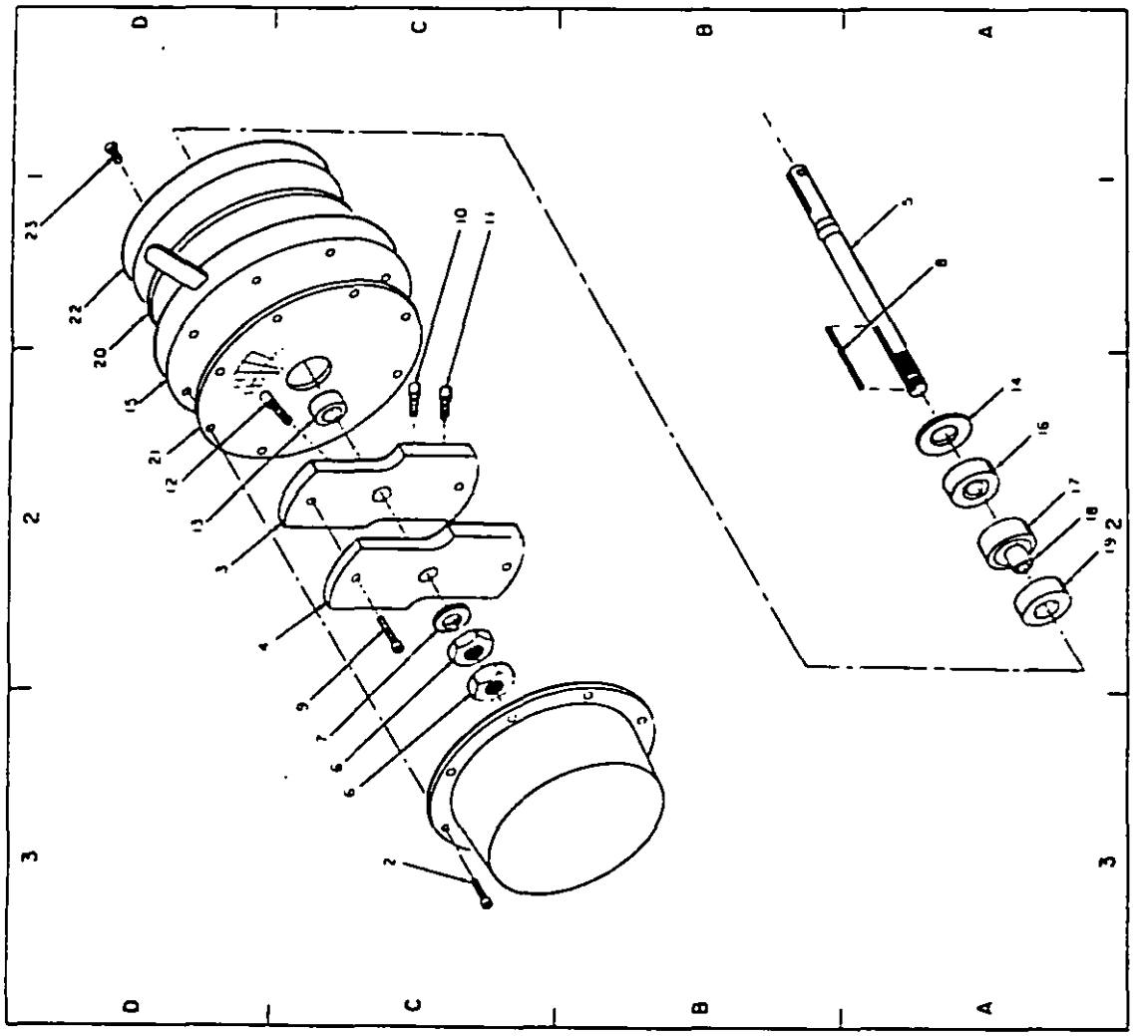
TABLE ZZ. RADAR SET AN/SPS-XX, PARTS LIST

Amplifier-Oscillator Group OA-2815/SPS-XX (Unit 2)

Reference Designation	Notes	Name and Description	Figure Number (Item)
2		AMPLIFIER-OSCILLATOR GROUP OA-2815/SPS-XX: Provides drive power for AN/SPS-XX final stage, made in three cabinets which can be separated; mfr 89661, part no. 478D-800G02.	1-1
2AT1	1	ATTENUATOR, FIXED: 50 ohms, 1 watt, bnc type connector; mfr 91578, type RT3-M-51. (Attaching Parts) B (1), T (4)	6-119(17)
2B1		FAN, CENTRIFUGAL: Cw rotation, 3 o'clock blast, aluminum case, 6-13/32 in. long, 6-7/16 in. w; mfr 82877, type DRFPKS406, 89661, dwg 331C158H03. (Attaching Parts) L(8), T(8), X(8)	6-119(42)
2C1 thru 2C8		CAPACITOR, FIXED, PAPER DIELECTRIC: 0.01 μF ± 10%, 600 Vdc working; mfr 56289, part no. 102P15, 89661, dwg 54B7098H07.	6-119

FIGURE 33. Parts List Table (Type II Manual).

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Index No	Zone	Part Number	Qty	*Reference Designation
1	3B	Commutator Cover	...	MP3
2	3C	Socket-Head Capscrew	4	MP14
3	2D	Wiper	...	MP17
4	..	Not Used	..	..
5	1A	Commutator Shaft	...	MP13
6	3C	Hex Nut	2	MP8
7	3C	Washer	...	MP18
8	1A	Key	...	MP14
9	3C	Socket-Head Capscrew	4	MP20
10**	1C	Brush	2	MP9
11**	1C	Brush	4	MP11
12**	2D	Brush	8	MP10
13	2D	Wiper Spacer	...	MP16
14	1A	Bearing Cap	...	MP12
15	2D	Commutator Hub	...	MP7
16	2A	Ball Bearing	...	MP4
17, 18	2A	Bearing Spacer Assembly	...	MP15
19	2A	Ball Bearing	...	MP4
20	2D	Insulator	...	MP21
21	2D	Printed Circuit	...	MP22
22	2D	Rear Cover	...	MP23
23	1D	Flat-Head Screw	2	MP24

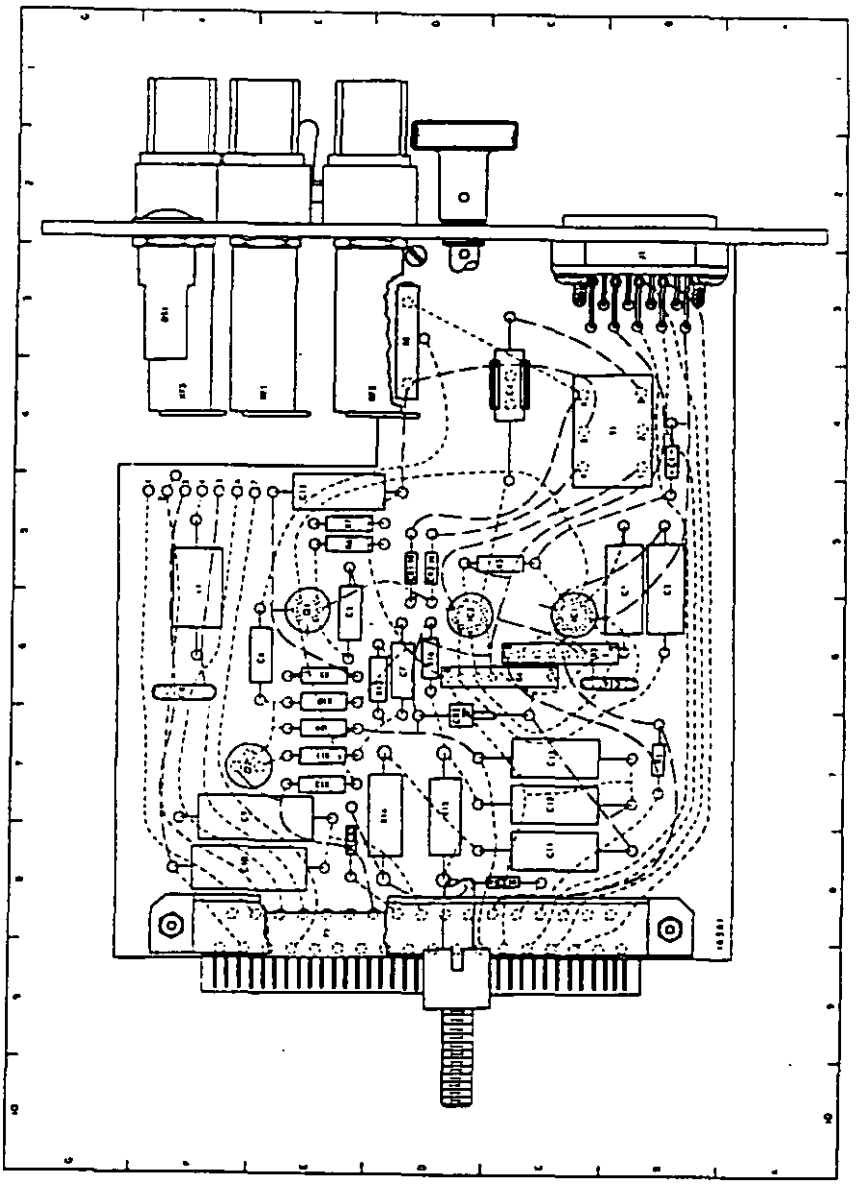
* Reference Designation Prefix is 12A17A1A1 unless otherwise stated

**Items to be segregated when removed for replacement at reassembly
Refer to Parts List for procurement information

Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data normally appears on an apron.

FIGURE 34. Exploded view.

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PARTS LOCATION INDEX

ASSEMBLY 1A3A1

REFERENCE DESIGNATION	LOCATION	REFERENCE DESIGNATION	LOCATION
C1	4-B	P1	8-E
C2	3-B		
C3	3-D	Q1	6-E
C4	4-C	Q2	7-F
C5	8-E		
C6	8-E	R1	7-B
C7	8-D	R2	6-C
C8	8-F	R3	5-C
C9	7-F	R4	6-C
C10	8-F	R5	6-E
C11	8-C	R6	5-E
C12	7-C	R7	5-E
C13	7-C	R8	3-D
C14	7-D	R9	2-E
C15	8-E	R10	6-E
		R11	7-E
CR1	8-D	R12	6-D
CR2	8-D	R13	7-E
CR3	8-E	R14	7-D
CR4	8-C	R15	7-D
CR5	7-D	R16	6-D
DS1	3-F	T1	4-B
IC1	3-C	XF1	4-E
IC2	8-D	XF2	4-D
		XF3	4-F
J1	3-B		
L1	6-F		

——— COMPONENT SIDE

- - - - - BACK SIDE

Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 35. Engineering Drawing.

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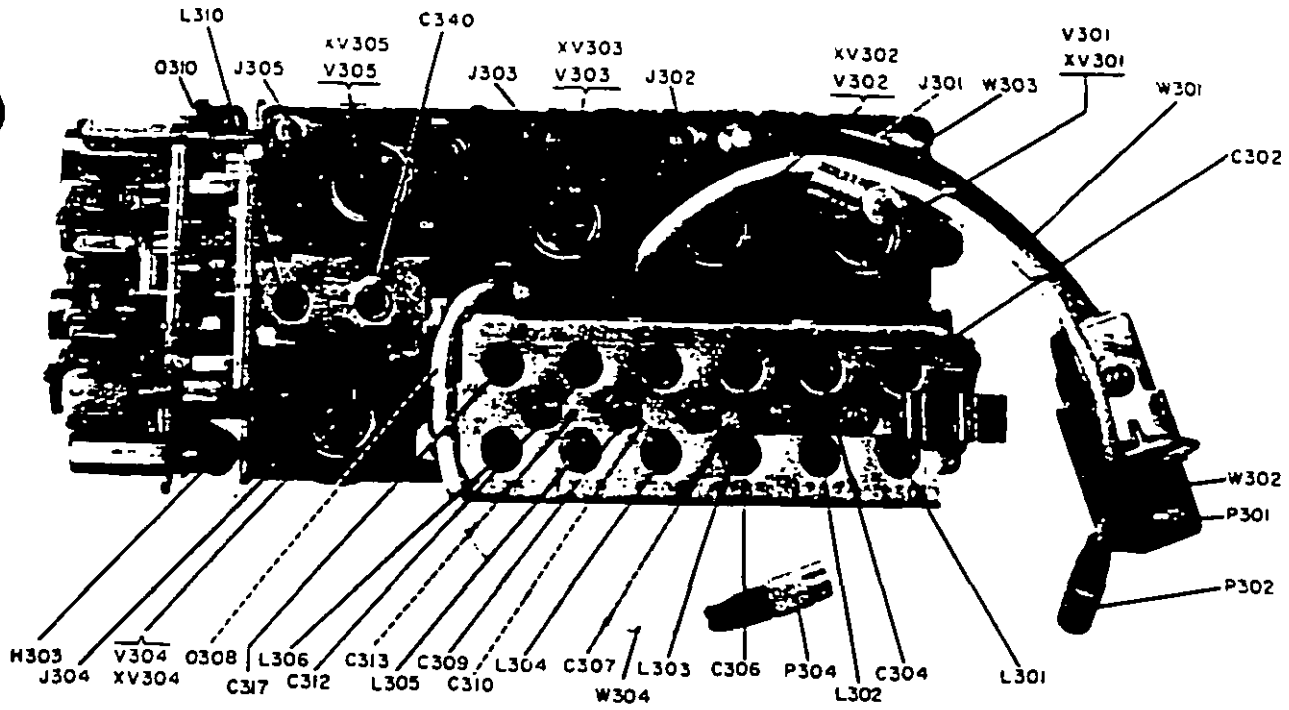


FIGURE 36. Photograph.

RADIO RECEIVER R-XXX/URR

NAVSEA

INSTALLATION STANDARDS SUMMARY

Input Voltage _____ Vac
Input Frequency _____ Hz
(When reference standard tests are made)

Date _____
Serial Number _____
of Model _____
Installed in (ship or station)
_____
Length of transmission line
_____

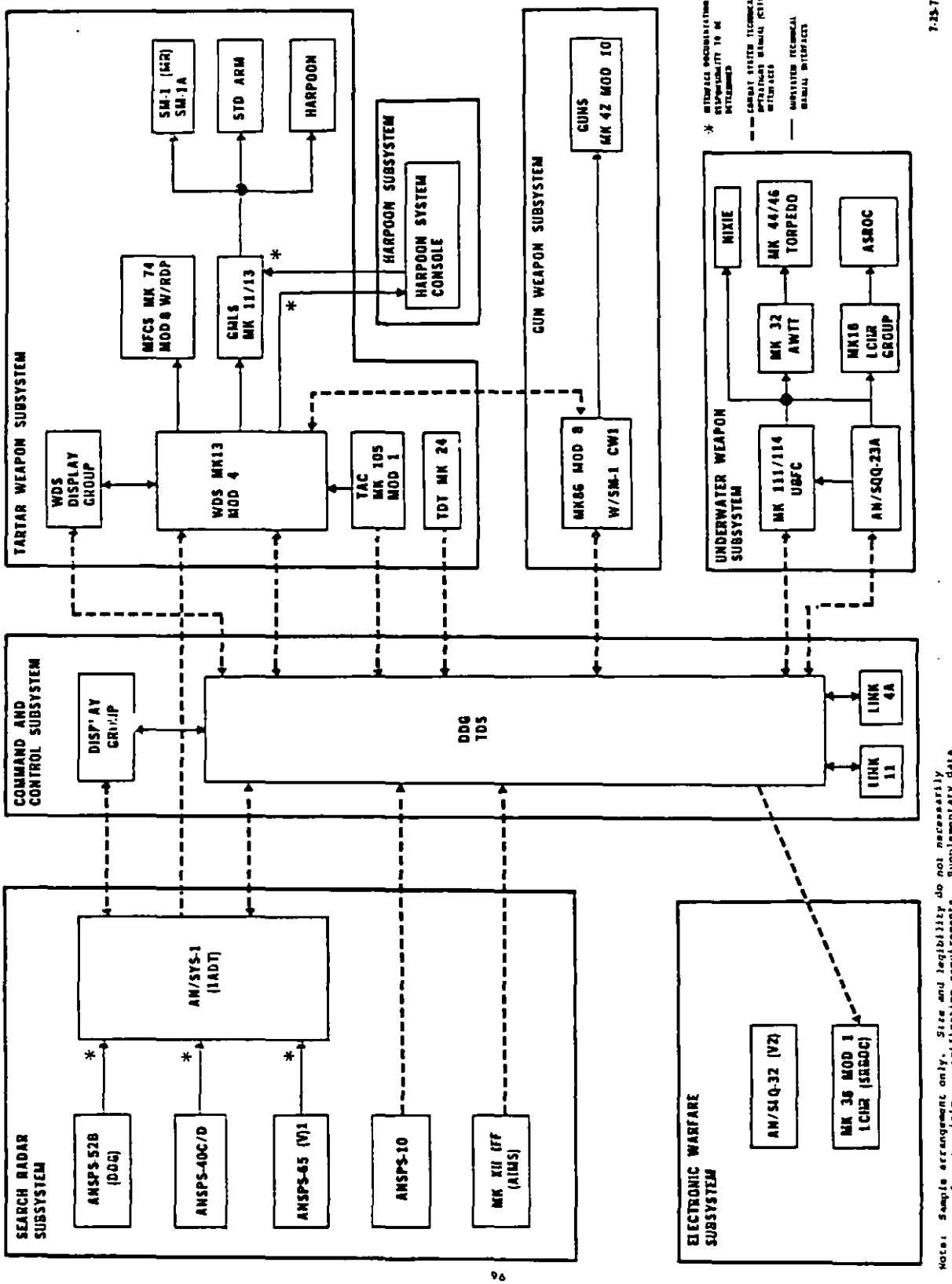
Record on this summary sheet the test indications which have been obtained during the installation verification test.

Paragraph No.	Ref. Std.	Paragraph No.	Ref. Std.
8-10	a. _____ Check	8-46	a. _____ μV
			b. _____ μV
8-21	a. _____ Vdc		c. _____ Check
	b. _____ Vdc		d. _____ μV
	c. _____ Vdc		e. _____ μV
	d. _____ Vdc		f. _____ μV
		8-51	a. _____ Sec
8-33	a. _____ Check		b. _____ Check
	b. _____ Check		c. _____ Check
	c. _____ Check		d. _____ Hz
	d. _____ Check		e. _____ Hz
			f. _____ Check
			g. _____ Check

NOTE: This sheet may be reproduced locally to size of manual page.

FIGURE 37. Installation Standards Summary Sheet.

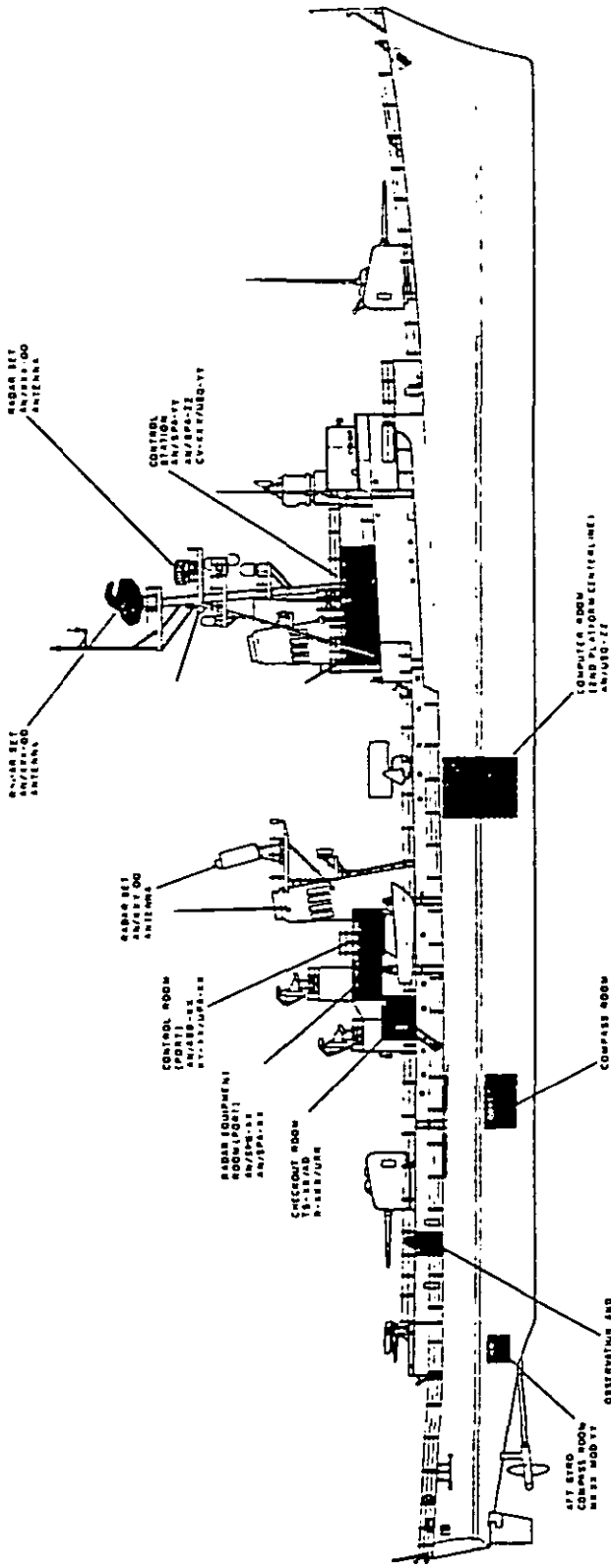
Source: https://assist.dla.mil -- Downloaded: 202640-18T05:27Z
Check the source to verify that this is the current version before use.



Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data appears on an apron.

FIGURE 18. System Block Diagram Interface Documentation.

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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements. Supplementary data normally appears on an apron.

FIGURE 39. System Compartments.

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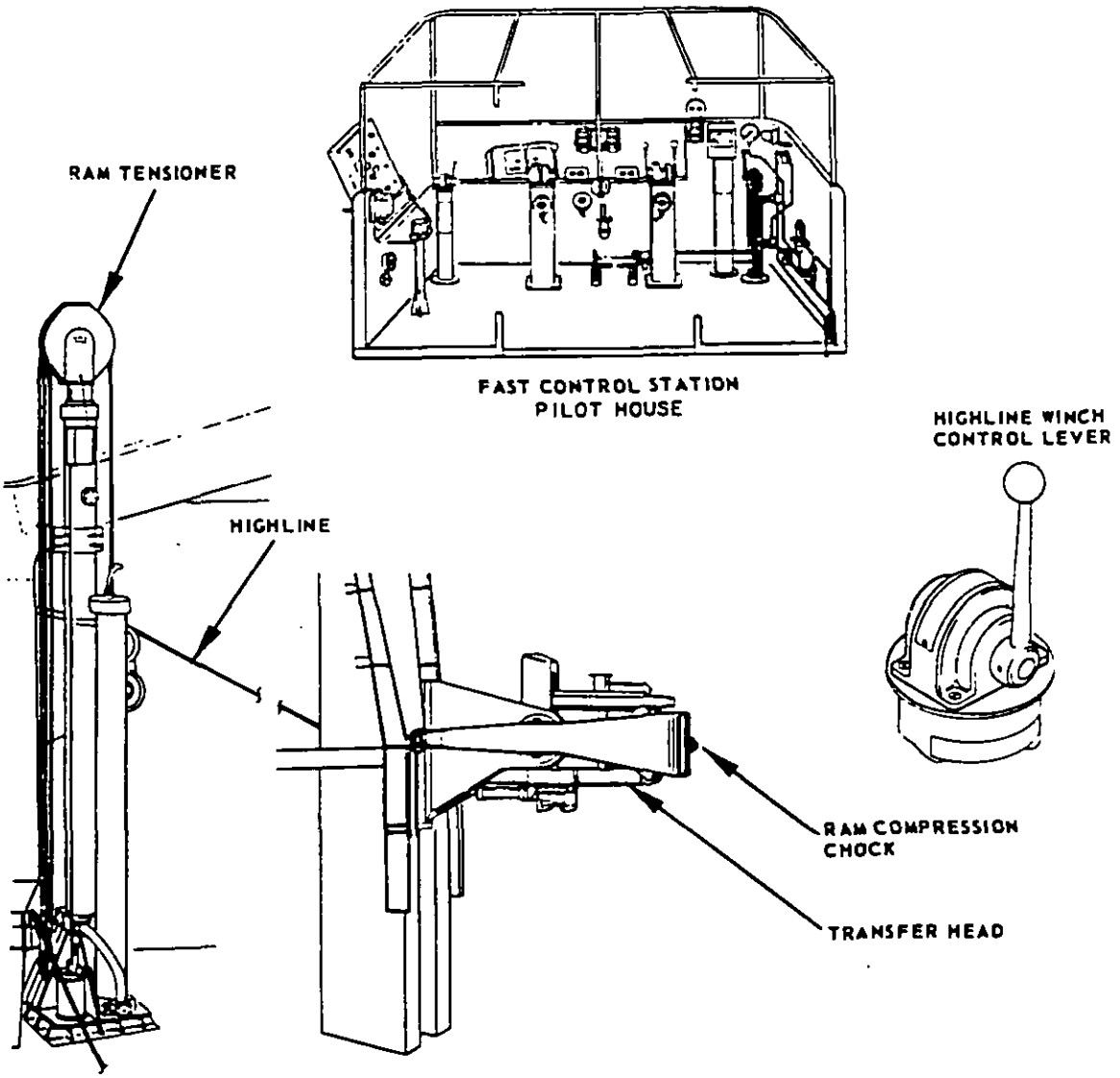


FIGURE 40. Compartment Area.

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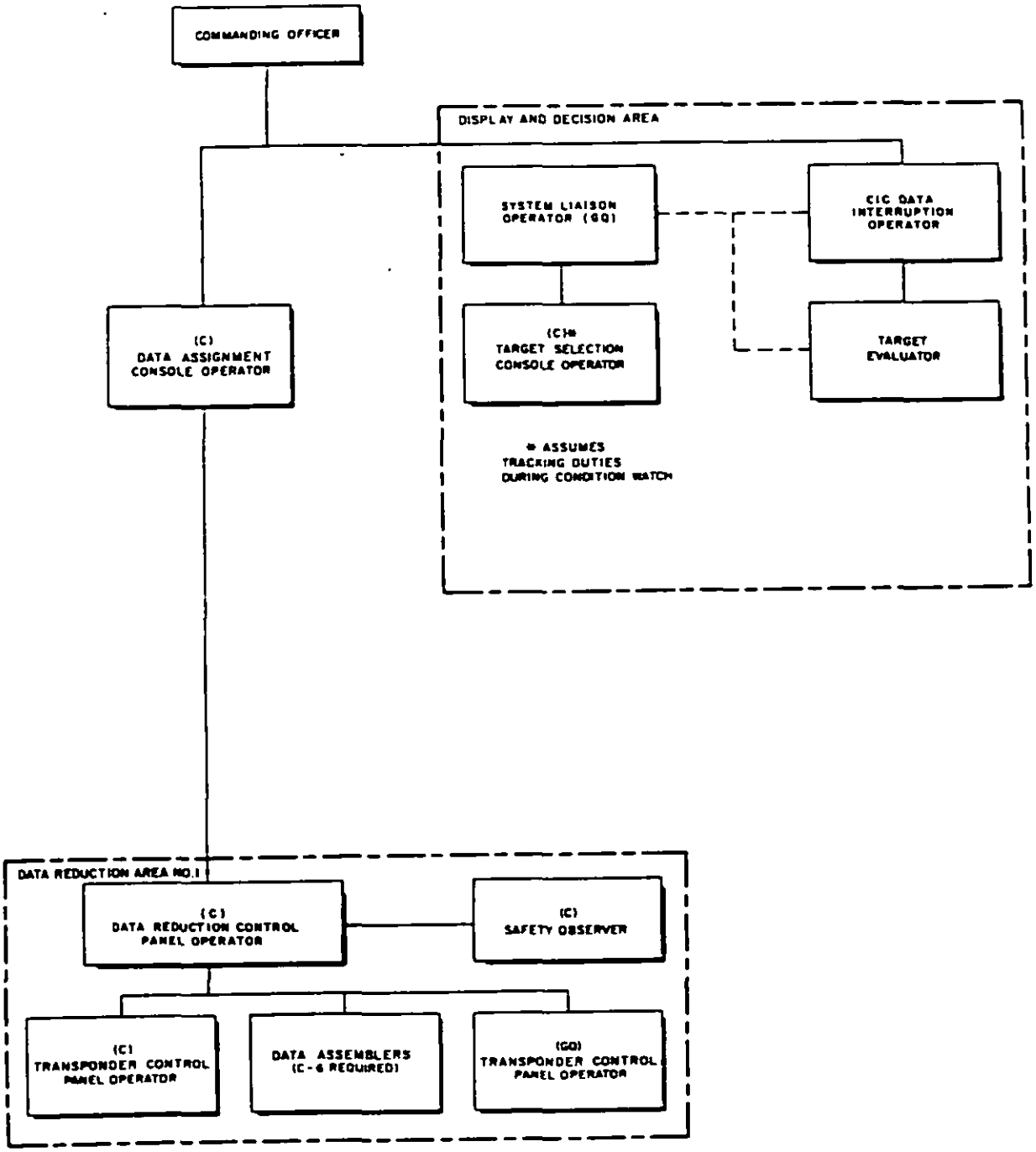


FIGURE 41. Conditions of Readiness, General Quarters Conditions.

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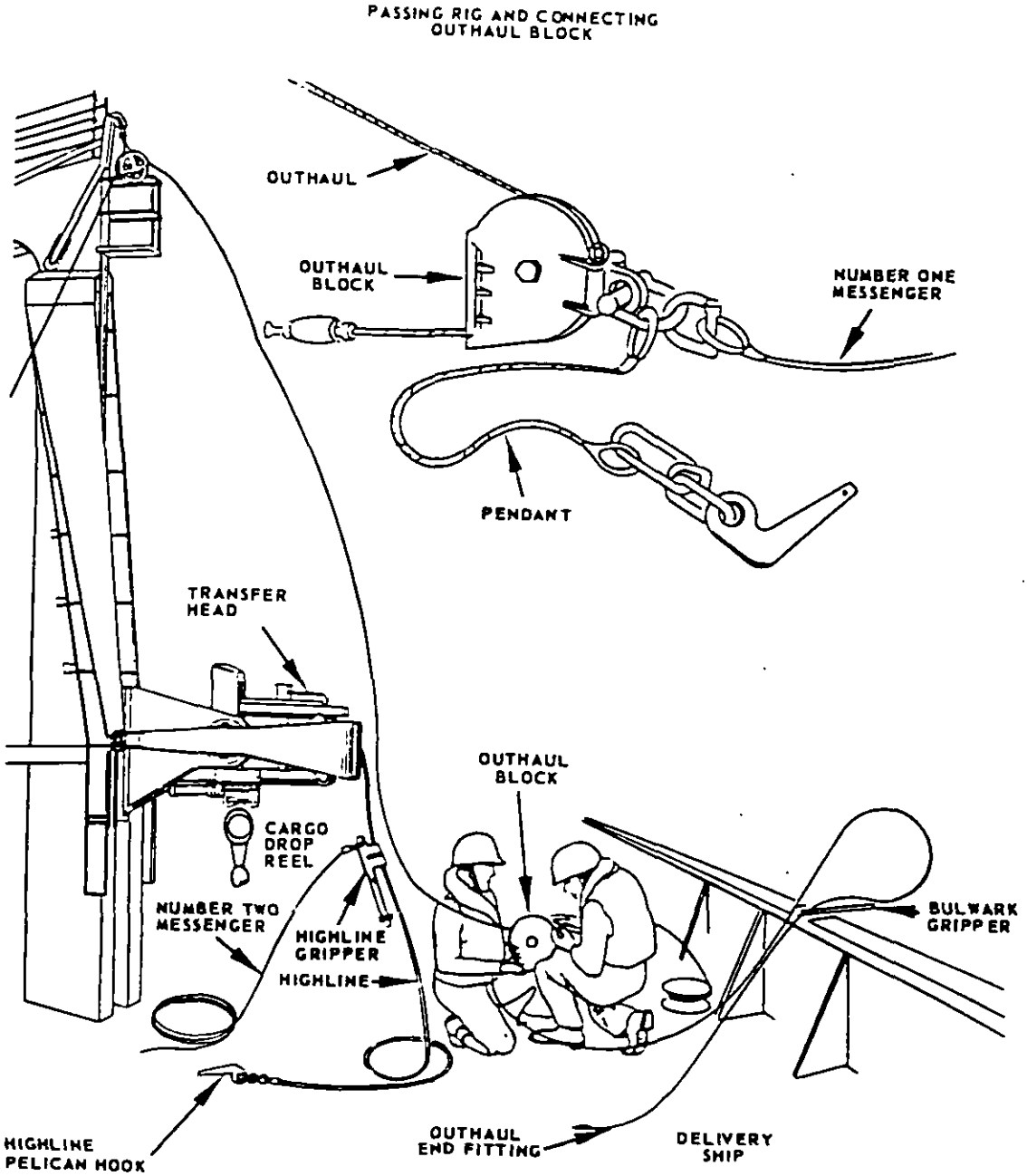
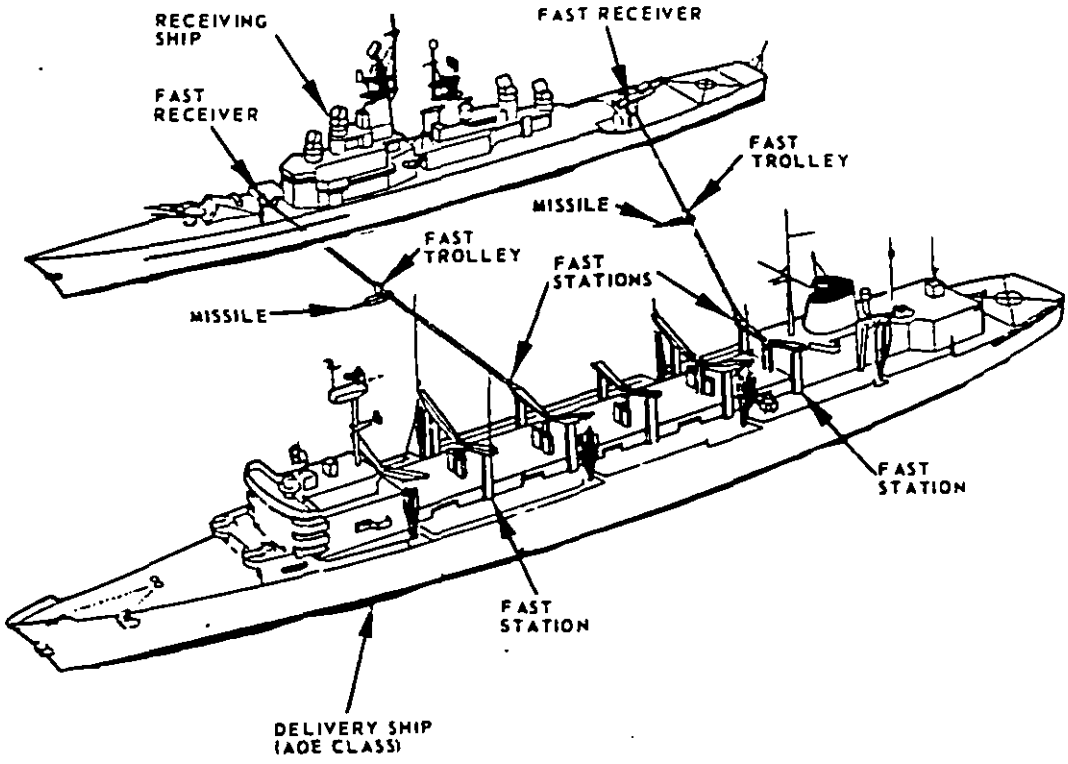


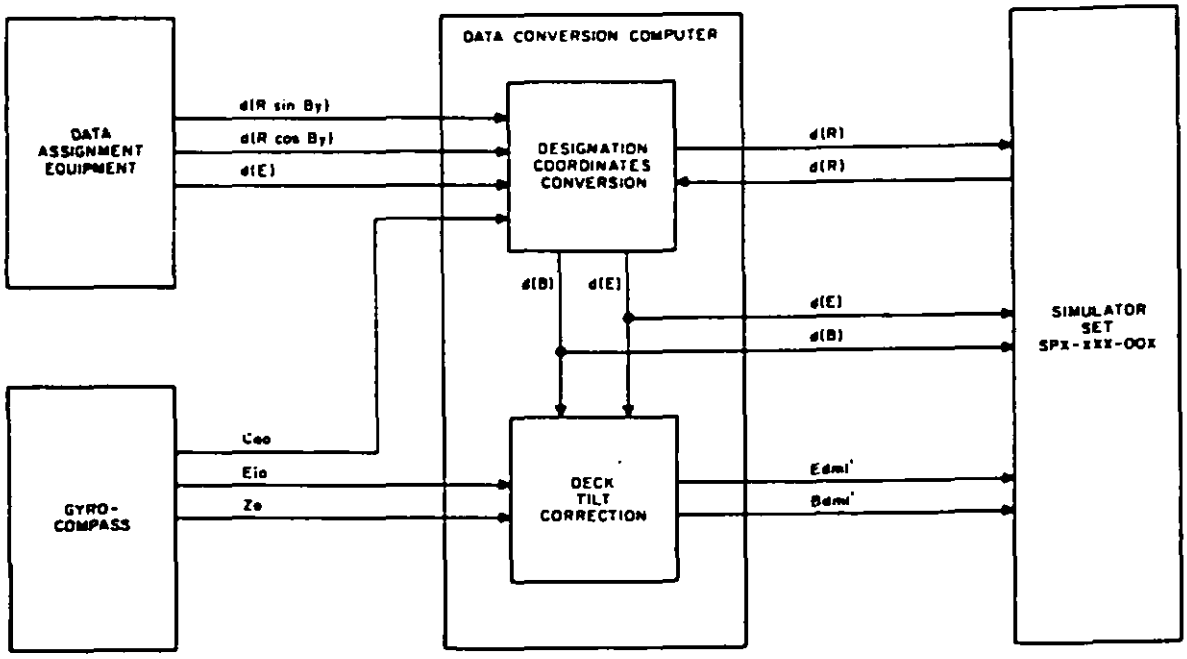
FIGURE 42. Operational Sequence.

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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements.

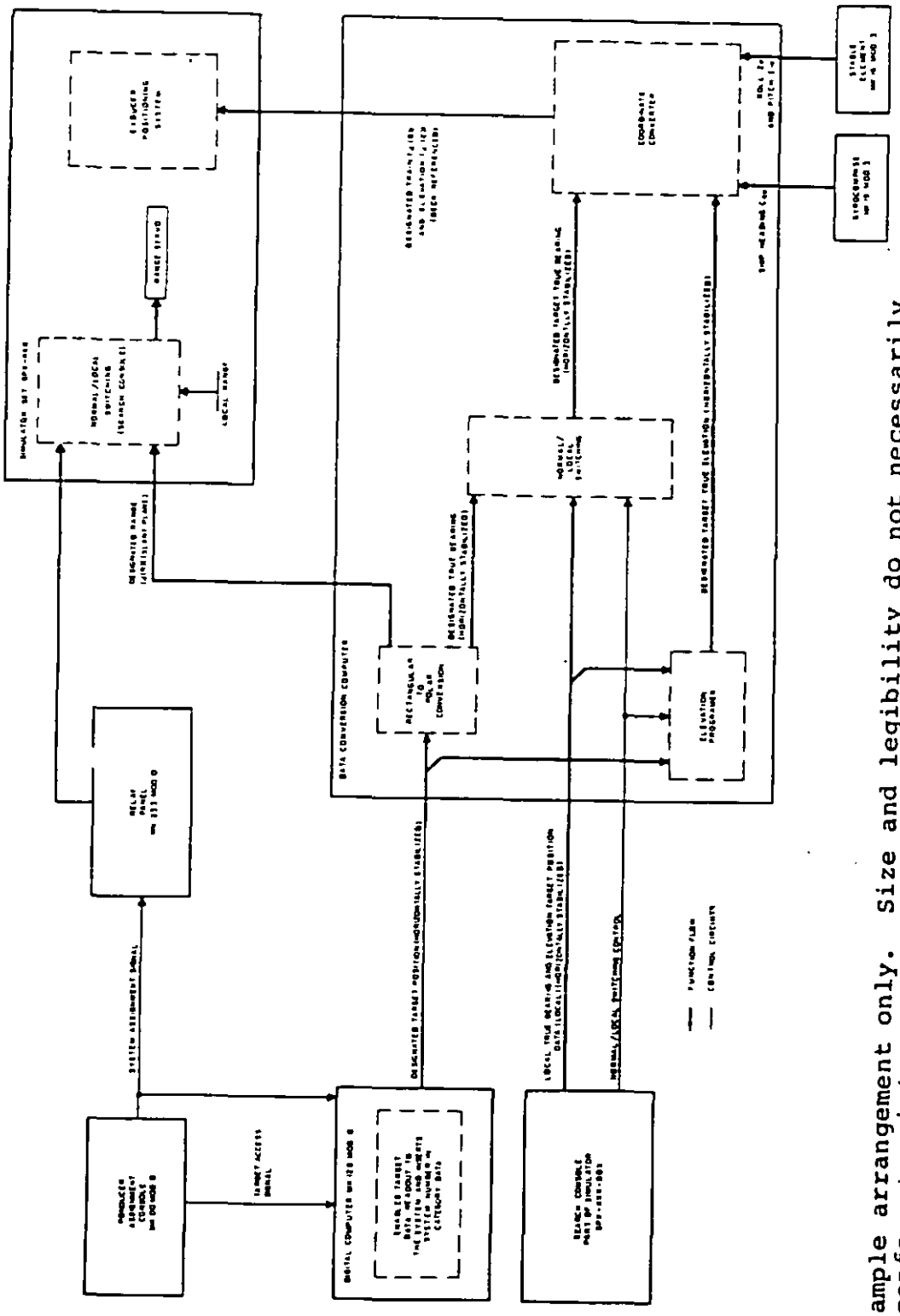
FIGURE 43. System Illustration Showing Interfaces.



Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements.

FIGURE 44. System Functional Block Diagram.

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Note: Sample arrangement only. Size and legibility do not necessarily conform to minimum specification requirements.

FIGURE 45. First Level Functional Description Electronic System.

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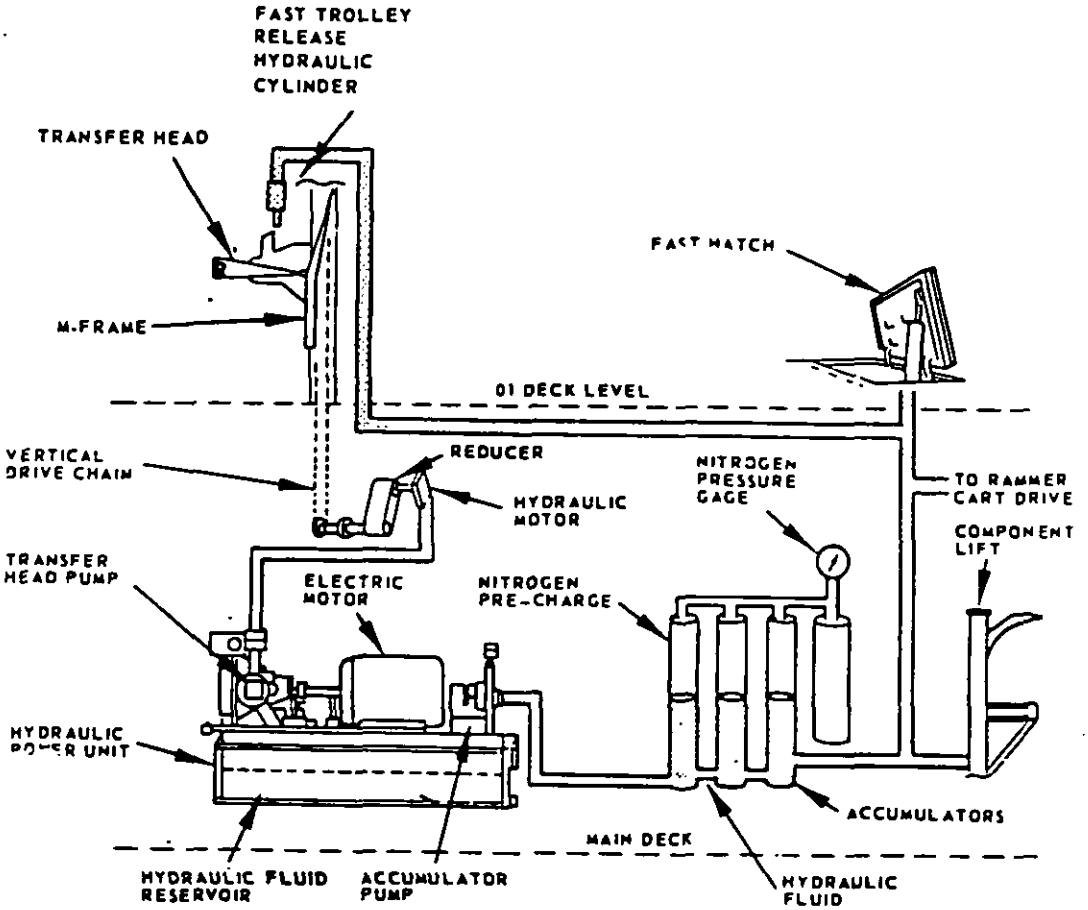


FIGURE 46. Electrical/Mechanical First Level Functional Description Diagram.

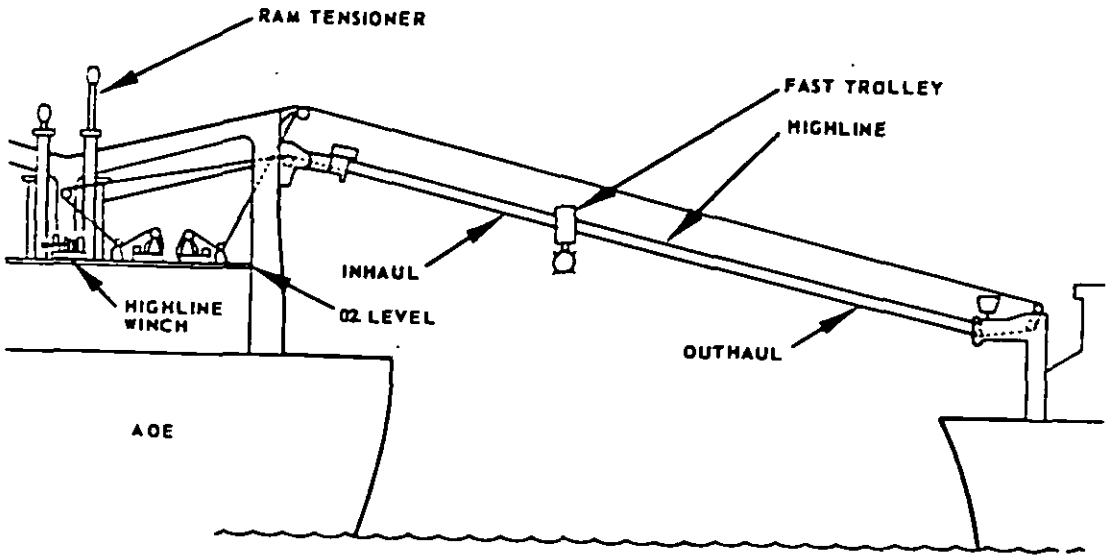


FIGURE 47. Second Level Functional Description Diagram f Electrical/Mechanical System.

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TABLE ZZ. Operation-based symptom fault directory.

Operating Procedure Step	Functional Description	FIP	Alinement Procedure	Fault Isolation Diagrams	Equipment Document
Normal Operating Mode - Paragraph 4- 5					
List the step numbers associated with the fault indications	Noun or Symbolic name of function being monitored	Fault isolation procedure contained in chapter 7	Applicable alinement procedure and paragraph number required to restore normal operation	List associated fault isolation diagram	Reference to applicable equipment technical manual number and other TM reference if available
(Example) 4-3. R dial	dCR	Figure 7-1 Para 7-68	Para - 8-4	Figure 7-38 Sheet 2	NAVSEA 0000 Chapter 4
Next Operating Mode - Paragraph 4- 6					
(CONTINUE INFORMATION AS ABOVE)					

FIGURE 48. Operation-based symptom fault directory.

TABLE ZZ. List of effective pages (showing changes).

PAGE NUMBERS	CHANGE IN EFFECT	PAGE NUMBERS	CHANGE IN EFFECT
Title Page	2	3-6B	1
A	2	3-7 & 3-8	Orig
B & C	Orig	4-1 thru 4-4	Orig
i	2	4-4A	2
ii thru vii	Orig	5-1 thru 5-18	Orig
1-0 thru 1-5	Orig	5-18A	1
2-0 thru 2-17	Orig	6-0 thru 6-20	Orig
3-0 thru 3-6	Orig	7-1 thru 7-4	1
3-6A	2	i-1 thru i-3	2

FIGURE 49. List of effective pages.

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UNCLASSIFIED

Interim Change T-1 NAVELEX 0967-LP-173-6011

INSTRUCTION SHEET

Interim Change T-1 to Technical Manual for Teletypewriter Set AN/UGC-16 NAVELEX 0967-LP-173-6011 (formerly NAVSHIPS 94104)

General Instructions:

This interim change revises the manual to reflect the equipment changes made by Field Change 5-AN/UGC-16. When this change is included in the manual, the manual shall cover the equipment as though Field Change 5, NAVELEX 0967-LP-173-6050, has been accomplished on the equipment. This change does not supersede any other changes or corrections.

Maintenance support activities shall make this change in the technical manual immediately but shall keep the superseded data intact for support of equipments that have not been modified.

Holders of equipment accompanied by technical manuals shall not make this change in the manual until accomplishment of the field change referenced above.

Insert this interim change in the manual immediately after the front cover preceding the title page, prior changes, or interim corrections in effect.

Specific Instructions:

1. Remove the following pages and insert the corrected T-1 pages:

REMOVE	INSERT
8-16	8-16 T-1
8-18	8-18 T-1

2. Add the following page:

Insert 6-2A between pages 6-2 and 6-3.

DATED: 1 July 1978

UNCLASSIFIED

FIGURE 50. Sample of an Instruction Sheet to be Used with Interim (Temporary) and Permanent Changes.

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CONTRACTOR'S CHECK LIST FOR MANUSCRIPTS OF TECHNICAL MANUALS
TYPE II, TYPE IIS, AND TYPE IIX MANUALS

EQUIPMENT _____ SUBMITTAL DATE _____

NOMENCLATURE _____ SECURITY CLASSIFICATION _____

CONTRACT NO _____ NAVSEA NO _____

CONTRACTOR _____ TYPE MANUAL _____

PRELIMINARY ☐ BASIC ☐

PERMANENT CHANGE ☐ COMPLETE REVISION ☐
TEMPORARY CHANGE ☐ UPDATED REVISION ☐

PUBLICATION SPECIFICATIONS _____

	YES ( )	NO ( )	NOT APPL ( )
1. The manuscript consists of the following subdivisions in the order indicated:			
Cover	( )	( )	( )
Front Matter	( )	( )	( )
Chapter 1 General Information	( )	( )	( )
Chapter 2 Operation	( )	( )	( )
Chapter 3 Functional Description	( )	( )	( )
Chapter 4 Scheduled Maintenance	( )	( )	( )
Chapter 5 Troubleshooting	( )	( )	( )
Chapter 6 Corrective Maintenance	( )	( )	( )
Chapter 7 Parts List	( )	( )	( )
Chapter 8 Installation	( )	( )	( )
Index	( )	( )	( )
Appendixes, as applicable	( )	( )	( )
User activity comment sheets	( )	( )	( )
2. Front Matter			
Content and format conform	( )	( )	( )
Cover and Title Page	( )	( )	( )
List of Effective Pages	( )	( )	( )

1 (of 5 pages)

FIGURE 51. Certification Check-off List (Types II, IIS and IIX Manuals).

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TYPE II, TYPE IIS, AND TYPE IIX (Cont)

	YES	NO	NOT APPL
Change Record	( )	( )	( )
Content and Assurance Pages	( )	( )	( )
Table of Contents	( )	( )	( )
List of Illustrations	( )	( )	( )
List of Tables	( )	( )	( )
3. Chapter I - General Information			
Purpose and scope of manual	( )	( )	( )
Meanings of abbreviations and symbols	( )	( )	( )
Warranty/guarantee information	( )	( )	( )
Supersedure data	( )	( )	( )
Interface relationships of technical manual to other publications	( )	( )	( )
Interface relationships of equipments to other equipments	( )	( )	( )
Equipment description	( )	( )	( )
Table defining differences in equipment configuration	( )	( )	( )
Equipment illustration showing relationship of units	( )	( )	( )
Reference data	( )	( )	( )
Tables:	( )	( )	( )
Equipment	( )	( )	( )
Equipment and publications required but not supplied	( )	( )	( )
Field changes and factory changes	( )	( )	( )
Equipment similarities	( )	( )	( )
4. Chapter 2 - Operation			
Introduction description of operator's relationship to equipment	( )	( )	( )
Description of controls, indicators, protective devices, and jacks	( )	( )	( )
Operating procedures:	( )	( )	( )
Operator's turn-on	( )	( )	( )
Modes of operation	( )	( )	( )
Operation under interfering conditions	( )	( )	( )

2 (of 5 pages)

FIGURE 51. Certification Check-off List (Types II, IIS, and IIX Manuals) (Sheet 2 of 5).

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Check the source to verify that this is the current version before use.

TYPE II, TYPE IIS, AND TYPE IIX (Cont)

	YES	NO	NOT APPL
Emergency operation and turn-off	( )	( )	( )
Safety requirements	( )	( )	( )
Operating checks and adjustments	( )	( )	( )
5. Chapter 3 - Functional Description			
Detailed analysis of details of operation	( )	( )	( )
Development of equipment outputs in each mode of operation	( )	( )	( )
Level 1 (over-all) description including over-all block diagram	( )	( )	( )
Level 2 (major functional level) including functional block diagram for each equipment output	( )	( )	( )
Level 3 (circuit level) including simplified schematic diagrams and detailed descriptions not covered in Handbook of Electronic Circuits	( )	( )	( )
6. Chapter 4 - Scheduled Maintenance			
Introduction	( )	( )	( )
Scheduled maintenance statement	( )	( )	( )
Scheduled maintenance action index	( )	( )	( )
Preventive maintenance procedures	( )	( )	( )
Performance tests	( )	( )	( )
7. Chapter 5 - Troubleshooting			
Introduction	( )	( )	( )
Relay, lamp, and protective device indexes	( )	( )	( )
Maintenance turn-on procedure	( )	( )	( )
Reference to troubleshooting procedures	( )	( )	( )
Troubleshooting procedures	( )	( )	( )
Signal flow diagrams	( )	( )	( )
Control diagrams	( )	( )	( )
Distribution diagrams	( )	( )	( )
Maintenance schematic diagrams:	( )	( )	( )
Zones	( )	( )	( )
Zoning	( )	( )	( )

3 (of 5 pages)

FIGURE 51. Certification Check-off List (Types II, IIS, and IIX Manuals) (Sheet 3 of 5).

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Check the source to verify that this is the current version before use.

TYPE II, TYPE IIS, AND TYPE IIX (Cont)

	YES	NO	NOT APPL
Major and minor signal paths	( )	( )	( )
Significant voltages	( )	( )	( )
Functional names of controls	( )	( )	( )
Source and destination of all circuits identified	( )	( )	( )
Current and voltage values of power transformers	( )	( )	( )
Resistance and voltage charts on aprons	( )	( )	( )
Logic, timing circuit, and control cycle diagrams for digital devices	( )	( )	( )
Module logic diagrams, flow charts, coding instruction sheets, and test programs	( )	( )	( )
Troubleshooting dependency diagrams	( )	( )	( )
8. Chapter 6 - Corrective Maintenance			
Introduction	( )	( )	( )
Adjustments and alinement section	( )	( )	( )
Repair section	( )	( )	( )
9. Chapter 7 - Parts List			
Introduction	( )	( )	( )
List of major units	( )	( )	( )
Parts list:	( )	( )	( )
Parts, descriptions (Type II only) with reference to parts location diagrams	( )	( )	( )
Parts divided by units in alpha-numerical order	( )	( )	( )
List of attaching hardware	( )	( )	( )
List of manufacturers	( )	( )	( )
Parts location illustrations for all parts	( )	( )	( )
10. Chapter 8 - Installation			
Mounting dimension diagrams	( )	( )	( )
Interconnecting wiring and cable	( )	( )	( )
Site selection data	( )	( )	( )
List of referenced publications	( )	( )	( )
Tools and material	( )	( )	( )
Unpacking and repacking	( )	( )	( )

4 (of 5 pages)

FIGURE 51. Certification Check-off List (Types II, IIS, and IIX Manuals) (Sheet 4 of 5).

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Check the source to verify that this is the current version before use.

TYPE II, TYPE IIS, AND TYPE IIX (Cont)

	YES	NO	NOT APPL
Preparation of foundations	( )	( )	( )
Input requirements	( )	( )	( )
Installation procedures	( )	( )	( )
Installation checkout:	( )	( )	( )
Phase 1, Installation inspection and pre-energizing procedures	( )	( )	( )
Phase 2, Turn-on and preliminary tests	( )	( )	( )
Phase 3, Installation verification test	( )	( )	( )
Test procedures	( )	( )	( )
Installation standard summary sheet	( )	( )	( )
11. General Requirements			
Information accurately covers all variations of equipment identified by the same nomenclature	( )	( )	( )
Consistent use of terminology through book	( )	( )	( )
Sufficient notes, cautions, and warnings	( )	( )	( )
Security classification of sections conform	( )	( )	( )

I certify that the manuscript conforms to the statements as checked above, and meets the requirements of the specification.

_____
Head of Publications Department

_____
Head of Technical Engineering Group

5 (of 5 pages)

FIGURE 51. Certification Check-off List (Types II, IIS, and IIX Manuals) (Sheet 5 of 5).

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